

Merged research report for the Surreal project

# Omnific-Preserving Automorphisms

Convex-scale stabilizers, definable constants,  
nondefinable monomials, and algebraic-parameter rigidity

Built from eight manuscripts dated September 23, 2026

## Abstract

Which symmetries of a Hahn field preserve its omnific-type integer part, and what does that integer part remember? Let  $k$  have characteristic zero, let  $\Gamma \neq 0$  be a divisible ordered abelian group, write  $C_\delta$  for the principal convex subgroup generated by  $\delta > 0$ , and let  $\mathfrak{o} \subseteq k$  be a unital subring. A strong coefficient-fixing 1-automorphism  $\sigma$  of  $k((t^\Gamma))$  preserves  $\mathfrak{o} \oplus \Pi$ , where  $\Pi$  is the purely infinite ideal, exactly when, for every  $\delta > 0$ , the shift- $\delta$  coefficient of its logarithmic character  $\log(t^{-g}\sigma(t^g))$  vanishes on  $C_\delta$ . The proof uses no exponential–logarithm correspondence and applies verbatim to the surreal field and  $\mathbf{Oz}$ . Where that correspondence is available, the same condition for the operator logarithm  $\text{Log } \sigma$  is equivalent; the two coefficient functionals are different.

Consequences: the stabilizer is trivial exactly in Archimedean rank one; its common fixed field is the Hahn field on the bottom convex subgroup, and for  $(\mathbf{No}, \mathbf{Oz})$  it is  $\mathbb{R}$ ; in finite rank  $r$  its derived length is at most  $r-1$ , with equality for  $\mathbb{Q}^r$ ; explicit successive-scale flows generate a faithful  $\mathbb{Z} \wr \mathbb{Z}$ ; for every set of surreal parameters the pointwise stabilizer is nonsolvable and carries a faithful action of  $\mathbb{R}^\kappa$  for every set cardinal  $\kappa$ . Consequently no set of parameters makes the Conway monomials, the omega-map, any valuation-representative rule, the simplicity relation or the Gonshor exponential definable in  $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ , although the pair defines the three zero-cut projections and the floor map; a monomial can even be sent to a two-term omnific integer. An equivalent form of the criterion bounds the support of the principal unit  $t^{-g}\sigma(t^g)$  itself, and every element of the stabilizer keeps each monomial in the signed Archimedean block of its exponent. A second part proves algebraic-parameter rigidity of  $\mathbf{Oz}$ : embeddings into polynomial, Laurent and finite-type domains land in the coefficients, strong cancellation holds, and  $\text{ML}(\mathbf{Oz}[T_1, \dots, T_n]) = \mathbf{Oz}$ , while explicit commuting derivations integrate to formal flows that no polynomial family realizes. A third part proves that every ring automorphism of  $\mathbf{Oz}$  extends to a strong automorphism of  $\mathbf{No}$ , as constant-term tests detect Hahn summability, and classifies the strong monomial-preserving embeddings that preserve the integer parts; proper self-embeddings of  $\mathbf{No}$  preserving and reflecting  $\mathbf{Oz}$  can move a real number.

This report merges eight independently written manuscripts, prints shared results once with every source named, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 09 (the base text), 04 and 02, pinned to repository commit [fb5c4b5](#), 08, pinned to [befe739](#), the later addition 10, pinned to [58cd8e1](#), and the batch-30 additions 11, 12 and 13, pinned to [cf56b89](#), [a6c68ac](#) and [0865f04](#); their titles are in [sections 1.2](#) and [20.1](#). Results already printed in the omnific Diophantine report [docs/surreal/omnific-diophantine-geometry](#) are cited, not reprinted. The provenance of every section is recorded in [section A](#); non-claims are collected in [section B](#). Finite checks accompany the report and check finite identities only ([section C](#)).

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# 1 Scope, results, and conventions

A Conway normal form carries three kinds of information: coefficients, exponents, and the monomials that represent the exponents. The omnific integers keep the purely infinite part of a normal form and an ordinary integer constant coefficient. This report asks how much of the normal-form presentation is determined by the surreal field together with this distinguished ring, and which automorphisms preserve it.

The answer is asymmetric. The coefficient field, the cut between negative and nonnegative Hahn exponents, the constant coefficient and the floor map are recoverable from the pair  $(\mathbf{No}, \mathbf{Oz})$ ; much of this was already proved in the omnific Diophantine report (section 8). The monomial cross-section is not recoverable. Even an arbitrary *set* of named parameters leaves enough automorphisms to move a monomial to an infinite series and to defeat every definable choice of valuation representatives. The mechanism is a family of derivations that change a higher Archimedean scale by a lower one without ever moving a purely infinite term across the constant-term boundary. Part I classifies these automorphisms and derives the consequences. Part II, from a separate manuscript, shows the complementary rigidity:  $\mathbf{Oz}$  admits no nonconstant *algebraic* family of embeddings, although it admits formal flows of all orders. Part III, from three later manuscripts, removes the strongness hypothesis from the classification for  $(\mathbf{No}, \mathbf{Oz})$  and shows that the rigidity of automorphisms does not pass to proper embeddings.

## 1.1 Principal results

Bracketed numbers name the source manuscripts (section 1.2); [merge] marks statements assembled or generalized by the merge, each with a complete proof. The Hahn conventions are fixed in section 1.5:  $K = k((t^\Gamma))$  with well-ordered supports,  $t^\gamma = \omega^{-\gamma}$  on the surreals,  $\Pi$  the series supported in  $\Gamma^{<0}$ ,  $\mathcal{A}_k = k \oplus \Pi$  and  $\mathcal{R}_o = o \oplus \Pi$ , so that  $\mathcal{R}_{\mathbb{Z}} = \mathbf{Oz}$  when  $k = \mathbb{R}$  and  $\Gamma = \mathbf{No}$ .

- (1) **The convex-support criterion** [04, 09][merge]. For every characteristic-zero  $k$ , every unital  $o \subseteq k$ , every nonzero divisible  $\Gamma$  (set-sized, or  $\Gamma = \mathbf{No}$ ), and every strong  $k$ -linear 1-automorphism  $\sigma$ :  $\sigma(\mathcal{R}_o) = \mathcal{R}_o$  if and only if the logarithmic-character coefficients  $a_{\sigma, \delta}$  vanish on  $C_\delta$  for every  $\delta > 0$ , if and only if  $\sigma$  commutes with the three zero-cut projections (theorem 3.3). The proof is 04's and imports no exponential–logarithm correspondence. For contracting strong derivations the analogous criterion holds with 09's coefficients  $c_\delta$  (theorem 3.4); where the correspondence is available the operator form follows (corollary 3.5). The functionals  $a_{\sigma, \delta}$  and  $c_\delta$  differ; their vanishing conditions are equivalent (example 3.2). For single fixed-shift flows 08 proves the criterion without divisibility (theorem 4.10). Source 10 proves it again in an equivalent form, on the support of the principal unit  $t^{-g}\sigma(t^g)$  rather than its logarithm, with a least-forbidden-shift proof kept as a second route; the inclusion  $\sigma(\mathcal{R}_o) \subseteq \mathcal{R}_o$  already suffices, every signed Archimedean block of exponents is preserved, and the necessity half needs no strongness (theorem 3.8, proposition 3.10, and remark 3.9).
- (2) **Rank and fixed fields** [04, 08, 09, 10]. The stabilizer  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$  is trivial exactly when  $\Gamma$  is Archimedean (theorem 4.13); for set-sized  $\Gamma$  its common fixed field is the Hahn field on the bottom convex subgroup (theorem 4.15); and  $\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$ , also for all field automorphisms of  $\mathbf{No}$  preserving  $\mathbf{Oz}$  (theorem 7.2). For every set  $A$  of parameters the elementary shears fixing  $A$  have common fixed field  $\mathbb{R}((t^{V_A}))$ ,  $V_A$  the rational span of the supports (theorem 7.4). Binomial twists  $t^g \mapsto t^g F(t^\delta)^{\theta(g)}$  with an arbitrary profile  $F$  have fixed field  $k((t^{\ker \theta}))$ , no nontrivial finite orbits and a relatively algebraically closed fixed

- field (theorems 4.18 and 4.20 and corollary 4.21), and in set-sized fields definable elements obey a support bound (proposition 7.8) [10].
- (3) **Group structure** [04, 08, 09]. In finite ordered rank  $r$  the stabilizer and its Lie algebra have derived length at most  $r - 1$ , with equality for  $\mathbb{Q}^r$  (theorems 6.3 and 6.6). Two successive-scale flows generate a faithful  $\mathbb{Z} \wr \mathbb{Z}$  with exact commutator leading terms (theorems 6.8 and 6.10). For every set  $P$  of surreal parameters, the pointwise stabilizer of  $P$  in  $\mathcal{U}_{\mathbf{Oz}}$  carries a faithful action of the full product  $\mathbb{R}^\kappa$  for every set cardinal  $\kappa$  and is nonsolvable (theorems 7.6 and 7.7).
  - (4) **What the pair remembers** [08, 09]. Beyond the reconstruction package of the Diophantine report, the three zero-cut projections  $P_-, P_0, P_+$  on all of  $\mathbf{No}$ , the floor map and truncation at a named monomial are parameter-free definable in  $(\mathbf{No}, \mathbf{Oz})$ , and every field automorphism preserving  $\mathbf{Oz}$  commutes with them, without strongness (theorem 8.2 and corollary 5.2).
  - (5) **What it forgets** [04, 08, 09, 10]. For every set  $P$  of parameters, some strong 1-automorphism preserving  $\mathbf{Oz}$  fixes  $P$  pointwise and moves a Conway monomial to an infinite series; so neither the monomial class nor the omega-map is  $P$ -definable (theorem 9.1), and no  $P$ -definable rule selects a representative of each natural-valuation class (theorem 9.4). Source 10 moves  $\omega^{\omega^\beta}$  to the two-term omnific integer  $\omega^{\omega^\beta} + \omega^{\omega^\beta - \delta}$  and shows that neither the simplicity relation nor the Gonshor exponential is  $P$ -definable, even with all reals and the zero-cut projections named (theorems 9.8 and 9.10).
  - (6) **Surcomplex consequences** [04, 08, 09, 10]. The criterion holds for  $\mathbf{Oz}[i]$ ; real shears commute with conjugation, and 10's two-term witnesses also preserve the order of the real axis and the projections (proposition 10.5); an imaginary-time shear fixing any set moves the real axis inside the leading-term kernel (theorem 10.2); a wild coefficient lift moves an ordinary real (proposition 10.3); and the common fixed field of all automorphisms preserving  $\mathbf{Oz}[i]$  is  $\mathbb{Q}$  (corollary 10.4).
  - (7) **Algebraic-parameter rigidity** [02]. Every nonzero element of  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  divides a monomial whose roots of all orders lie in the ring (corollary 13.3). Consequently injective homomorphisms into polynomial and Laurent rings over a domain land in the coefficients (theorems 15.1 and 15.3), polynomial cylinders over  $\mathbf{Oz}$  recover  $\mathbf{Oz}$  under every isomorphism (theorem 15.4),  $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$  (theorem 16.2), and embeddings of set-sized root-covered rings into finite-type domains land in a finite field of algebraic constants (theorem 14.4). Commuting derivations of  $\mathbf{Oz}$  integrate to formal flows of all orders that no polynomial family realizes, with an exact transcendence degree for their monomial orbit fields (theorems 17.3 and 17.5).
  - (8) **Automatic strongness and proper embeddings** [11, 12, 13]. A set-indexed family in a full Hahn field is summable exactly when, for every  $y$ , only finitely many  $\text{ct}(x_j y)$  are nonzero, and countable  $\{0, 1\}$ -valued tests suffice; so a field isomorphism of full Hahn fields commuting with  $\text{ct}$  up to a coefficient isomorphism is strong (theorems 21.4 and 22.1). Every ring automorphism of  $\mathbf{Oz}$  extends uniquely to a strongly  $\mathbb{R}$ -linear automorphism of  $\mathbf{No}$  (theorem 23.2), which answers question 11.1; every automorphism of  $(\mathbf{No}, \mathbf{Oz})$ , or of a full real Hahn pair with divisible exponent group, is  $M_{\chi, \tau} \circ u$  with  $u$  as in theorem 3.3 (theorem 24.1). Strong exact-monomial embeddings preserving the integer parts correspond to coefficient sections into the Hahn field on the bottom gap, and move reals exactly when the positive value image is not coinitial (theorem 26.3 and corollary 26.7); explicit proper embeddings preserve and reflect  $\mathbf{Oz}$  and send a real  $b$  to  $b + \omega^{-1}$  (theorem 27.3). Further: constant-term adjoints at set size and a proper-class obstruction, closed images and a

continuity dichotomy, nonelementarity, continuum many nonconjugate copies, parameter-fixed copies and surcomplex versions (sections 25 and 28 to 30).

These statements are proved here, but proving a statement here is not a claim that nobody proved it earlier. The general theory of strong Hahn-field automorphisms and of the exponential-derivation correspondence is prior work (section 1.3); each source names its most distinctive contributions, and no exhaustive priority claim is made (section B).

## 1.2 Provenance of this report

The report is built from eight manuscripts written independently: five for parts I and II, described here, and three for part III, described in section 20.1. Four were placed together in commit `f4c9504` (batch 26, items 02, 04, 08 and 09; they keep these numbers here, as their shipped file prefixes do). Manuscripts 02, 04 and 09 pin repository commit `fb5c4b53` (12 commits before the placement); manuscript 08 pins `befe739b` (10 commits before). Both pins contain the batch-24 placement `be06fc8` but precede the batch-24 and batch-25 writes, so no source saw the reconstruction package now in the omnific Diophantine report (its section `odg:def:sec:reconstruction`, placed after both pins). The fifth, manuscript 10, is item 05 of batch 28, placed as an addition in commit `c6359e4` under the file prefix `10-support-cut-`. It pins `58cd8e1`, a commit on a line that does not contain `f4c9504`: its author never saw this report, and its repository comparison predates it. Its material carries the label sub-prefix `opa:sc:.`

| Source | Manuscript                                                                                                                              | Principal contribution to this report                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09     | <i>Omnific-Preserving Automorphisms: convex-scale stabilizers, definable constants, and nondefinable monomials</i>                      | Base text and structure; the criterion for any characteristic-zero $k$ and unital $\mathfrak{o}$ ; the derivation form with $c_\delta$ ; the Archimedean dichotomy; the strong factorization; the finite-rank derived-length bound and its sharpness for the abstract group; set-parameter nondefinability and the absence of any definable representative rule; $\mathbb{R}^\kappa$ actions and nonsolvable stabilizers; projections and truncation. |
| 04     | <i>Automorphisms Preserving the Omnific Integers: a convex-support criterion, fixed fields, and nondefinability of Conway monomials</i> | The import-free proof of the criterion through the logarithmic character (the proof of record here); exact displacement and fixed fields; the bottom subgroup and the relative Hahn hull; the common-positive-shift exponential-logarithm correspondence; the explicit commutator formula; the one-term class and the RV sort; the real axis inside the leading-term kernel.                                                                          |

| Source | Manuscript                                                                                                                                                     | Principal contribution to this report                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08     | <i>What the Omnific Integer Part Remembers: coefficient reconstruction, convex-scale automorphisms, nondefinable monomials, and an explicit wreath product</i> | The divisibility-free threshold for single fixed-shift flows; phase shears; the faithful $\mathbb{Z} \wr \mathbb{Z}$ and exact commutator leading terms; the floor map and the order formula on $\mathbf{Oz}$ ; the wild coefficient lift and the Gaussian fixed field $\mathbb{Q}$ .                                                                                                                                                     |
| 02     | <i>Algebraic-Parameter Rigidity of the Omnific Integers: strong cancellation, finite-type embeddings, and nonalgebraizable formal flows</i>                    | All of <a href="#">part II</a> .                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10     | <i>Omnific Integers Do Not Determine Surreal Monomials: exact support-cut stabilizers, fixed fields, and parameterwise nondefinability</i>                     | A fifth proof of the criterion, in the support-cut form, with the least-forbidden-shift argument as a second route; necessity without strongness; Archimedean blocks; twists with a general profile, their fixed fields, finite orbits and relative algebraic closedness; the set-sized definable-closure bound; the two-term witness; nondefinability of simplicity and of the Gonshor exponential; the question on other integer parts. |
| 11     | <i>Coefficient Gaps and Proper Self-Embeddings of the Surreal Field: omnific preservation, topological dichotomies, and nonconjugate copies</i>                | Base of the embedding sections of <a href="#">part III</a> : the bottom-gap classification, Taylor admissibility, closed images and the continuity dichotomy, nonelementarity, nonconjugate copies, parameter-fixed copies, conjugation-compatible surcomplex embeddings.                                                                                                                                                                 |
| 12     | <i>Automatic Strongness and Proper Embeddings of the Omnific Integers: constant-term duality, full stabilizers, and the surreal-surcomplex boundary</i>        | Base of the automatic-strongness sections of <a href="#">part III</a> : scalar detection, the constant-term isomorphism theorem, automatic strongness of $\text{Aut}(\mathbf{Oz})$ , the factorization at set size and the Archimedean corollary, adjoints and retractions, the proper-class functional, regular cardinals.                                                                                                               |
| 13     | <i>Automatic Hahn Linearity of Omnific Isomorphisms: residue duality, full stabilizers, and coefficient-drifting self-embeddings</i>                           | The same main theorems as 12, printed once; its countable binary witness, isomorphisms between different groups, the coefficient-matrix formula, the example that constant-term preservation does not preserve $\mathbf{Oz}$ , the small-integer-part example and the failure of pair homogeneity.                                                                                                                                        |

Manuscripts 04, 08 and 09 prove one classification at three generalities ([section 3](#)) and the same consequences: rank triviality, fixed field  $\mathbb{R}$ , set-parameter nondefinability and a nonnilpotent stabilizer. They are printed once. 09 is the base because its hypotheses on  $(k, \mathfrak{o})$

are the weakest and its group-theoretic results the widest; 04’s proof is used for the criterion itself because it needs no imported correspondence, and 08’s single-flow version is kept as a marked case because it needs no divisibility. Manuscript 10, added later, is a fifth derivation of the same classification under the same hypotheses (section 3.3). Their reconstruction and definability sections re-derive results now in the Diophantine report; those are cited, and only the remaining extras are printed (section 8). Manuscript 02 has a different subject, algebraic families rather than automorphisms preserving a support condition, and forms part II with the sub-prefix `opa:par:`; its relation to part I is recorded in section 19. None of the four contradicts another or any report in the collection. Section A records, section by section, what each source proved, where the merge had to choose, and every symbol renamed from a source.

Manuscript 10 re-derives the criterion, the rank dichotomy, the strong factorization, the fixed field of a shear, set-parameter nondefinability of the monomials and the Gaussian witnesses. It has the hypotheses of theorem 3.3 exactly: characteristic-zero  $k$ , an arbitrary unital  $\mathfrak{o}$ , divisible  $\Gamma$ , a set or **No**. None of this is a strengthening. These results are printed once, with 10 added to their sources. Its genuinely new material is appended at the end of the sections it belongs to, so that no existing number changes: section 3.3 (support-cut form, second proof, blocks), section 4.7 (general twists and their fixed fields), example 6.13, section 7.4, section 9.2 (two-term witnesses, simplicity and the exponential), proposition 10.5 and question 11.7. 10 contradicts neither this report nor any other; its novelty statement, written before this report existed, is corrected in section 11.1.

Manuscripts 11, 12 and 13, items 03, 04 and 05 of batch 30, were placed as an addition in commit 21375f8. They form part III, placed after part II and before the appendices, so that no existing theorem, section or equation number changes; their labels carry the sub-prefix `opa:as:`. 12 and 13 prove the same main theorems and are printed once, with 12 as the base; 11 is the base of the embedding sections, which contain the embeddings of 12 and 13 as instances. Where the merge chose is recorded in section 20.1. None of the three contradicts this report or another; they answer question 11.1, and the passages of part I that recorded it as open now carry status notes.

### 1.3 Imported results

Conway normal forms, real closedness of **No** and the omnific integers are classical [5, 7]; factorization theory of **Oz** is due to L’Innocente and Mantova [11]. The monomial/character/kernel decomposition of valued Hahn automorphisms is due to Kuhlmann and Serra [10], and its surreal form, with the class-size conventions used here, to Kaplan, Krapp and Serra [9]; their rigidity of order together with simplicity [9, Proposition 5.1] is used in theorem 9.10. Ehrlich and Kaplan’s initial subdomains and canonical integer part [6] are a different notion from a truncation-closed subring or a subring stabilized by an automorphism, and none is substituted for another here [10]. The collection’s surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`) already constructs the fixed-shift derivations and flows used below, in greater generality (its `saut:thm:shiftflow`), and the four-layer decomposition of valued automorphisms (`saut:thm:decomp`).

The correspondence between contracting strong derivations and strong 1-automorphisms is due to Bagayoko, Krapp, Kuhlmann, Panazzolo and Serra.

**Cited theorem 1.1** ([1, Theorem 3.13]; [9, Fact 4.2]). *Let  $k$  have characteristic zero and  $\Gamma$  be an ordered abelian group. If  $\partial$  is a contracting strongly  $k$ -linear derivation of  $k((t^\Gamma))$ , then*

$$\text{Exp}(\partial)(f) = \sum_{n \geq 0} \frac{\partial^n(f)}{n!}$$

is a strongly  $k$ -linear 1-automorphism, and  $\partial \mapsto \text{Exp}(\partial)$  is a bijection between contracting strongly  $k$ -linear derivations and strongly  $k$ -linear 1-automorphisms, with inverse  $\text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$ . Composition corresponds to the Baker–Campbell–Hausdorff operation.

*Remark 1.2* (What is proved where). [1] proves the correspondence for set-sized  $\Gamma$ . Source 09 attributes the surreal case to [9, Fact 4.2]; the placement dossier, reading an HTML rendering, doubted that this Fact states the bijection. The merge re-read the PDF of [9] (version 3): Fact 4.2 does state the bijection, cites [1, Theorem 3.13], and adds that the correspondence “was established in the case that  $G$  is a set, but it also holds for  $G$  a proper class”, without a separate proof. So 09’s citation is accurate, and for  $\Gamma = \mathbf{No}$  the operator-logarithm statements rest on that assertion. The criterion itself does not: its proof (theorem 3.3) and 04’s correspondence on the common-positive-shift class (theorem 4.8) are complete and elementwise, and the explicit constructions on  $\mathbf{No}$  use one set-sized well-ordered set of shifts (proposition 4.4). Every statement below that uses cited theorem 1.1 says so.

## 1.4 Class size and foundations

Unless  $\Gamma = \mathbf{No}$  is explicitly under discussion, exponent groups are sets, so fields and automorphism groups are ordinary sets. For  $\Gamma = \mathbf{No}$  we work in NBG with global choice [04, 08, 09]; every individual Hahn support and every summation family is a set. An automorphism of  $\mathbf{No}$  is an individual class map; no set or NBG class whose members are all such proper-class graphs is formed. A statement about  $\mathcal{U}_{\mathbf{Oz}}$  is a statement about individual maps, their compositions, or a set-indexed action encoded by one class relation; this is the convention of [9]. A common-fixed-field statement is pointwise; nonsolvability means that no ordinary finite derived length annihilates all iterated commutators; valuation representatives are treated through an equivalence relation on nonzero elements, not through proper-class cosets [09]. A quotient notation  $G/H$  in a full-class application abbreviates an additive map on  $G$  vanishing on  $H$  [08]. Bases are chosen only for set-sized vector spaces (lemma 7.3).

## 1.5 Notation and conventions

Part I uses the small- $t$  Hahn convention of sources 04 and 09. Part II keeps its source’s large-monomial convention, which is the convention of the other omnific reports; see section 12.1. Part III uses the  $t$ -convention; its further symbols are fixed in section 20.3. For actual surreals

$$t^\gamma = \omega^{-\gamma}. \quad (1.1)$$

Thus negative  $t$ -exponents are purely infinite terms and positive ones are infinitesimal. The tempting false reading is the opposite: in this Part, II is supported at *negative*  $t$ -exponents, and a “positive shift”  $\delta > 0$  moves a term towards the infinitesimal side. Source 08 wrote in the  $\omega$ -convention with decreasing supports; its statements have been translated, and its derivation  $D_{\varphi, \delta}$  with  $D_{\varphi, \delta}(\omega^g) = \varphi(g)\omega^{g-\delta}$  is  $D_{\delta, \theta}$  below with  $\theta(\gamma) = \varphi(-\gamma) = -\varphi(\gamma)$ .

| Notation            | Meaning in part I, and the sources’ symbols                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $K = k((t^\Gamma))$ | Hahn field, $k$ of characteristic zero; $v(f) = \min \text{supp } f$ , $v(0) = \infty$ ; $\text{lc}(f)$ the coefficient at $v(f)$ . |
| $\Pi$               | Series supported in $\Gamma^{<0}$ , zero included (the purely infinite ideal). Sources: $\mathcal{J}$ (08, 09), $\mathcal{I}$ (04). |

| Notation                                                                      | Meaning in <a href="#">part I</a> , and the sources' symbols                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{A}_k = k \oplus \Pi$                                                | Series supported in $\Gamma^{\leq 0}$ , the boundary ring. Sources: $B$ (09), $\mathcal{A}$ (04), $\mathcal{B}_k$ (08); the Diophantine report's $\mathcal{A}_{\mathbb{k}}(\Gamma)$ .                                                                                                                                                                                                                                  |
| $\mathcal{R}_{\mathfrak{o}} = \mathfrak{o} \oplus \Pi$                        | The integer-part ring for a unital subring $\mathfrak{o} \subseteq k$ ; $\mathcal{R}_{\mathbb{Z}} = \mathbf{Oz}$ for $(\mathbb{R}, \mathbf{No})$ and $\mathcal{R}_{\mathbb{Z}[i]} = \mathbf{Oz}[i]$ for $(\mathbb{C}, \mathbf{No})$ . Sources: $A_D$ (09), $O_D$ (04), $A_k$ (08); the Diophantine report's $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . The letter $D$ of the sources is freed for derivations. |
| $\mathcal{O}, \mathfrak{m}$                                                   | Series supported in $\Gamma^{\geq 0}$ and $\Gamma^{>0}$ : the valuation ring and its maximal ideal.                                                                                                                                                                                                                                                                                                                    |
| $P_-, P_0, P_+$                                                               | Projections onto $\Pi$ , $k$ and $\mathfrak{m}$ in $K = \Pi \oplus k \oplus \mathfrak{m}$ ; $P_0$ restricted to $\mathcal{A}_k$ is the retraction $\text{ct}$ .                                                                                                                                                                                                                                                        |
| $C_{\delta}$                                                                  | Principal convex subgroup $\{\gamma :  \gamma  \leq n\delta \text{ for some } n \geq 1\}$ of $\delta > 0$ . Sources: $C_{\delta}$ (04, 09), $H_{\delta}$ (08), $H(a)$ (02); the transcendence report's $C(a)$ .                                                                                                                                                                                                        |
| $\delta \ll \epsilon$                                                         | $n\delta < \epsilon$ for every ordinary $n \geq 1$ .                                                                                                                                                                                                                                                                                                                                                                   |
| $H_g$                                                                         | $\{h : n h  <  g  \text{ for all } n \geq 1\}$ for $g \neq 0$ [04]; the same convex subgroup as the quotient report's $H_a$ with $a =  g $ .                                                                                                                                                                                                                                                                           |
| $a_{\sigma, \delta}$                                                          | Logarithmic-character coefficient $[t^{\delta}] \log(t^{-g} \sigma(t^g))$ of a 1-automorphism [04].                                                                                                                                                                                                                                                                                                                    |
| $c_{\delta}$                                                                  | Operator coefficient $[t^{\gamma+\delta}] \partial(t^{\gamma})$ of a derivation $\partial$ [09]; for $\partial = \text{Log } \sigma$ it differs from $a_{\sigma, \delta}$ ( <a href="#">example 3.2</a> ).                                                                                                                                                                                                             |
| $D_{\delta, \theta}$                                                          | The derivation $t^{\gamma} \mapsto \theta(\gamma) t^{\gamma+\delta}$ for additive $\theta : \Gamma \rightarrow k$ and $\delta > 0$ . Sources: $D_{\delta, \theta}$ (09), $D_{\delta, a}$ (04), $D_{\varphi, \delta}$ (08, sign as above).                                                                                                                                                                              |
| $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma), \mathcal{U}_{\mathbf{Oz}}$ | Strong $k$ -linear 1-automorphisms preserving $\mathcal{R}_{\mathfrak{o}}$ (independent of $\mathfrak{o}$ by <a href="#">theorem 3.3</a> ). Sources: $\mathcal{U}_{A_D}$ (09), $\mathcal{U}_D(G)$ (04).                                                                                                                                                                                                                |
| $q_{\beta}(\gamma)$                                                           | The coefficient of $\omega^{\beta}$ in the Conway normal form of the <i>exponent</i> $\gamma \in \mathbf{No}$ . Sources: $q_{\eta}$ (09), $[\omega^{\beta}]g$ (04), $c_a$ (08), $\chi_b$ (02), $\ell_{\eta}$ (quotient report).                                                                                                                                                                                        |
| $\exp(u), \text{Exp}, \text{Log}$                                             | The Hahn series $\sum u^n/n!$ for $v(u) > 0$ (E in 08); operator exponential and logarithm.                                                                                                                                                                                                                                                                                                                            |

Two further collisions are resolved in the new material. Source 09's parameter monomial  $H = \omega^{\eta}$  and its family  $H_j$  are written  $\eta = \omega^{\beta}$  and  $\eta_j$  here, leaving  $H_g$  to 04's convex subgroup; and 08's automorphisms  $A_1, B_1$  are written in 04's normalization  $A_s, B_u$ , with  $A_1^{(08)} = A_{-1}$  and  $B_1^{(08)} = B_{-1}$  ([section 6](#)). A 1-automorphism is a field automorphism that preserves the valuation and the leading coefficient of every nonzero series. *Strong* means that every Hahn-summable family is sent to a summable family with the same sum; a strong automorphism is one whose inverse is also strong (04's convention); *k-linear* includes fixing  $k$  pointwise.

**Source 10's symbols.** Source 10 writes in the large-monomial convention  $X^g = \omega^g$  with reverse-well-ordered supports, that is  $X^g = t^{-g}$ ; its statements are translated into the  $t$ -convention, and a  $t$ -exponent  $\gamma < 0$  is its exponent  $g = -\gamma > 0$ . Two of its letters collide with this report's, and the tempting false readings are wrong:

- 10's  $H_h$ , the principal convex subgroup generated by  $h$ , is  $C_{\delta}$  here, while 10's  $C_g$ , the

elements infinitesimal relative to  $g$ , is  $H_g$  here. The letters  $C$  and  $H$  are *swapped*: 10's  $C_g$  is not a principal convex subgroup, and  $H_\delta$  here is not 10's  $H_h$ .

- 10's projection  $\text{pr}_+$  onto the positive  $\omega$ -exponents, the purely infinite part, is  $P_-$  here, and its  $\text{pr}_-$  is  $P_+$ ; its  $\text{ct}$  is  $P_0$ . The subscripts are reversed by the sign convention.

Further renamings: 10's positive-exponent ideal  $\mathcal{I}$  is  $\Pi$ ; its  $\mathfrak{m}$  is  $\mathfrak{m}$ , the same set; its  $R_D = D + \mathcal{I}$  is  $\mathcal{R}_o$  with  $D = o$ ; its multiplier ring  $B$  is  $\mathcal{A}_k$ ; its stabilizer  $\mathcal{U}_D$  is  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ ; its shift  $h$  is  $\delta$ , and its condition “ $h \ll g$ ” is  $\delta \ll |\gamma|$ ; its principal unit  $u_g = 1 + \varepsilon_g$  is  $1 + \varepsilon_\sigma(-g)$  (section 3.3); its exponent functional  $\lambda_b = [\omega^b] \cdot$  is  $q_b$ ; its additive profile  $P$  and its formal variable  $T$  are  $\Theta$  and  $\xi$ , since  $P$  and  $T$  are taken here; its twist  $\sigma_{\lambda, F, h}$  is  $\sigma_{h, -\lambda}^F$  of (4.4), the sign coming from  $g = -\gamma$ ; its class twist  $\Sigma_{b, h}$  is  $\Sigma_{b, \delta}$  of (9.2); its expanded structure  $\mathcal{N}^\dagger$  is  $\mathfrak{S}^\dagger$  of theorem 9.10.

## Part I

# Automorphisms preserving the omnific integers

## 2 Hahn support, the boundary ring, and strong maps

### 2.1 Summability

An element of  $K = k((t^\Gamma))$  is a formal sum  $f = \sum_{\gamma \in S} a_\gamma t^\gamma$  with  $S \subseteq \Gamma$  well ordered. A set-indexed family  $(f_i)_{i \in I}$  is *Hahn summable* if the union of its supports is well ordered and each exponent lies in only finitely many of the supports; its sum is formed coefficientwise with finite sums in  $k$ . No convergence in the fine topology of the surreals is involved; a strong sum is a support-controlled algebraic construction [04, 08, 09].

We use the Neumann support principle in its word form: if  $S$  is well ordered and  $T \subseteq \Gamma^{>0}$  is well ordered, then  $S + \langle T \rangle_{\mathbb{N}}$  is well ordered, and each exponent receives only finitely many contributions from a choice of an element of  $S$  and a finite word in  $T$ , counted across all word lengths [1]. The one-shift case is proved directly.

**Lemma 2.1** (One-shift summation [04, 08, 10]). *If  $T \subseteq \Gamma$  is well ordered and  $\delta > 0$ , then  $T + \mathbb{N}\delta$  is well ordered, and each  $\gamma \in \Gamma$  has only finitely many representations  $\gamma = g + n\delta$  with  $g \in T$  and  $n \in \mathbb{N}$ .*

*Proof.* An infinite strictly decreasing sequence  $g_j + n_j\delta$  has an infinite subsequence on which the  $n_j$  are nondecreasing; on it the  $g_j$  strictly decrease, contradicting well-ordering of  $T$ . Infinitely many representations of one  $\gamma$  would have unbounded, hence eventually increasing, values of  $n$ , and the corresponding  $g = \gamma - n\delta$  would strictly decrease in  $T$ .  $\square$

For  $h \in \mathfrak{m}$  the series  $\exp(h) = \sum_{n \geq 0} h^n/n!$  and  $\log(1+h) = \sum_{n \geq 1} (-1)^{n+1} h^n/n$  are Hahn summable, and they are inverse group isomorphisms between  $(\mathfrak{m}, +)$  and  $(1 + \mathfrak{m}, \cdot)$ . This local exponential is the only exponential used in the criterion; neither  $\exp$  nor  $\text{Exp}$  asserts anything about the Gonshor exponential of **No**.

**Lemma 2.2** (Coefficient polynomial [04]). *For  $h \in \mathfrak{m}$  and  $\delta > 0$  the expression  $P_{h, \delta}(X) = \sum_{n \geq 1} \frac{X^n}{n!} [t^\delta] h^n$  is a polynomial over  $k$ . For every  $q \in \mathbb{Q}$  one has  $P_{h, \delta}(q) = [t^\delta] \exp(qh)$ , and its coefficient of  $X$  is  $[t^\delta] h$ .*

*Proof.* The support principle gives only finitely many contributions to the exponent  $\delta$  across all powers  $h^n$ , not merely within each power. So only finitely many displayed terms are nonzero. The rest is coefficientwise.  $\square$

## 2.2 The rings and the zero cut

The additive decompositions

$$K = \Pi \oplus k \oplus \mathfrak{m}, \quad \mathcal{A}_k = k \oplus \Pi, \quad \mathcal{R}_{\mathfrak{o}} = \mathfrak{o} \oplus \Pi, \quad \mathcal{O} = k \oplus \mathfrak{m} \quad (2.1)$$

define the projections  $P_-, P_0, P_+$  [04, 09]. Membership in  $\Pi$  means that *all* exponents are negative, which is stronger than  $v(f) < 0$ : the series  $t^{-1} + t$  has negative valuation and is not in  $\Pi$  [04].  $\Pi$  is a nonunital ideal of  $\mathcal{A}_k$  and of  $\mathcal{R}_{\mathfrak{o}}$ ;  $\mathcal{A}_k$  is not a valuation ring;  $P_0$  is a ring retraction  $\mathcal{A}_k \rightarrow k$  restricting to  $\mathcal{R}_{\mathfrak{o}} \rightarrow \mathfrak{o}$ , but not a ring map on  $K$ :  $P_0(t^{-1})P_0(t) = 0 \neq 1 = P_0(1)$ . For  $k = \mathbb{R}$ ,  $\mathfrak{o} = \mathbb{Z}$  the ring  $\mathcal{R}_{\mathbb{Z}}$  is an integer part of  $K$ ; for  $\Gamma = \mathbf{No}$  it is  $\mathbf{Oz}$  and its floor function is the Diophantine report's `odg:thm:floor`.

The units  $\mathcal{A}_k^\times = k^\times$ , and more generally the fact that a product of two elements of  $\mathcal{A}_k$  lying in  $k^\times$  has constant factors, are the Diophantine report's constant-product lemma (`odg:lem:constantproduct`, `odg:def:lem:units`); sources 04, 08 and 09 reprove it, and it is not reprinted.

**Lemma 2.3** (The divisible part [04, 08]). *If  $\bigcap_{n \geq 1} n\mathfrak{o} = 0$  (for example  $\mathfrak{o} = \mathbb{Z}$  or  $\mathbb{Z}[i]$ ), then  $\Pi = \bigcap_{n \geq 1} n\mathcal{R}_{\mathfrak{o}}$ , the largest divisible subgroup of  $\mathcal{R}_{\mathfrak{o}}$ ; it is preserved by every ring automorphism of  $\mathcal{R}_{\mathfrak{o}}$ . This holds for every ordered abelian  $\Gamma$ , divisible or not.*

*Proof.* A series in  $\Pi$  is  $n$  times a series with the same support. Conversely the constant coefficient of an element of the intersection lies in  $\bigcap_n n\mathfrak{o} = 0$ . Every divisible subgroup lies in the intersection.  $\square$

For  $\mathbf{Oz}$  this is the identity `odg:eq:divkernel` of the Diophantine report; it is printed because the threshold theorem uses it for non-divisible  $\Gamma$ .

**Definition 2.4** (Principal convex scale). For  $\delta > 0$  let  $C_\delta = \{\gamma \in \Gamma : |\gamma| \leq n\delta \text{ for some } n \geq 1\}$ , and write  $\delta \ll \epsilon$  if  $n\delta < \epsilon$  for all  $n \geq 1$ . Then  $\gamma \notin C_\delta$  if and only if  $\delta \ll |\gamma|$ ;  $C_\delta \subsetneq C_\epsilon$  if and only if  $\delta \ll \epsilon$ ; and  $C_\delta$  is divisible when  $\Gamma$  is. For  $g \neq 0$  the set  $H_g = \{h : n|h| < |g| \text{ for all } n \geq 1\}$  is a convex subgroup, and a finite sum of positive elements of  $H_g$  is still smaller than  $|g|$  [04].

## 3 The convex-support criterion

### 3.1 Two coefficient functionals

Let  $\sigma$  be a strong  $k$ -linear 1-automorphism. Since  $t^{-g}\sigma(t^g)$  has leading term 1, its local logarithm is defined [04]:

$$h_\sigma(g) = \log(t^{-g}\sigma(t^g)) \in \mathfrak{m}, \quad a_{\sigma,\delta}(g) = [t^\delta]h_\sigma(g) \quad (\delta > 0), \quad (3.1)$$

so that  $\sigma(t^g) = t^g \exp(h_\sigma(g))$ . For a contracting strong  $k$ -derivation  $\partial$  (that is,  $v(\partial f) > v(f)$  for  $f \neq 0$ ) put [09]

$$c_\delta(\gamma) = [t^{\gamma+\delta}]\partial(t^\gamma) \quad (\delta > 0), \quad \text{so} \quad \partial(t^\gamma) = t^\gamma \sum_{\delta > 0} c_\delta(\gamma)t^\delta. \quad (3.2)$$

For fixed  $\gamma$  the nonzero terms have well-ordered set support; no assumption is made that all shifts together form one well-ordered set.

**Lemma 3.1** (Additivity [04, 09]). *The maps  $h_\sigma : \Gamma \rightarrow (\mathfrak{m}, +)$ ,  $a_{\sigma, \delta}$  and  $c_\delta$  are additive, hence  $\mathbb{Q}$ -linear when  $\Gamma$  is divisible. If  $\Gamma$  is divisible and an additive  $\theta : \Gamma \rightarrow k$  vanishes on  $(-\delta, 0)$ , it vanishes on  $C_\delta$ .*

*Proof.* The quotient at  $g + g'$  is the product of the quotients at  $g$  and  $g'$ , and log takes products to sums. For  $c_\delta$  the product rule gives  $t^{-\gamma-\eta}\partial(t^{\gamma+\eta}) = t^{-\gamma}\partial(t^\gamma) + t^{-\eta}\partial(t^\eta)$ . For the last claim, given  $0 \neq h \in C_\delta$  choose  $N$  with  $|h|/N < \delta$ ; one of  $\pm h/N$  lies in  $(-\delta, 0)$ , and additivity gives  $\theta(h) = 0$ .  $\square$

**Example 3.2** (The functionals differ [merge]). Let  $\Gamma = \mathbf{No}$ ,  $k = \mathbb{R}$ , and  $\partial = D_{1, q_1} + D_{\omega, q_2}$  (notation of section 1.5; the  $t$ -exponent  $\gamma$  has  $q_1(\gamma)$  and  $q_2(\gamma)$  as its coefficients at  $\omega$  and  $\omega^2$ ). Take  $\sigma = \text{Exp}(\partial)$ , an element of  $\mathcal{U}_{\mathbf{Oz}}$  by proposition 4.4, and  $g = \omega^2$ . Then  $\partial(t^g) = t^{g+\omega}$  and  $\partial^2(t^g) = t^{g+\omega+1} + t^{g+2\omega}$ , and the only word in the shifts  $\{1, \omega\}$  reaching the exponent  $\omega + 1$  applies the shift  $\omega$  first;  $\log(1 + u) = u - u^2/2 + \dots$  adds nothing at  $\omega + 1$ , since  $u^2$  starts at  $2\omega$ . Hence

$$t^{-g}\sigma(t^g) = 1 + t^\omega + \frac{1}{2}t^{\omega+1} + (\text{terms at exponents } > \omega + 1), \quad a_{\sigma, \omega+1}(\omega^2) = \frac{1}{2},$$

whereas  $c_{\omega+1}(\omega^2) = 0$ : the operator coefficients of  $\partial$  at  $g$  are  $c_\omega(g) = 1$  and zero at every other shift. So  $a_{\sigma, \delta} \neq c_\delta$  in general; 04's condition concerns the logarithm of the monomial image, 09's the logarithm of the operator. Both vanishing conditions hold here, and by theorem 3.3 and corollary 3.5 each is equivalent to preservation of  $\mathbf{Oz}$ , hence to the other. The placement dossier checked the example by exact computation.

### 3.2 The criterion

The following statement combines 09's generality with 04's proof. 04 proved it for  $k = \mathbb{R}, \mathbb{C}$  and  $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$ , passing through lemma 2.3; 09 proved it for every characteristic-zero  $k$  and unital  $\mathfrak{o}$ , but through the operator logarithm (cited theorem 1.1). The first step below is 09's and removes the need for  $\bigcap n\mathfrak{o} = 0$ ; the rest is 04's.

**Theorem 3.3** (Convex-support criterion [04, 09, 10][merge]). *Let  $k$  have characteristic zero, let  $\mathfrak{o} \subseteq k$  be a unital subring, let  $\Gamma \neq 0$  be divisible (a set, or  $\Gamma = \mathbf{No}$ ), and let  $\sigma$  be a strong  $k$ -linear 1-automorphism of  $K = k((t^\Gamma))$ . The following are equivalent:*

- (i)  $\sigma(\mathcal{R}_\mathfrak{o}) = \mathcal{R}_\mathfrak{o}$ ;
- (ii)  $\sigma(\mathcal{A}_k) \subseteq \mathcal{A}_k$ ;
- (iii)  $\sigma(\Pi) \subseteq \Pi$ ;
- (iv)  $a_{\sigma, \delta}(C_\delta) = 0$  for every  $\delta > 0$ ;
- (v) for every  $g \neq 0$ , every exponent  $\delta \in \text{supp } h_\sigma(g)$  satisfies  $\delta \ll |g|$ ;
- (vi)  $\sigma$  commutes with  $P_-$ ,  $P_0$  and  $P_+$ .

When they hold,  $\sigma(\Pi) = \Pi$ ,  $\sigma(\mathcal{A}_k) = \mathcal{A}_k$ ,  $\sigma(\mathfrak{m}) = \mathfrak{m}$ , the constant coefficient of every series is preserved, and every purely infinite monomial stays purely infinite, though not necessarily a monomial. In particular the group  $\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)$  does not depend on  $\mathfrak{o}$ .

*Proof.* (i) $\Rightarrow$ (ii) [09]. For  $\gamma < 0$ ,  $t^\gamma \in \mathcal{R}_\mathfrak{o}$ , so  $\sigma(t^\gamma) \in \mathcal{R}_\mathfrak{o} \subseteq \mathcal{A}_k$ . An element of  $\mathcal{A}_k$  is  $c + \sum_{\gamma < 0} a_\gamma t^\gamma$  with  $c \in k$ ; strong  $k$ -linearity writes its image as  $c + \sum a_\gamma \sigma(t^\gamma)$ , a summable family of series supported in  $\Gamma^{\leq 0}$ , whose sum is supported there.

(ii) $\Rightarrow$ (iv) [04]. Fix  $\delta > 0$  and  $0 \neq g \in C_\delta$ ; put  $x = -|g| < 0$ , so  $h_\sigma(x) = \pm h_\sigma(g)$ , and choose  $N \geq 1$  with  $|g| \leq N\delta$ . By lemma 2.2,  $P(X) = [t^\delta] \exp(Xh_\sigma(x))$  is a polynomial. For rational

$0 < q < 1/N$ , divisibility gives  $qx \in \Gamma^{<0}$  and additivity gives  $\sigma(t^{qx}) = t^{qx} \exp(qh_\sigma(x))$ , whose coefficient at  $qx + \delta$  is  $P(q)$ . Since  $q|g| < \delta$ , this exponent is strictly positive, while  $\sigma(t^{qx}) \in \mathcal{A}_k$  has no positive exponents. So  $P$  has infinitely many zeros and vanishes; its linear coefficient is  $a_{\sigma,\delta}(x)$ , so  $a_{\sigma,\delta}(g) = 0$ . This tests every coefficient of the logarithm; testing only the first nonzero one would not suffice.

(iv) $\Leftrightarrow$ (v) [04].  $g \notin C_\delta$  means  $\delta \ll |g|$ ; consider each nonzero coefficient.

(v) $\Rightarrow$ (iii) [04]. For  $g < 0$ ,  $\text{supp } h_\sigma(g) \subseteq H_g^{>0}$ , so every exponent of  $\exp(h_\sigma(g))$  is 0 or a finite sum of elements of  $H_g^{>0}$ , hence smaller than  $|g|$ ; thus  $\sigma(t^g) = t^g \exp(h_\sigma(g)) \in \Pi$ . For  $f \in \Pi$  write  $\sigma(f) = \sum c_g \sigma(t^g)$  by strongness; all supports are negative.

(iii) $\Rightarrow$ (vi) [04]. A 1-automorphism maps  $\mathfrak{m}$  into  $\mathfrak{m}$ , and  $\sigma$  fixes  $k$ ; so each summand of (2.1) is mapped into itself and  $\sigma$  commutes with the projections.

(vi) $\Rightarrow$ (i) [04]. If  $\sigma(y) \in \Pi$  with  $y = y_- + c + y_+$ , then  $0 = P_0 \sigma(y) = c$  and  $0 = P_+ \sigma(y) = \sigma(y_+)$ , so  $y \in \Pi$ . With (iii) this gives  $\sigma(\Pi) = \Pi$ , and since  $\sigma$  fixes  $\mathfrak{o} \subseteq k$ ,  $\sigma(\mathcal{R}_\mathfrak{o}) = \mathcal{R}_\mathfrak{o}$ . The same argument gives  $\sigma(\mathcal{A}_k) = \mathcal{A}_k$  and  $\sigma(\mathfrak{m}) = \mathfrak{m}$ , and (vi) preserves constant coefficients. Condition (ii) does not mention  $\mathfrak{o}$ , which proves the last assertion.  $\square$

Only (ii) $\Rightarrow$ (iv) uses divisibility; it divides an *exponent*, while characteristic zero separately permits rational scalars [09]. The argument is elementwise with set-sized supports, so it applies verbatim to  $\Gamma = \mathbf{No}$ ,  $k = \mathbb{R}$ ,  $\mathfrak{o} = \mathbb{Z}$ , that is to  $\mathbf{Oz}$ , and to  $\mathbf{Oz}[i]$  with  $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$  [04]. The theorem classifies *existing* strong maps: an arbitrary additive  $h : \Gamma \rightarrow \mathfrak{m}$  need not define a strong surjective field map by  $t^g \mapsto t^g \exp h(g)$  [04] (question 11.2).

**Theorem 3.4** (Boundary-preserving derivations [09]). *Let  $\Gamma \neq 0$  be divisible and let  $\partial$  be a contracting strong  $k$ -derivation of  $K$ . Then  $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$  if and only if  $c_\delta(C_\delta) = 0$  for every  $\delta > 0$ . In that case  $\partial(\mathcal{A}_k) \subseteq \Pi$ ,  $\partial(\Pi) \subseteq \Pi$ , and  $\partial$  commutes with  $P_-$ ,  $P_0$ ,  $P_+$ .*

*Proof.* If  $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$  and  $-\delta < \gamma < 0$ , then  $t^\gamma \in \mathcal{A}_k$  and  $\gamma + \delta > 0$ , so  $c_\delta(\gamma) = 0$ ; lemma 3.1 extends this to  $C_\delta$ . Conversely, if  $\gamma < 0$  and  $c_\delta(\gamma) \neq 0$ , then  $\gamma \notin C_\delta$ , so  $-\gamma > n\delta$  for all  $n$  and in particular  $\gamma + \delta < 0$ . Every term of  $\partial(t^\gamma)$  is negative; constants are killed; strongness sums over any negative support. Positive exponents stay positive under positive shifts, which gives the projection identities.  $\square$

The condition excludes a boundary term as well as a crossing term: testing  $-\delta/2$  gives  $c_\delta(-\delta) = 0$ . This matters when constant coefficients are restricted to  $\mathfrak{o}$  rather than  $k$  [09]. Source 04 proves the same criterion for derivations with common positive shifts (theorem 4.7), testing  $t^{-u}$  for  $0 < u \leq \delta$ .

**Corollary 3.5** (Operator form [09]). *Let  $\sigma$  be a strong  $k$ -linear 1-automorphism of the form  $\sigma = \text{Exp}(\partial)$  for a contracting strong  $k$ -derivation  $\partial$  with  $\text{Log}(\sigma) = \partial$ . Then the conditions of theorem 3.3 are equivalent to  $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$  and to  $c_\delta(C_\delta) = 0$  for all  $\delta > 0$ . For set-sized  $\Gamma$  every strong  $k$ -linear 1-automorphism has this form, so*

$$\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma) = \text{Exp}(\mathfrak{L}_A), \quad \mathfrak{L}_A = \{\partial \text{ contracting strong } k\text{-derivation} : c_\delta(C_\delta) = 0 \text{ for all } \delta > 0\}.$$

For  $\Gamma = \mathbf{No}$  the same equality holds if the correspondence of cited theorem 1.1 is granted for proper-class  $\Gamma$ , as [9, Fact 4.2] asserts (remark 1.2); on the common-positive-shift class it is proved in theorem 4.8.

*Proof.* If the conditions hold,  $\sigma(\mathcal{A}_k) = \mathcal{A}_k$ ; then each  $(\sigma - \text{id})^n$  maps  $\mathcal{A}_k$  into  $\mathcal{A}_k$ , and  $\text{Log}(\sigma)$  is a strongly summable family of series supported in  $\Gamma^{\leq 0}$ , so  $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ . This step is taken in the  $k$ -algebra  $\mathcal{A}_k$ , not in  $\mathcal{R}_\mathfrak{o}$ , so no division by integers inside  $\mathfrak{o}$  is needed. Theorem 3.4 converts this into the coefficient condition. Conversely, under the coefficient condition every

iterate of  $\partial$  maps  $\mathcal{A}_k$  into  $\Pi$  and kills  $k$ , so  $\text{Exp}(\pm\partial)$  preserve  $\mathcal{R}_\circ$ . The last assertions are [cited theorem 1.1](#).  $\square$

The two routes are genuinely different. 04's functional is read off the monomial images and needs no correspondence; 09's is read off the generator and gives the Lie-theoretic structure of [section 6](#). Neither condition is stated in terms of the other: each is equivalent to preservation of  $\mathcal{R}_\circ$  ([example 3.2](#)).

*Remark 3.6* (What has and has not been classified [04, 09]). The classification is inside the *strong, coefficient-fixing, leading-term-preserving* automorphisms. It does not assert that every abstract field automorphism preserving  $\mathcal{R}_\circ$  is strong; for the real pair, fixing of  $\mathbb{R}$  is automatic ([corollary 5.2](#)), strongness is not known to be. The coefficient functionals come from an existing strong map; arbitrary functions satisfying the vanishing condition cannot be assembled into an operator without checking summability and invertibility.

*Status* (batch 30). For the real pair, strongness is now known to be automatic as well: every field automorphism of **No** preserving **Oz**, and every automorphism of a full real Hahn pair with divisible exponent group, is strong ([theorem 23.2](#) and [corollary 23.4](#)), so the classification covers them all ([theorem 24.1](#)). The remark stands as written for a general  $(k, \circ)$  ([question 32.13](#)), and its last sentence is unaffected ([remark 24.2](#)).

### 3.3 The support-cut form and a second route [10]

Source 10 proves [theorem 3.3](#) a fifth time, without knowledge of this report and under the same hypotheses. Its condition bounds the support of the principal unit itself rather than of its logarithm, and its necessity proof isolates the least forbidden shift instead of testing a polynomial in a rational parameter. The two forms are equivalent, so the statement is printed once, as further equivalent conditions, and 10's proof is kept as a marked second route. For a strong  $k$ -linear 1-automorphism  $\sigma$  put

$$\varepsilon_\sigma(\gamma) = t^{-\gamma}\sigma(t^\gamma) - 1 \in \mathfrak{m}, \quad 1 + \varepsilon_\sigma(\gamma) = \exp(h_\sigma(\gamma)). \quad (3.3)$$

Multiplicativity makes  $\gamma \mapsto 1 + \varepsilon_\sigma(\gamma)$  a character  $\Gamma \rightarrow (1 + \mathfrak{m}, \cdot)$  [10]. For  $u \in 1 + \mathfrak{m}$  and  $q \in \mathbb{Q}$  let  $u^q = \sum_{r \geq 0} \binom{q}{r} (u - 1)^r = \exp(q \log u)$ ; the sum is Hahn summable by the support principle, and  $(u^q)^n = u^{qn}$  for integers  $n$  is a formal identity of binomial series [10].

**Lemma 3.7** (Leading-one roots [10]). *If  $u, w \in 1 + \mathfrak{m}$  and  $u^n = w^n$  for some integer  $n \geq 1$ , then  $u = w$ . Hence, for a character  $c : \Gamma \rightarrow 1 + \mathfrak{m}$  on a divisible  $\Gamma$ ,  $c(q\gamma) = c(\gamma)^q$  for all  $q \in \mathbb{Q}$  and  $\gamma \in \Gamma$ .*

*Proof.* Otherwise  $u/w = 1 + \eta$  with  $0 \neq \eta \in \mathfrak{m}$ , and the term of least exponent of  $(1 + \eta)^n - 1$  is  $n$  times that of  $\eta$ , which is nonzero in characteristic zero. For  $q = a/b$  with  $b \geq 1$ , both  $c(q\gamma)$  and  $c(\gamma)^q$  have  $b$ th power  $c(a\gamma) = c(\gamma)^a$ .  $\square$

**Theorem 3.8** (Support-cut form [10]). *Under the hypotheses of [theorem 3.3](#), each of the following conditions is equivalent to the conditions (i)–(vi) there:*

- (vii)  $\sigma(\mathcal{R}_\circ) \subseteq \mathcal{R}_\circ$ ;
- (viii) *for every  $\gamma < 0$ , every  $\delta \in \text{supp } \varepsilon_\sigma(\gamma)$  satisfies  $\delta \ll |\gamma|$ , that is  $\text{supp } \varepsilon_\sigma(\gamma) \subseteq H_\gamma$ ;*
- (ix)  $\text{supp } \varepsilon_\sigma(\gamma) \subseteq H_\gamma$  for every  $\gamma \neq 0$ .

*In source 10's notation, (viii) reads: every shift  $h > 0$  occurring in  $\varepsilon_g$ , for  $g > 0$ , satisfies  $nh < g$  for every  $n \geq 1$ .*

*First route, from the proof of [theorem 3.3](#).* (i) $\Rightarrow$ (vii) is trivial, and (vii) $\Rightarrow$ (ii) is the proof of (i) $\Rightarrow$ (ii) there, which uses only  $\sigma(t^\gamma) \in \mathcal{R}_\mathfrak{o}$  for  $\gamma < 0$ . For fixed  $\gamma \neq 0$  the series  $\varepsilon_\sigma(\gamma) = \exp(h_\sigma(\gamma)) - 1$  has its support among the finite sums of elements of  $\text{supp } h_\sigma(\gamma)$ , and  $h_\sigma(\gamma) = \log(1 + \varepsilon_\sigma(\gamma))$  has its support among the finite sums of elements of  $\text{supp } \varepsilon_\sigma(\gamma)$ . Since  $H_\gamma$  is a subgroup, (ix) is equivalent to condition (v) of [theorem 3.3](#); this is 10's logarithmic corollary. Finally (ix) $\Rightarrow$ (viii) is trivial, and (viii) $\Rightarrow$ (ix) because, for  $\gamma > 0$ ,  $1 + \varepsilon_\sigma(\gamma) = (1 + \varepsilon_\sigma(-\gamma))^{-1} = \sum_{n \geq 0} (-\varepsilon_\sigma(-\gamma))^n$  has its support among the finite sums of elements of  $\text{supp } \varepsilon_\sigma(-\gamma) \subseteq H_{-\gamma} = H_\gamma$ .  $\square$

*Second route: least forbidden shift [10].* We prove (vii) $\Rightarrow$ (viii) directly. Fix  $\gamma < 0$  and write  $\varepsilon_\sigma(\gamma) = \sum_{\delta \in \mathcal{S}} b_\delta t^\delta$  with  $\mathcal{S} \subseteq \Gamma^{>0}$  well ordered and all  $b_\delta \neq 0$ . Suppose some  $\delta \in \mathcal{S}$  lies outside  $H_\gamma$ , and let  $\delta$  be the *least* such element; the elements of  $\mathcal{S}$  below  $\delta$  lie in  $H_\gamma$ . As  $\delta \notin H_\gamma$  there is  $N \geq 1$  with  $|\gamma| \leq N\delta$ . Choose a rational  $0 < q < 1/N$ ; divisibility gives  $q\gamma \in \Gamma$  with  $0 < q|\gamma| < \delta$ , and [lemma 3.7](#) gives  $t^{-q\gamma}\sigma(t^{q\gamma}) = (1 + \varepsilon_\sigma(\gamma))^q$ . In this binomial series the linear term contributes  $qb_\delta$  at the exponent  $\delta$ . A term of degree  $r \geq 2$  contributes there only through  $\delta = \delta_1 + \dots + \delta_r$  with  $\delta_j \in \mathcal{S}$ ; each  $\delta_j$  is then smaller than  $\delta$ , hence in  $H_\gamma$ , and so is their sum, which is impossible. So the coefficient of  $\sigma(t^{q\gamma})$  at the exponent  $q\gamma + \delta > 0$  is exactly  $qb_\delta \neq 0$ , whereas  $t^{q\gamma} \in \Pi \subseteq \mathcal{R}_\mathfrak{o}$  and no element of  $\mathcal{R}_\mathfrak{o}$  has a positive exponent. This contradicts (vii).

Conversely, under (viii) every exponent of  $\sigma(t^\gamma)$ ,  $\gamma < 0$ , is  $\gamma$  or  $\gamma + \delta$  with  $\delta \ll |\gamma|$ , hence negative; strongness gives  $\sigma(\Pi) \subseteq \Pi$ , which is (iii) of [theorem 3.3](#), and 10 concludes by the same injectivity and surjectivity argument as (iii) $\Rightarrow$ (vi) $\Rightarrow$ (i) there.  $\square$

The two necessity proofs are genuinely different. That of [theorem 3.3](#) (from 04) shows, for each shift, that the coefficient of  $\exp(qh_\sigma(x))$  is a polynomial in  $q$  with infinitely many zeros, and so tests every coefficient of the logarithm. 10's proof needs one rational  $q$ : at the least shift outside  $H_\gamma$  no nonlinear term of the binomial series can contribute. Both use divisibility of  $\Gamma$  only to form  $q\gamma$ , and neither uses  $\mathfrak{o}$  beyond  $\mathfrak{o} \subseteq k$ . The inclusion form (vii) is implicit in the proof of [theorem 3.3](#) but not in its statement; 10 states it.

*Remark 3.9* (Necessity without strongness [10]). The second route applies  $\sigma$  only to monomials, and uses only that  $\sigma$  is a field automorphism with  $\sigma(t^\gamma) \in t^\gamma(1 + \mathfrak{m})$  for every  $\gamma$ : multiplicativity on monomials, [lemma 3.7](#), and  $\sigma(t^{q\gamma}) \in \mathcal{R}_\mathfrak{o}$ . So every  $k$ -fixing 1-automorphism  $\sigma$  of  $K$  with  $\sigma(\mathcal{R}_\mathfrak{o}) \subseteq \mathcal{R}_\mathfrak{o}$ , strong or not, satisfies (viii), and (ix) by inversion of units. What fails without strongness is the passage from monomials to arbitrary series, which the converse and all consequences for nonmonomial elements use. For **(No, Oz)** fixing of  $\mathbb{R}$  is automatic ([corollary 5.2](#)), so every field automorphism of **No** that preserves **Oz** and the leading terms of monomials satisfies the monomial condition [10]. The merge notes that the necessity half of 04's route is equally monomial: it evaluates  $\sigma$  only at the monomials  $t^{q^x}$ . This is partial information on [question 11.1](#), not an answer.

*Status* (batch 30). The remark is subsumed: [question 11.1](#) is now answered ([theorem 23.2](#)), every such automorphism of **(No, Oz)** is strong, and the passage from monomials to arbitrary series is automatic ([corollary 24.4](#)).

**Proposition 3.10** (Signed Archimedean blocks [10]). *Call nonzero  $\gamma, \gamma' \in \Gamma$  block-equivalent if they have the same sign and  $|\gamma| \leq n|\gamma'|$ ,  $|\gamma'| \leq n|\gamma|$  for some  $n \geq 1$ ; the exponent 0 forms a block of its own. If  $\sigma$  satisfies the conditions of [theorem 3.3](#), every exponent of  $\sigma(t^\gamma)$  lies in the block of  $\gamma$ . Consequently  $\sigma$  commutes with the projection onto the terms whose exponents lie in any prescribed union of blocks, and  $\sigma(k((t^\Delta))) = k((t^\Delta))$  for every convex subgroup  $\Delta \subseteq \Gamma$ .*

*Proof.* By (ix) of [theorem 3.8](#), the exponents of  $\sigma(t^\gamma)$  are  $\gamma + \delta$  with  $\delta = 0$  or  $\delta \in H_\gamma^{>0}$ , and then  $2\delta < |\gamma|$ . For  $\gamma < 0$  this gives  $|\gamma|/2 < |\gamma + \delta| \leq |\gamma|$ , and for  $\gamma > 0$  it gives  $\gamma \leq \gamma + \delta < 3\gamma/2$ ;

either way  $\gamma + \delta$  is in the block of  $\gamma$ . If  $\Lambda$  is a union of blocks and  $f = f_\Lambda + f'$  splits the terms of  $f$  according to  $\Lambda$ , strongness puts the support of  $\sigma(f_\Lambda)$  inside  $\Lambda$  and that of  $\sigma(f')$  outside it, so the projection of  $\sigma(f)$  is  $\sigma(f_\Lambda)$ . A convex subgroup is a union of blocks, so  $\sigma$  maps  $k((t^\Delta))$  into itself;  $\sigma^{-1}$  satisfies the same conditions, which gives equality.  $\square$

The zero cut is a union of blocks, so this recovers commutation with  $P_-, P_0, P_+$ ; [theorem 4.10](#) fixes  $k((t^{C_\delta}))$  pointwise for single flows, whereas here the whole stabilizer preserves every convex Hahn subfield setwise. This is not preservation of every truncation: a twist may send  $t^\gamma$  to  $t^\gamma + t^{\gamma+\delta}$  ([theorem 9.8](#)), and a cut strictly between the two exponents separates the two terms. The preserved blocks are determined by Archimedean scale, not by individual exponents [10].

## 4 Shears, flows, and the rank dichotomy

### 4.1 Homogeneous shears

Let  $\theta : \Gamma \rightarrow (k, +)$  be additive and  $\delta > 0$ . The formula

$$D_{\delta, \theta} \left( \sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} \theta(\gamma) t^{\gamma+\delta} \quad (4.1)$$

defines a contracting strong derivation: translation keeps supports well ordered and families summable, and additivity of  $\theta$  gives the product rule on monomials, hence on all series. This is the fixed-shift derivation of the surcomplex automorphism report, `saut:thm:shiftflow`, which constructs its exponential for every additive  $\theta$  and every  $\delta > 0$ ; its  $n$ th iterate has coefficient  $\prod_{j < n} \theta(\gamma + j\delta)$ . Sources 04, 08 and 09 re-derive it; only 04 credited it. The integer-part condition is the additional restriction  $\theta(C_\delta) = 0$ , not merely  $\theta(\delta) = 0$ .

**Theorem 4.1** (Homogeneous shears [04, 09]; after `saut:thm:shiftflow`). *Suppose  $\theta(C_\delta) = 0$ . Then  $D_{\delta, \theta}^n(t^\gamma) = \theta(\gamma)^n t^{\gamma+n\delta}$  and*

$$\sigma_{\delta, \theta}(t^\gamma) := \text{Exp}(D_{\delta, \theta})(t^\gamma) = t^\gamma \exp(\theta(\gamma) t^\delta). \quad (4.2)$$

*The strong extension  $\sigma_{\delta, \theta}$  lies in  $\mathcal{U}_{\mathcal{R}_\circ}(\Gamma)$  for every  $\circ$ , with inverse  $\sigma_{\delta, -\theta}$ , and  $\sigma_{\delta, \theta} \sigma_{\delta, \psi} = \sigma_{\delta, \theta+\psi}$  when  $\theta, \psi$  both vanish on  $C_\delta$ . With  $\Phi_s = \sigma_{\delta, s\theta}$  ( $s \in k$ ),  $s \mapsto \Phi_s$  is a homomorphism, injective if  $\theta \neq 0$ . Conversely  $\text{Exp}(D_{\delta, \theta})$  preserves  $\mathcal{R}_\circ$  only if  $\theta(C_\delta) = 0$ .*

*Proof.*  $\theta(\delta) = 0$  gives  $\theta(\gamma + n\delta) = \theta(\gamma)$  and the iterate formula. For a series supported on  $S$ , all terms of the image lie in  $S + \mathbb{N}\delta$ ; [lemma 2.1](#) gives well-ordering and finite contributions across all  $n$ , for single inputs and for summable families. On monomials the prescription is multiplicative by additivity of  $\theta$  and the exponential law; strongness extends this to all series. The map fixes  $t^\delta$  and every coefficient of  $\exp(\cdot)$ , so applying it termwise gives  $\sigma_{\delta, \theta} \sigma_{\delta, \psi}(t^\gamma) = t^\gamma \exp((\theta + \psi)(\gamma) t^\delta)$ , which gives the group law and the inverse. Every correction has larger exponent, so leading terms are fixed. Here  $h_\sigma(\gamma) = \theta(\gamma) t^\delta$ , whose only coefficient  $a_{\sigma, \delta} = \theta$  annihilates  $C_\delta$ ; apply [theorem 3.3](#). The coefficient at  $\gamma + \delta$  of  $\Phi_s(t^\gamma)$  is  $s\theta(\gamma)$ , which gives injectivity. The converse is the same criterion, since  $a_{\sigma, \delta} = \theta$  is the only functional.  $\square$

The summability above is proved directly; no appeal to [cited theorem 1.1](#) is made. For  $\gamma < 0$  with  $\theta(\gamma) \neq 0$ , every  $\gamma + n\delta$  stays negative, because  $-\gamma > n\delta$  for all ordinary  $n$ : the whole exponential, not only a Taylor polynomial, stays on the correct side of the boundary [09].

**Theorem 4.2** (Exact displacement and fixed field [04, 08, 09, 10]). *Let  $\theta(C_\delta) = 0$ ,  $s \neq 0$ , and  $f = \sum c_\gamma t^\gamma$ . If  $T_\theta(f) = \{\gamma \in \text{supp } f : \theta(\gamma) \neq 0\}$  is empty,  $\Phi_s(f) = f$ . Otherwise let  $\gamma_0 = \min T_\theta(f)$ ; then*

$$v(\Phi_s(f) - f) = \gamma_0 + \delta, \quad [t^{\gamma_0 + \delta}](\Phi_s(f) - f) = s\theta(\gamma_0)c_{\gamma_0}.$$

*In particular  $\text{Fix}(\Phi_s) = k((t^{\ker \theta}))$ .*

*Proof.* Only exponents with  $\theta(\gamma) \neq 0$  and powers  $n \geq 1$  contribute to the difference, at  $\gamma + n\delta \geq \gamma_0 + \delta$ , with equality only for  $\gamma = \gamma_0$ ,  $n = 1$ .  $\ker \theta$  is a divisible subgroup, so the Hahn field on it is a subfield.  $\square$

This first-displacement computation appears three times in the sources: as 04's exact displacement, as 08's leading term of  $T(x) - x$ , and inside 09's fixed-field proof. It is printed once.

**Example 4.3** (A nonmonomial image [04, 09]). Let  $\Gamma = \mathbf{No}$ ,  $\delta = 1$ ,  $\theta = q_1$ , the coefficient of  $\omega$  in an exponent. Every member of  $C_1$  is finite, so  $q_1(C_1) = 0$ . The flow fixes  $\mathbb{R}$  and  $\omega = t^{-1}$ , and

$$\Phi_s(\omega^\omega) = \omega^\omega \exp(-s\omega^{-1}) = \sum_{n \geq 0} \frac{(-s)^n}{n!} \omega^{\omega-n}.$$

For  $s \neq 0$  this is not a monomial; every exponent  $\omega - n$  is positive, so it is an omnific integer. The sum is a Hahn sum with strictly decreasing exponents, not a limit in the fine topology.

## 4.2 Families of shifts and a proved correspondence

**Proposition 4.4** (Uniformly supported shifts [09]). *Let  $S \subseteq \Gamma^{>0}$  be a set-sized well-ordered set and  $(\theta_s)_{s \in S}$  additive maps  $\Gamma \rightarrow k$ . Then  $D(f) = \sum_\gamma \sum_{s \in S} a_\gamma \theta_s(\gamma) t^{\gamma+s}$  is a contracting strong derivation, the terms of  $\text{Exp}(D)(f)$  are supported in  $\text{supp } f + \langle S \rangle_{\mathbb{N}}$  and form a summable family, and if  $\theta_s(C_s) = 0$  for all  $s$  then  $\text{Exp}(D) \in \mathcal{U}_{\mathcal{R}_0}(\Gamma)$ .*

*Proof.* The support principle applies to  $\text{supp } f + S$  and to all finite words in  $S$ ; every coefficient of  $D$ , its iterates and products is a finite sum. Additivity of each  $\theta_s$  gives the product rule; replacing one input support by the union of the supports of a summable input family gives strongness; each output exponent exceeds its input exponent. Preservation of  $\mathcal{R}_0$  follows from the no-crossing argument of [theorem 3.4](#) applied at each step of a word in  $S$ , which shows that  $\text{Exp}(\pm D)$  preserve  $\Pi$  and fix  $k$ .  $\square$

All explicit constructions on  $\mathbf{No}$  below use one set-sized well-ordered set of shifts, so they do not depend on the proper-class case of [cited theorem 1.1](#) [09]. Source 04 proves the correspondence itself on a class of this kind.

**Definition 4.5** (Common positive shifts [04]). A strong  $k$ -linear operator  $L$  has *common positive shifts* if there is a well-ordered set  $S \subseteq \Gamma^{>0}$  with  $\text{supp } L(t^g) \subseteq g + S$  for every  $g$ . Write  $S^*$  for the additive monoid generated by  $S$ .

**Lemma 4.6** (Iterated operator summability [04]). *If  $L$  has common positive shifts, then  $(L^n f)_{n \geq 0}$  is Hahn summable for every  $f$ , formal power series in  $L$  define strong operators, and coefficientwise formal identities between such series hold under composition.*

*Proof.*  $\text{supp } L^n f \subseteq \text{supp } f + nS$ . Translate  $\text{supp } f$  by its minimum and apply the support principle to the positive well-ordered set  $S \cup ((\text{supp } f - \min \text{supp } f) \setminus \{0\})$ ; a base exponent is

used once and later shifts come from  $S$ , so every exponent receives finitely many contributions across all paths and all  $n$ . For a summable input family use the union of supports; regrouping is justified because every coefficient has finitely many contributions.  $\square$

**Theorem 4.7** (Derivations with common shifts [04]). *Let  $S \subseteq \Gamma^{>0}$  be well ordered and  $a_\delta : \Gamma \rightarrow k$  additive for  $\delta \in S$ . Then  $D = \sum_{\delta \in S} D_{\delta, a_\delta}$  is a strong  $k$ -linear derivation with common positive shifts, and every such derivation has a unique expression of this form. When  $\bigcap_n n\mathfrak{o} = 0$ ,  $D$  maps  $\mathcal{R}_\mathfrak{o}$  into itself if and only if  $a_\delta(C_\delta) = 0$  for all  $\delta \in S$ .*

*Proof.* Summability and the Leibniz rule are termwise. Conversely put  $a_\delta(g) = [t^{g+\delta}]D(t^g)$ ; the Leibniz rule for  $t^{g+h}$  gives additivity, and strong linearity the expression. If  $D(\mathcal{R}_\mathfrak{o}) \subseteq \mathcal{R}_\mathfrak{o}$  and  $x \in \Pi$ , then  $D(x) = nD(x/n) \in n\mathcal{R}_\mathfrak{o}$  for all  $n$ , so  $D(x) \in \Pi$  by lemma 2.3. For  $0 < u \leq \delta$  the coefficient of  $D(t^{-u})$  at the nonnegative exponent  $\delta - u$  is  $a_\delta(-u)$ , with no cancellation between distinct shifts, so it vanishes; additivity and divisibility extend this to  $C_\delta$ . The converse is the no-crossing argument.  $\square$

In this theorem  $D$  is a derivation; the sources' coefficient ring  $D$  is  $\mathfrak{o}$  here, which removes the collision 04 flagged.

**Theorem 4.8** (Exponential and logarithm on common shifts [04]). *On operators with common positive shifts,  $D \mapsto \text{Exp}(D)$  and  $\sigma \mapsto \text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$  are inverse bijections between strong  $k$ -linear derivations and strong  $k$ -linear field automorphisms whose difference from the identity has common positive shifts; the shift set may enlarge to  $S^* \setminus \{0\}$ . They restrict to a bijection between derivations preserving  $\mathcal{R}_\mathfrak{o}$  and automorphisms preserving  $\mathcal{R}_\mathfrak{o}$  (for  $\bigcap_n n\mathfrak{o} = 0$ ).*

*Proof.* All series are justified by lemma 4.6. For a derivation  $D$ , the iterated Leibniz rule gives multiplicativity of  $\text{Exp } D$  after regrouping, and  $\text{Exp } D \text{Exp } (-D) = \text{id}$  gives bijectivity; leading terms are fixed since all other shifts are positive. Conversely let  $L = \sigma - \text{id}$  and  $\sigma^{[X]} = \sum_{n \geq 0} \binom{X}{n} L^n$ . For each ordinary  $m \geq 0$  this is the binomial expansion of  $\sigma^m$ , so for every exponent  $\gamma$  the  $t^\gamma$  coefficient of  $\sigma^{[X]}(fg) - \sigma^{[X]}(f)\sigma^{[X]}(g)$ , a polynomial in  $X$  by lemma 4.6, vanishes at all  $m \geq 0$  and is zero. Differentiating at  $X = 0$ , with  $\frac{d}{dX} \binom{X}{n} \big|_{X=0} = (-1)^{n-1}/n$ , shows that  $\text{Log } \sigma$  is a derivation; one-variable formal identities give the inverse correspondence. If  $D$  preserves  $\mathcal{R}_\mathfrak{o}$  it maps  $\Pi$  into  $\Pi$ , so  $\text{Exp}(\pm D)$  do; if  $\sigma(\Pi) = \Pi$ , then  $(\sigma - \text{id})(\Pi) \subseteq \Pi$  and the strong logarithmic sum has negative support.  $\square$

### 4.3 Single-shift flows without divisibility

Source 08 treats one flow at a time but allows any ordered abelian group and does not require  $\theta(\delta) = 0$ . Its statements are translated from the  $\omega$ -convention (section 1.5). For an ordered abelian group  $G$ ,  $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$  or  $(\mathbb{C}, \mathbb{Z}[i])$ , let  $\Pi_G$  be the series supported in  $G^{<0}$  and  $\mathcal{R}_G = \mathfrak{o} \oplus \Pi_G$ .

**Proposition 4.9** (Falling-factorial iterates [08]). *For any additive  $\theta : G \rightarrow k$  and  $\delta > 0$ ,  $D_{\delta, \theta}^n(t^\gamma) = \prod_{j=0}^{n-1} (\theta(\gamma) + j\theta(\delta)) t^{\gamma+n\delta}$ , and  $T^s = \sum_{n \geq 0} s^n D_{\delta, \theta}^n / n!$  ( $s \in k$ ) is a strong  $k$ -linear 1-automorphism with  $T^s T^u = T^{s+u}$ . If  $\theta(\delta) = 0$ ,  $T^s(t^\gamma) = t^\gamma \exp(s\theta(\gamma)t^\delta)$ ; if  $c = \theta(\delta) \neq 0$ ,  $T^s(t^\gamma) = t^\gamma (1 + sct^\delta)^{\theta(\gamma)/c}$  as a formal binomial series. In either case  $T^s(t^{-\delta}) = t^{-\delta} - s\theta(\delta)$ .*

*Proof.* The iterate formula is induction. Summability, strongness and the group law follow from lemma 2.1 as in theorem 4.1, and the pairs  $(n, m)$  with  $n + m = r$  are finite in number; multiplicativity follows from the iterated Leibniz rule. The two monomial formulas are the substitutions of the iterate. For  $\gamma = -\delta$  every product of length at least two has the factor  $\theta(-\delta) + \theta(\delta) = 0$ .  $\square$

This is `saut:thm:shiftflow` once more, over an arbitrary ordered abelian  $G$ ; 08 reproves the summability for that reason.

**Theorem 4.10** (Threshold for single flows [08]). *Let  $G$  be any ordered abelian group,  $\delta > 0$ ,  $\theta : G \rightarrow k$  additive and  $s \in k \setminus \{0\}$ . The following are equivalent: (a)  $T^s(\mathcal{R}_G) \subseteq \mathcal{R}_G$ ; (b)  $T^s(\mathcal{R}_G) = \mathcal{R}_G$ ; (c)  $\theta(C_\delta) = 0$ . When they hold,  $T^s$  fixes every series supported in  $C_\delta$  and preserves the constant coefficient of every series.*

*Proof.* Assume (a). By lemma 2.3,  $\Pi_G = \bigcap_n n\mathcal{R}_G$ , so  $T^s(\Pi_G) \subseteq \Pi_G$ . The constant term of  $T^s(t^{-\delta})$  is  $-s\theta(\delta)$ , so  $\theta(\delta) = 0$ . If  $0 > \gamma \in C_\delta$  and  $\theta(\gamma) \neq 0$ , choose  $n$  with  $\gamma + n\delta > 0$ ; the coefficient of  $T^s(t^\gamma)$  there is  $(s\theta(\gamma))^n/n! \neq 0$ , and the exponents  $\gamma + m\delta$  are distinct, contradicting  $T^s(t^\gamma) \in \Pi_G$ . So  $\theta$  vanishes on the negative part of  $C_\delta$ , hence on  $C_\delta$ . Conversely, under (c), monomials with exponents in  $C_\delta$  are fixed; if  $\gamma < 0$  lies outside  $C_\delta$ , then  $\gamma < -n\delta$  for all  $n$ , so all  $\gamma + n\delta$  stay negative; positive exponents outside  $C_\delta$  stay positive. So no term enters or leaves exponent 0,  $T^{\pm s}(\mathcal{R}_G) \subseteq \mathcal{R}_G$ , and (b) follows. (b) $\Rightarrow$ (a) is trivial.  $\square$

Only divisibility of *coefficients* by ordinary integers is used, so  $G$  need not be divisible. The theorem covers the elementary family only: even at rank one, exponent rescalings and characters preserve the integer part, and it does not assert that every leading-term-preserving automorphism is the exponential of one fixed-shift derivation [08].

*Remark 4.11* (Divisibility and the ring  $\mathcal{A}_k$  [09]; corrected). Source 09 remarks that divisibility of  $\Gamma$  is essential, because with  $\Gamma = \mathbb{Z}$  the translation below preserves  $\mathcal{A}_k(\mathbb{Z}) = k[t^{-1}]$ . That is right for the boundary ring  $\mathcal{A}_k$  and wrong if read for the integer-part ring: the translation  $t^{-1} \mapsto t^{-1} - 1$  sends  $\frac{1}{2}t^{-1} \in \mathcal{R}_{\mathbb{Z}}(\mathbb{Z}) = \mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$  to  $\frac{1}{2}t^{-1} - \frac{1}{2} \notin \mathcal{R}_{\mathbb{Z}}(\mathbb{Z})$ , as theorem 4.10 predicts. So for  $\Gamma = \mathbb{Z}$  the stabilizers of  $\mathcal{A}_k$  and of  $\mathcal{R}_{\mathbb{Z}}$  differ, and the independence of  $\mathfrak{o}$  in theorem 3.3 genuinely needs divisibility, whereas the  $\mathcal{R}_{\mathbb{Z}}$ -form of the criterion holds without divisibility for single flows. Whether the general criterion for  $\mathcal{R}_{\mathbb{Z}}$  holds for every non-divisible  $\Gamma$  is left open (question 11.6).

#### 4.4 Why translations fail

**Example 4.12** (Translations [04, 08, 09]; the saut example). Let  $\Gamma = \mathbb{Q}$ ,  $\delta = 1$ ,  $\theta(\gamma) = \gamma$ . Then  $D^n(t^\gamma) = \gamma(\gamma+1)\cdots(\gamma+n-1)t^{\gamma+n}$  and  $\text{Exp}(D)(t^\gamma) = t^\gamma(1-t)^{-\gamma}$ ; on  $t$  this is  $t \mapsto t/(1-t)$ , and on  $t^{-1}$  it is  $t^{-1} \mapsto t^{-1} - 1$ , which preserves  $k[t^{-1}]$ . But

$$\text{Exp}(D)(t^{-1/2}) = t^{-1/2}(1-t)^{1/2} = t^{-1/2} - \frac{1}{2}t^{1/2} - \frac{1}{8}t^{3/2} - \dots$$

has positive support, so neither  $\mathcal{A}_k$  nor  $\mathcal{R}_{\mathfrak{o}}$  is preserved:  $\theta(C_1) \neq 0$ . The inverse  $t \mapsto t/(1+t)$  is 04's example and  $X \mapsto X + s$ ,  $X = t^{-1}$ , is 08's. This is exactly the transparent example  $T_1(t) = t/(1-t)$  of the surcomplex automorphism report (its  $\lambda = \ell$ , the constant-coefficient functional, with  $\delta = 1$ , after `saut:thm:shiftflow`); on the surreals  $\ell(1) = 1 \neq 0$ , so  $T_1$  does *not* preserve **Oz**: it sends  $\omega^{1/2}$  to  $\omega^{1/2} - \frac{1}{2}\omega^{-1/2} - \dots \notin \mathbf{Oz}$ . Preservation of a few familiar infinite integers is not enough; all rational monomial powers must be tested.

The Taylor coefficient flows of that report (`saut:thm:coeffflow`) move real constants and never preserve **Oz**: this follows from corollary 5.2, and directly, for  $d(b) = 1$  they send  $bt^{-\delta/2}$  to  $bt^{-\delta/2} + t^{\delta/2}$ . Composed with a proper exponent embedding whose positive values dominate every multiple of  $\delta$ , they do preserve and reflect **Oz**, as embeddings (theorems 26.6 and 27.3).

#### 4.5 The rank dichotomy and the bottom subgroup

**Theorem 4.13** (Rank dichotomy [04, 09, 10]; [08] for single flows). *For a nonzero divisible ordered abelian group  $\Gamma$ , any characteristic-zero  $k$  and unital  $\mathfrak{o} \subseteq k$ :  $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma) = \{\text{id}\}$  if and only*

if  $\Gamma$  is Archimedean. For every ordered abelian  $G$ , the single flows of [theorem 4.10](#) preserving  $\mathcal{R}_G$  are trivial if  $G$  is Archimedean, and some are nontrivial if  $G$  is a set with a proper  $C_\delta$ .

*Proof.* If  $\Gamma$  is Archimedean,  $C_\delta = \Gamma$  for all  $\delta$ , so by [theorem 3.3](#) every  $h_\sigma(g)$  vanishes;  $\sigma$  fixes all monomials, hence everything. If some  $C_\delta$  is proper, choose (by a basis of the set-sized  $\mathbb{Q}$ -space  $\Gamma/C_\delta$ , or of the rationalization of the torsion-free  $G/C_\delta$  for 08's case) a nonzero  $\mathbb{Q}$ -linear  $\theta$  into  $\mathbb{Q} \subseteq k$  vanishing on  $C_\delta$ ; [theorem 4.1](#) or [theorem 4.10](#) gives a nonidentity element. For  $\Gamma = \mathbf{No}$  explicit functionals  $q_\beta$  replace the choice ([section 7](#)).  $\square$

For  $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$  and a set-sized Archimedean  $\Gamma$ , every field automorphism of  $\mathbb{R}((t^\Gamma))$  preserving  $\mathcal{R}_\mathbb{Z}$ , strong or not, is therefore a monomial map  $M_{\chi, \tau}$  ([corollary 24.3](#)).

**Definition 4.14** (Bottom convex subgroup [04]).  $B(\Gamma) = \bigcap_{\delta > 0} C_\delta$ , possibly zero.

**Theorem 4.15** (Common fixed field at arbitrary set-sized rank [04]). *For a set-sized nonzero divisible  $\Gamma$ ,  $\text{Fix}(\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)) = k((t^{B(\Gamma)}))$ .*

*Proof.* If  $g \in B(\Gamma)$ , every  $a_{\sigma, \delta}(g)$  vanishes by [theorem 3.3](#), so  $t^g$  is fixed, and strong linearity fixes the Hahn field on  $B(\Gamma)$ . If  $g \in \text{supp } f$  lies outside  $B(\Gamma)$ , choose  $\delta$  with  $g \notin C_\delta$  and a  $\mathbb{Q}$ -linear  $\theta$  with  $\theta(C_\delta) = 0$  and  $\theta(g) = 1$ ; [theorem 4.2](#) shows  $\Phi_1$  moves  $f$ .  $\square$

**Example 4.16** ([04, 09]). Let  $\Gamma = \mathbb{Q}^r$ , ordered so that the *last* nonzero coordinate decides the sign (09's reverse-lexicographic order; 04 orders by the first coordinate, which is the same group with coordinates reversed). Then the coordinate vectors satisfy  $0 < e_1 \ll e_2 \ll \dots \ll e_r$ ,  $C_{e_j} = \mathbb{Q}e_1 \oplus \dots \oplus \mathbb{Q}e_j$ , and  $B(\Gamma) = \mathbb{Q}e_1$ . For  $r = 1$  the stabilizer is trivial; for  $r \geq 2$  it is nontrivial and fixes exactly  $k((t^{\mathbb{Q}e_1}))$ . The presence of a smallest Archimedean layer, not the number of layers, controls the fixed field. For  $\Gamma = \mathbf{No}$ ,  $B(\mathbf{No}) = 0$ : for  $g \neq 0$  take  $\delta = |g|/\omega$ .

## 4.6 Phase shears

**Theorem 4.17** (Finite-dimensional phase shears [08]). *Let  $H$  be a convex subgroup of an ordered abelian group  $G$ , let  $\mathfrak{m}_H$  be the series supported in  $H^{>0}$ , and let  $L$  be an additive map on  $G$  vanishing on  $H$ , with values in  $\mathfrak{m}_H$  and image in a finite-dimensional  $k$ -subspace. Then  $T_L(t^g) = t^g \exp(L(g))$  extends strongly to a  $k$ -linear 1-automorphism of the Hahn field that fixes  $k((t^H))$  pointwise, preserves  $\mathcal{R}_G$  and constant coefficients, and satisfies  $T_L T_M = T_{L+M}$ ,  $T_L^{-1} = T_{-L}$ , and  $T_L = \text{id} \Leftrightarrow L = 0$ .*

*Proof.* Let  $u_1, \dots, u_d$  span the image of  $L$  and  $U = \bigcup_j \text{supp } u_j$ , a well-ordered subset of  $H^{>0}$ . All supports of all powers in all  $\exp(L(g))$  lie in the monoid generated by  $U$ , which is well ordered with finite representations; with the input support this gives one summable bound, hence the strong extension. Additivity of  $L$  gives multiplicativity. Exponents in  $H$  have zero phase, so  $k((t^H))$  is fixed; in particular each  $M(g)$  and its exponential are fixed, which gives  $T_L T_M = T_{L+M}$  on monomials and then strongly. Injectivity follows from injectivity of  $\exp$  on  $\mathfrak{m}$ . If  $g \notin H$ , every exponent of  $t^g \exp(L(g))$  lies in  $g + H$ , which has constant strict sign by convexity; so positive and negative supports are preserved and no term reaches exponent 0.  $\square$

The finite-dimensionality is a sufficient uniform support condition and must not be dropped silently: independent phases for infinitely many exponents need not form a summable family [08]. Taking  $H = C_\delta$  and  $L(g) = s\theta(g)t^\delta$  recovers the flows of [theorem 4.10](#).

## 4.7 Twists with a general profile [10]

Source 10 builds its explicit automorphisms directly, without derivations. Fix  $\delta > 0$ , a formal variable  $\xi$ , and an additive map  $\Theta : \Gamma \rightarrow \xi k[[\xi]]$  into the power series without constant term. Define

$$\sigma_{\delta, \Theta} \left( \sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} t^{\gamma} \exp(\Theta(\gamma)(t^{\delta})), \quad (4.3)$$

where  $\Theta(\gamma)(t^{\delta})$  is  $\Theta(\gamma)$  evaluated at  $\xi = t^{\delta}$ . For  $\Theta(\gamma) = \theta(\gamma)\xi$  this is the shear  $\sigma_{\delta, \theta}$  of (4.2). Writing  $\Theta(\gamma) = \sum_{n \geq 1} \theta_n(\gamma) \xi^n$  with additive  $\theta_n$ , the map is Exp of  $\sum_n D_{n\delta, \theta_n}$ , a derivation with the one well-ordered shift set  $\mathbb{N}_{>0}\delta$  as in [proposition 4.4](#) (an identification by the merge); 10's proof below is direct.

**Theorem 4.18** (Twists [10]). *If  $\Theta(\delta) = 0$ , then  $\sigma_{\delta, \Theta}$  is a strong  $k$ -linear 1-automorphism of  $K$  with strong inverse  $\sigma_{\delta, -\Theta}$ , and  $\sigma_{\delta, \Theta} \sigma_{\delta, \Psi} = \sigma_{\delta, \Theta + \Psi}$  for every second such  $\Psi$ ; the map  $\Theta \mapsto \sigma_{\delta, \Theta}$  is injective. If  $\Theta(C_{\delta}) = 0$ , then  $\sigma_{\delta, \Theta} \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$  for every unital  $o \subseteq k$ ; conversely [merge]  $\sigma_{\delta, \Theta} \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$  forces  $\Theta(C_{\delta}) = 0$ . For set-sized  $\Gamma$ , the additive group  $\text{Hom}_{\mathbb{Q}}(\Gamma/C_{\delta}, \xi k[[\xi]])$  thus embeds in  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ .*

*Proof.* The support of  $t^{\gamma} \exp(\Theta(\gamma)(t^{\delta}))$  lies in  $\gamma + \mathbb{N}\delta$ . For an input supported in  $S$ , [lemma 2.1](#) makes  $S + \mathbb{N}\delta$  well ordered with finitely many pairs  $(\gamma, n)$  at each exponent, which defines (4.3); for a summable family of inputs use the union of their supports. So the map is strong. Additivity of  $\Theta$  gives multiplicativity on monomials, and strongness extends it to products of series, which are summable sums of monomial products;  $k$  is fixed. Since  $\Theta(\delta) = 0$ , the map fixes  $t^{\delta}$  and hence every series in  $t^{\delta}$  with coefficients in  $k$ , so  $\sigma_{\delta, \Theta} \sigma_{\delta, \Psi}(t^{\gamma}) = t^{\gamma} \exp(\Theta(\gamma)(t^{\delta})) \exp(\Psi(\gamma)(t^{\delta})) = \sigma_{\delta, \Theta + \Psi}(t^{\gamma})$ , and strongness extends the identity;  $\Psi = -\Theta$  gives the inverse. Every correction lies at a larger exponent, and no term arising from a larger input exponent reaches the least one, so leading terms are fixed. Evaluation at  $\xi = t^{\delta}$  and exp are injective, which gives injectivity. If  $\Theta(C_{\delta}) = 0$  and  $\Theta(\gamma) \neq 0$ , then  $\gamma \notin C_{\delta}$ , so every shift  $n\delta$  satisfies  $n\delta \ll |\gamma|$ ; thus  $\text{supp } h_{\sigma}(\gamma) \subseteq H_{\gamma}$  for every  $\gamma \neq 0$ , and [theorem 3.3](#) applies. Conversely, if  $0 \neq \gamma \in C_{\delta}$ , say  $|\gamma| \leq N\delta$ , no shift  $n\delta$  lies in  $H_{\gamma}$ , so condition (v) there forces  $h_{\sigma}(\gamma) = \Theta(\gamma)(t^{\delta}) = 0$ .  $\square$

The field-automorphism part needs only  $\Theta(\delta) = 0$ ; the stronger  $\Theta(C_{\delta}) = 0$  is exactly the integer-part condition [10]. (For a single linear profile, [proposition 4.9](#) even allows  $\theta(\delta) \neq 0$ .) For additive  $\theta : \Gamma \rightarrow k$  with  $\theta(\delta) = 0$  and a profile  $F \in 1 + \xi k[[\xi]]$ , the choice  $\Theta = \theta \log F$  gives the *binomial twist*

$$\sigma_{\delta, \theta}^F \left( \sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} t^{\gamma} F(t^{\delta})^{\theta(\gamma)}, \quad F^a = \exp(a \log F) = \sum_{n \geq 0} \binom{a}{n} (F - 1)^n. \quad (4.4)$$

Its inverse is  $\sigma_{\delta, -\theta}^F = \sigma_{\delta, \theta}^{1/F}$ , composition adds  $\theta$  for fixed  $F, \delta$ , and  $\sigma_{\delta, \theta}^{\exp \xi} = \sigma_{\delta, \theta}$ . The most economical profile is  $F = 1 + \xi$ :

$$\sigma_{\delta, \theta}^{1+\xi}(t^{\gamma}) = \sum_{n \geq 0} \binom{\theta(\gamma)}{n} t^{\gamma + n\delta}, \quad (4.5)$$

whose coefficients are polynomials over  $\mathbb{Q}$  in  $\theta(\gamma)$ , with no choice of branch even for  $k = \mathbb{C}$  [10]. Source 10's  $\sigma_{\lambda, F, h}$  is  $\sigma_{h, -\lambda}^F$  ([section 1.5](#)).

**Example 4.19** (Two scales [10]). Let  $\Gamma = \mathbb{Q}^2$  as in [example 4.16](#), so  $e_1 \ll e_2$ , and let  $\theta$  be minus the  $e_2$ -coordinate, which vanishes on  $C_{e_1} = \mathbb{Q}e_1$ . Then  $\sigma = \sigma_{e_1, \theta}^{1+\xi}$  sends  $t^{xe_1 + ye_2}$  to  $t^{xe_1 + ye_2} (1 + t^{e_1})^{-y}$ ; for instance  $\sigma(t^{-e_2}) = t^{-e_2} + t^{-e_2 + e_1}$  and

$$\sigma(t^{-e_2/2}) = t^{-e_2/2} + \frac{1}{2} t^{-e_2/2 + e_1} - \frac{1}{8} t^{-e_2/2 + 2e_1} + \frac{1}{16} t^{-e_2/2 + 3e_1} - \dots$$

Every exponent  $-e_2/2 + ne_1$  is negative, however large  $n$  is. With  $\Gamma = \mathbb{Q}$  the prescription  $t^{-1} \mapsto t^{-1}(1+t)$  would force a positive exponent at  $t^{-1/2}$  (compare [example 4.12](#)).

**Theorem 4.20** (Fixed field and first displacement [10]). *Let  $\theta(C_\delta) = 0$ , let  $F = 1 + c_m \xi^m + c_{m+1} \xi^{m+1} + \dots$  with  $m \geq 1$  and  $c_m \neq 0$ , and put  $\sigma = \sigma_{\delta, \theta}^F$ . Then  $\text{Fix}(\sigma) = k((t^{\ker \theta}))$ . More precisely, if  $f = \sum a_\gamma t^\gamma$  is not supported in  $\ker \theta$  and  $\gamma_* = \min\{\gamma \in \text{supp } f : \theta(\gamma) \neq 0\}$ , then*

$$v(\sigma(f) - f) = \gamma_* + m\delta, \quad [t^{\gamma_* + m\delta}](\sigma(f) - f) = a_{\gamma_*} \theta(\gamma_*) c_m.$$

For every integer  $n \neq 0$ ,  $\text{Fix}(\sigma^n) = \text{Fix}(\sigma)$ .

*Proof.* Monomials with  $\theta(\gamma) = 0$  are fixed, hence, by strongness, all of  $k((t^{\ker \theta}))$ . For  $a \neq 0$  the first nonconstant term of  $F^a$  is  $a c_m \xi^m$ , so for  $\theta(\gamma) \neq 0$  the difference  $\sigma(t^\gamma) - t^\gamma$  has least term  $\theta(\gamma) c_m t^{\gamma+m\delta}$  and all other terms at exponents at least  $\gamma + (m+1)\delta$ . Exponents  $\gamma < \gamma_*$  of  $f$  contribute no difference, and exponents  $\gamma > \gamma_*$  contribute only beyond  $\gamma + m\delta > \gamma_* + m\delta$ . So nothing cancels the term at  $\gamma_* + m\delta$ . The composition law gives  $\sigma^n = \sigma_{\delta, n\theta}^F$ , and  $\ker(n\theta) = \ker \theta$  in characteristic zero.  $\square$

For  $F = \exp \xi$  this is [theorem 4.2](#); the case of a general profile and the statement for powers are 10's.

**Corollary 4.21** (No finite motion [10]). *Every finite orbit of  $\sigma$  in [theorem 4.20](#) is a fixed point, and  $\text{Fix}(\sigma)$  is relatively algebraically closed in  $K$ .*

*Proof.* A finite orbit gives  $\sigma^n(f) = f$  for some  $n \geq 1$ , hence  $\sigma(f) = f$ . If  $f$  is algebraic over  $\text{Fix}(\sigma)$ , all  $\sigma^n(f)$  are roots of its minimal polynomial, whose coefficients are fixed; so the orbit is finite.  $\square$

This is relative algebraic closedness only: with real coefficients the fixed field need not contain a square root of  $-1$  [10].

**Proposition 4.22** (Independent exponent cosets [10]). *Let  $V = \ker \theta$  and let  $(\gamma_j)_{j \in J}$  have  $\mathbb{Q}$ -linearly independent images in  $\Gamma/V$ . Then the monomials  $t^{\gamma_j}$  are algebraically independent over  $k((t^V))$ .*

*Proof.* A relation involves finitely many indices and has the form  $\sum_{\nu \in A} b_\nu t^{\nu_1 \gamma_1 + \dots + \nu_r \gamma_r} = 0$  with  $b_\nu \in k((t^V))$ , distinct  $\nu$  and some  $b_\nu \neq 0$ . For  $\nu \neq \mu$ ,  $\sum_j (\nu_j - \mu_j) \gamma_j \notin V$ , so the nonzero summands have supports in distinct cosets of  $V$  and cannot cancel.  $\square$

**Remark 4.23** (Rank, divisibility and what is classified [10]). Source 10 proves [theorem 4.13](#) with a binomial witness: if  $\delta \ll |\gamma_0|$  with  $\gamma_0 < 0$  and  $\theta$  is  $\mathbb{Q}$ -linear with  $\theta(C_\delta) = 0$  and  $\theta(\gamma_0) = 1$ , then  $\sigma_{\delta, \theta}^{1+\xi}(t^{\gamma_0}) = t^{\gamma_0} + t^{\gamma_0+\delta}$ , and the map fixes  $k((t^{C_\delta}))$  pointwise. Its divisibility example is stated for  $\mathfrak{o} = k$ :  $t^{-1} \mapsto t^{-1} + 1$  preserves  $\mathcal{A}_k(\mathbb{Z}) = k[t^{-1}]$  for  $\Gamma = \mathbb{Z}$ , and 10 says explicitly that it does not preserve  $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ . This agrees with [remark 4.11](#) as corrected there; since 10 assumes divisibility throughout, it does not bear on [question 11.6](#). At Archimedean rank the monomial maps  $t^\gamma \mapsto t^{2\gamma}$  and the character rescalings still preserve  $\mathcal{R}_{\mathfrak{o}}$ ; they lie outside the leading-term-fixing factor and are no counterexample to [theorem 4.13](#).

## 5 The strong stabilizer and all pair automorphisms

Call  $\sigma$  *valued* if  $v(\sigma f) = \tau(v(f))$  for an order automorphism  $\tau$  of  $\Gamma$ . For a character  $\chi : (\Gamma, +) \rightarrow (k^\times, \cdot)$  put

$$M_{\chi, \tau} \left( \sum a_\gamma t^\gamma \right) = \sum a_\gamma \chi(\gamma) t^{\tau(\gamma)}. \quad (5.1)$$

It is a strong field automorphism preserving  $\mathcal{R}_o$ , since  $\tau$  preserves signs of exponents and  $\chi(0) = 1$ ;  $M_{\chi,\tau} M_{\chi',\tau'} = M_{\chi' \cdot (\chi \circ \tau'), \tau \tau'}$  [04, 09].

**Theorem 5.1** (Strong valued factorization [04, 09, 10]). *Every strong valued  $k$ -fixing automorphism  $\sigma$  with  $\sigma(\mathcal{R}_o) = \mathcal{R}_o$  factors uniquely as  $\sigma = M_{\chi,\tau} \circ u$  with  $u \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$ , where  $\tau$  is the induced exponent automorphism and  $\chi(\gamma) = \text{lc}(\sigma(t^\gamma))$ . The monomial maps form a subgroup normalizing  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ , and*

$$M_{\chi,\tau} D_{\delta,\theta} M_{\chi,\tau}^{-1} = D_{\tau(\delta), \chi(\delta) \theta \circ \tau^{-1}}, \quad (5.2)$$

so the convex condition is invariant, since  $\tau(C_\delta) = C_{\tau(\delta)}$ .

*Proof.* Multiplicativity of leading coefficients gives that  $\chi$  is a character.  $u = M_{\chi,\tau}^{-1} \sigma$  fixes the leading term of each monomial; since the images of distinct monomial terms have strictly ordered valuations, strongness shows that  $u$  fixes the leading term of every series, and  $u$  preserves  $\mathcal{R}_o$ . The data  $\tau, \chi$  are forced by  $\sigma$ , hence so is  $u$ . Conjugating a 1-automorphism by a monomial map gives a 1-automorphism preserving  $\mathcal{R}_o$ . Formula (5.2) is checked on monomials.  $\square$

The ambient decomposition is due to Kuhlmann–Serra [10], and its surcomplex form is the four-layer decomposition `saut:thm:decomp`  $U \rtimes \text{Hom}(\Gamma, \mathbb{C}^\times) \rtimes (\text{Aut } \mathbb{C} \times \text{Aut } \Gamma)$  of the surcomplex automorphism report; [theorem 5.1](#) is its restriction to the  $\mathcal{R}_o$ -stabilizer with the residue layer trivial. Sources 04, 09 and 10 credit [10]; the new information is the description of  $u$  by [theorem 3.3](#). Source 10 states the factorization as  $\text{Stab}(\mathcal{R}_o) = \mathcal{U}_{\mathcal{R}_o}(\Gamma) \rtimes \{M_{\chi,\tau}\}$ , restricting to positive characters when  $k$  is ordered and only order-preserving maps are considered; it is not a new decomposition theorem for  $\text{Aut}(\mathbf{No})$ , and coefficient-moving or nonstrong automorphisms lie outside it [10]. For the real pair neither kind exists: coefficients are fixed by [corollary 5.2](#) and strongness is automatic by [theorem 23.2](#), so the splitting covers every automorphism ([theorem 24.1](#)). For  $k = \mathbb{R}$  and divisible  $\Gamma$  every character is positive,  $\chi(\gamma) = \chi(\gamma/2)^2$ ; for  $\Gamma = \mathbb{Q}$  the factor  $u$  is absent and every such real automorphism is  $t^q \mapsto c^q t^{\lambda q}$  with  $c > 0$ ,  $\lambda \in \mathbb{Q}_{>0}$ , so the higher-rank shears are a new phenomenon, not complicated one-variable substitutions [09]. The monomial maps include the dilations  $S_a$  of the Diophantine report (`odg:def:ex:dilation`) and the character twists ([remark 9.5](#)).

**Corollary 5.2** (All pair automorphisms [08, 09, 10]). *Every field automorphism of  $\mathbf{No}$  preserving  $\mathbf{Oz}$  setwise fixes  $\mathbb{R}$  pointwise, preserves order and the natural valuation ring, and commutes with  $P_-, P_0, P_+$  and with the floor map of  $\mathbf{Oz}$ ; this needs no strongness. Consequently, among strong automorphisms, the  $\mathbf{Oz}$ -stabilizer is exactly  $\{M_{\chi,\tau}\} \rtimes \mathcal{U}_{\mathbf{Oz}}$ , with  $\chi$  positive; by [theorem 23.2](#) every automorphism preserving  $\mathbf{Oz}$  is strong, so this is the whole stabilizer ([theorem 24.1](#)).*

*Proof.* By `odg:def:thm:autreal` of the Diophantine report, every automorphism of  $\mathbf{Oz}$  extends uniquely to  $\mathbf{No}$ , preserves order,  $\mathbb{R}$  and the finite ring, and fixes  $\mathbb{R}$  pointwise; restriction and extension are inverse correspondences [08, 09]. The sets and graphs of [theorem 8.2](#) are parameter-free definable, so they are preserved. Fixing  $\mathbb{R}$  pointwise makes a strong such map  $k$ -fixing, and preserving the valuation ring makes it valued, so [theorem 5.1](#) applies. The last clause is [theorem 23.2](#), proved in [part III](#) from the first sentence of this corollary and [theorem 22.1](#), with no circularity.  $\square$

## 6 Convex rank, solvability, and a wreath product

### 6.1 The bracket law

**Lemma 6.1** (Homogeneous brackets [04, 08, 09]). *For additive  $\theta, \psi : \Gamma \rightarrow k$  and  $\delta, \epsilon > 0$ ,*

$$[D_{\delta, \theta}, D_{\epsilon, \psi}] = D_{\delta + \epsilon, \theta(\epsilon)\psi - \psi(\delta)\theta}. \quad (6.1)$$

*If  $\theta(C_\delta) = 0$  and  $\psi(C_\epsilon) = 0$ , the bracket is zero when  $C_\delta = C_\epsilon$ ; if  $C_\delta \subsetneq C_\epsilon$  it is a multiple of  $D_{\delta + \epsilon, \psi}$ , whose shift lies in the  $C_\epsilon$  scale and whose functional vanishes on  $C_{\delta + \epsilon} = C_\epsilon$ .*

*Proof.* On  $t^\gamma$  the two compositions differ at  $\gamma + \delta + \epsilon$  by  $\psi(\gamma)\theta(\gamma + \epsilon) - \theta(\gamma)\psi(\gamma + \delta) = \theta(\epsilon)\psi(\gamma) - \psi(\delta)\theta(\gamma)$ ; strongness extends the identity. Equal scales: each functional kills the other's shift. If  $C_\delta \subsetneq C_\epsilon$ , then  $\psi(\delta) = 0$ .  $\square$

Source 08 records the same identity in its own normalization (its bracket  $[D_{\varphi, \delta}, D_{\psi, \epsilon}] = D_{\psi(\delta)\varphi - \varphi(\epsilon)\psi, \delta + \epsilon}$ , which is (6.1) after the sign change of [section 1.5](#)). Elementary derivations with shifts in the same Archimedean class commute; noncommutativity is an interaction between layers [04]. For general strong derivations the computation applies after expansion into shift coefficients: the two compositions are strongly summable and every output coefficient is a finite sum [09].

### 6.2 A filtration by convex scales

Say that  $\Gamma$  has *finite ordered rank*  $r$  if its distinct nonzero principal convex subgroups are exactly  $0 < C_1 < \dots < C_r = \Gamma$ . The layer of a shift  $\delta$  is the  $j$  with  $C_\delta = C_j$ . A shift in layer  $r$  contributes nothing to  $\mathfrak{L}_A$  ([corollary 3.5](#)), since its functional vanishes on  $\Gamma$ . Let  $\mathfrak{F}_j \subseteq \mathfrak{L}_A$  consist of the derivations whose nonzero shifts have layers at least  $j$ , so  $\mathfrak{L}_A = \mathfrak{F}_1 \supseteq \dots \supseteq \mathfrak{F}_r = 0$ . Selecting the terms of prescribed layers from the images of monomials keeps a summable subfamily, and the new coefficient functions stay additive, so the  $\mathfrak{F}_j$  are strongly closed Lie ideals [09].

**Lemma 6.2** ([09]).  $[\mathfrak{F}_j, \mathfrak{F}_j] \subseteq \mathfrak{F}_{j+1}$  and  $[\mathfrak{L}_A, \mathfrak{F}_j] \subseteq \mathfrak{F}_j$ .

*Proof.* Apply [lemma 6.1](#) to pairs of shifts: equal layers cancel and unequal layers give a term in the larger layer.  $\square$

**Theorem 6.3** (Finite-rank bound [09]). *If the set-sized group  $\Gamma$  has finite ordered rank  $r$ , then  $\mathfrak{L}_A$  and  $\mathcal{U}_{\mathcal{R}_0}(\Gamma)$  have derived length at most  $r - 1$ .*

*Proof.* Inductively  $\mathfrak{L}_A^{(n)} \subseteq \mathfrak{F}_{n+1}$ . For the group, put  $G_j = \text{Exp}(\mathfrak{F}_j)$ ; by the Baker–Campbell–Hausdorff part of [cited theorem 1.1](#) each  $G_j$  is a subgroup, and the logarithm of a commutator of two elements of  $G_j$  is a strongly summable Lie series each of whose terms contains a bracket of two elements of  $\mathfrak{F}_j$ , hence lies in  $\mathfrak{F}_{j+1}$ . So  $[G_j, G_j] \subseteq G_{j+1}$ , and  $G_r = \{\text{id}\}$ .  $\square$

The Lie bound is elementary; the group bound uses the imported correspondence, which [1] proves for set-sized  $\Gamma$ .

### 6.3 Sharpness, including the abstract group

In  $\mathbb{Q}^r$  ordered as in [example 4.16](#), let  $\theta_i$  extract the  $i$ th coordinate,  $X_{j,i} = D_{e_j, \theta_i}$  for  $j < i$ , and

$$a_0 = 0, \quad a_k = \sum_{h=1}^k 2^{k-h} e_h, \quad Y_{k;j,i} = D_{a_k + e_j, \theta_i} \quad (k < j < i \leq r).$$

These satisfy the criterion, since  $C_{e_j} = \mathbb{Q}e_1 \oplus \cdots \oplus \mathbb{Q}e_j$ .

**Lemma 6.4** (Balanced brackets [09]).  $Y_{k;j,i} \in \mathfrak{L}_A^{(k)}$ , and  $[Y_{k-1;k,j}, Y_{k-1;j,i}] = Y_{k;j,i}$  for  $k \geq 1$ .

*Proof.* At  $k = 0$  this is the definition. In the bracket,  $\theta_j(a_{k-1} + e_j) = 1$  and  $\theta_i(a_{k-1} + e_k) = 0$ , and the shift is  $2a_{k-1} + e_k + e_j = a_k + e_j$ ; apply (6.1) and induct.  $\square$

**Lemma 6.5** (Mixed-parameter detection [09]). *Take finitely many homogeneous positive-shift derivations and an iterated Lie bracket  $W$  of them. Replace each leaf occurrence  $D_\ell$  by  $\text{Exp}(s_\ell D_\ell)$  with independent scalar parameters and each bracket by the group word  $ghg^{-1}h^{-1}$ , giving  $\mathcal{W}(\mathbf{s})$ . In its action on a monomial, the coefficient of the product of all leaf parameters is the action of  $W$ . If  $W \neq 0$ , some specialization  $\mathbf{s} \in k^m$  gives a nonidentity group word.*

*Proof.* Formally,  $(1+sD+\cdots)(1+uE+\cdots)(1-sD+\cdots)(1-uE+\cdots) = 1 + su[D, E] + \text{terms of other multidegrees}$ ; induct on the bracket tree, using that the subtrees have disjoint parameter sets and that a group word is the identity when a required leaf parameter is zero. Choose a monomial and an output exponent on which  $W$  acts nontrivially. Only a finite set of shifts occurs, so by the support principle the coefficient at that output is a polynomial in the parameters, nonzero by the first part. An infinite field has a point where it does not vanish. Repeated leaves are handled by their separate parameters.  $\square$

**Theorem 6.6** (Sharp derived length [09]). *For  $\Gamma = \mathbb{Q}^r$  as above, every characteristic-zero  $k$  and every unital  $\mathfrak{o} \subseteq k$ , the group  $\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)$  and its Lie algebra  $\mathfrak{L}_A$  have derived length exactly  $r - 1$ .*

*Proof.* For  $r = 1$  use [theorem 4.13](#). For  $r \geq 2$ ,  $0 \neq Y_{r-2;r-1,r} \in \mathfrak{L}_A^{(r-2)}$  by [lemma 6.4](#), and [lemma 6.5](#) turns its balanced bracket into a nonidentity element of the  $(r - 2)$ nd derived subgroup. The upper bound is [theorem 6.3](#).  $\square$

The lower bound does not identify the abstract derived subgroup with the exponential of the derived Lie algebra; it uses independent parameters and a nonzero mixed coefficient. The upper bound uses BCH and strongly closed ideals [09].

### 6.4 Metabelian and not nilpotent: one group, printed once

On  $\Gamma = \mathbf{No}$  put  $D = D_{1,q_1}$ ,  $E = D_{\omega,q_2}$  and  $E_n = D_{\omega+n,q_2}$ . Both  $D$  and  $E$  satisfy the criterion:  $q_1$  kills the finite exponents and  $q_2$  kills every exponent bounded by an ordinary multiple of  $\omega$ .

**Proposition 6.7** ([04, 09]).  $[D, E_n] = E_{n+1}$  and  $[E_m, E_n] = 0$ ; the  $E_n$  are nonzero and linearly independent over  $\mathbb{R}$ . So the Lie algebra spanned by  $D, E_0, E_1, \dots$  is metabelian and not nilpotent. The same holds in rank three: in  $\mathbb{Q}^3$  with  $X = D_{e_1, \theta_2}$ ,  $Y = D_{e_2, \theta_3}$  one has  $(\text{ad } X)^n Y = D_{e_2 + ne_1, \theta_3} \neq 0$ .

*Proof.*  $q_1(\omega + n) = 1$ ,  $q_2(1) = q_2(\omega + n) = 0$ ; apply (6.1).  $E_n(t^{\omega^2}) = t^{\omega^2 + \omega + n}$  gives independence. The rank-three case is the same computation [09].  $\square$

Put  $A_s = \text{Exp}(sD)$  and  $B_u = \text{Exp}(uE)$  for  $s, u \in \mathbb{R}$ , and  $[F, G] = FGF^{-1}G^{-1}$ . Here  $e^{st}$  denotes the Hahn series  $\exp(st)$  in the monomial  $t = \omega^{-1}$ .

**Theorem 6.8** (Exact commutators [04, 08]).  $A_s B_u A_s^{-1} = \text{Exp}(ue^{st}E)$  and  $[A_s, B_u] = \text{Exp}(u(e^{st} - 1)E)$ . With  $C_0 = B_u$  and  $C_{n+1} = [A_s, C_n]$ ,

$$C_n = \text{Exp}(u(e^{st} - 1)^n E), \quad C_n(t^{-\omega^2}) = t^{-\omega^2} \exp(-u(e^{st} - 1)^n t^\omega),$$

so for  $s, u \neq 0$ ,  $C_n(t^{-\omega^2}) - t^{-\omega^2}$  has leading term  $-us^n t^{-\omega^2 + \omega + n}$ . Hence  $\langle A_s, B_u \rangle$  is metabelian and not nilpotent. At  $s = u = -1$ , the leading term is  $(-1)^n \omega^{\omega^2 - \omega - n}$ .

*Proof.*  $D$  fixes  $t$  (as  $q_1(1) = 0$ ) and  $A_s(t^\omega) = t^\omega e^{st}$ . On monomials,  $A_s E A_s^{-1}(t^\gamma) = q_2(\gamma) t^{\gamma + \omega} e^{st}$ , because  $E$  kills every series in  $t$  and  $q_1(\gamma + \omega) - q_1(\gamma) = 1$ ; so  $A_s E A_s^{-1} = e^{st} E$ , and exponentiating gives the first formula.  $E$  kills  $\mathbb{R}[[t]]$ , so the derivations  $h(t)E$ ,  $h \in \mathbb{R}[[t]]$ , commute and their exponentials add coefficients, which gives the second formula and, by induction,  $C_n$ . All shifts lie in the monoid generated by the well-ordered set  $\{1\} \cup (\omega + \mathbb{N})$ , so [theorem 4.8](#) justifies all sums. The maps  $\text{Exp}(h(t)E)$  form an abelian group normalized by  $A_s$ , which acts by  $h \mapsto e^{st}h$ ; so the generated group is abelian-by-abelian. Since  $q_2(-\omega^2) = -1$ ,  $C_n(t^{-\omega^2})$  is as displayed, and  $(e^{st} - 1)^n$  has leading term  $s^n t^n$ ; the leading term of the difference is therefore nonzero at every depth, so the lower central series never terminates.  $\square$

Source 08 works with  $A_1^{(08)} = A_{-1}$  and  $B_1^{(08)} = B_{-1}$  and proves faithfulness, which identifies the group. Write  $v = t = \omega^{-1}$ ,  $u_0 = t^\omega = \omega^{-\omega}$ ,  $P = t^{-\omega} = \omega^\omega$ ,  $M = t^{-\omega^2} = \omega^{\omega^2}$ , and for  $h$  supported in  $C_\omega^{>0}$  let  $N_h(t^\gamma) = t^\gamma \exp(-q_2(\gamma)h)$ , a phase shear ([theorem 4.17](#)) with  $N_h N_{h'} = N_{h+h'}$ , fixing  $v, u_0, P$  and with  $N_h(M) = M \exp(h)$ . Then  $B_{-1} = N_{u_0}$ ,  $A_{-1}(u_0) = u_0 e^{-v}$ ,  $A_{-1}(P) = P e^v$ , and  $A_{-1} N_h A_{-1}^{-1} = N_{A_{-1}(h)}$ .

**Lemma 6.9** (Finite exponential independence [08]). *For a Laurent polynomial  $p \in \mathbb{Z}[X, X^{-1}]$  put  $h_p = u_0 p(e^{-v})$  and  $\mathcal{N}_p = N_{h_p}$ . Then  $p \mapsto p(e^{-v})$  is injective,  $p \mapsto \mathcal{N}_p$  is an injective homomorphism, and  $A_{-1} \mathcal{N}_p A_{-1}^{-1} = \mathcal{N}_{Xp}$ .*

*Proof.* If  $\sum_{n \in F} k_n e^{-nv} = 0$  with  $|F| = r$  distinct integers, comparing the coefficients of  $v^j$ ,  $0 \leq j < r$ , gives a Vandermonde system with distinct nodes  $-n$ , so all  $k_n = 0$ . Injectivity of  $\mathcal{N}$  follows from  $\mathcal{N}_p(M) = M \exp(h_p)$ , and  $A_{-1}(h_p) = u_0 e^{-v} p(e^{-v}) = h_{Xp}$ .  $\square$

**Theorem 6.10** (A faithful  $\mathbb{Z} \wr \mathbb{Z}$  [08]). *The map  $\mathbb{Z}[X, X^{-1}] \rtimes \mathbb{Z} \rightarrow \mathcal{U}_{\mathbf{Oz}}$ ,  $(p, m) \mapsto \mathcal{N}_p A_{-1}^m$ , with  $(p, m)(q, n) = (p + X^m q, m + n)$ , is an injective homomorphism onto  $\langle A_{-1}, B_{-1} \rangle$ . So this group is the restricted wreath product  $\mathbb{Z} \wr \mathbb{Z}$ , of derived length exactly two. It extends  $\mathbb{C}$ -linearly to  $\mathbf{No}(i)$  preserving  $\mathbf{Oz}[i]$ , and it already lives on the set-sized exponent group  $\mathbb{Q}\omega^2 + \mathbb{Q}\omega + \mathbb{Q}$ .*

*Proof.* [Lemma 6.9](#) gives the multiplication law, and  $\mathcal{N}_{X^n} = A_{-1}^n B_{-1} A_{-1}^{-n}$  shows that the image is generated by  $A_{-1}, B_{-1}$ . If  $\mathcal{N}_p A_{-1}^m = \text{id}$ , applying it to  $P$  gives  $P e^{mv} = P$ , so  $m = 0$ , and then to  $M$  gives  $h_p = 0$ , so  $p = 0$ . Finite Laurent polynomials under multiplication by  $X$  are  $\bigoplus_{\mathbb{Z}} \mathbb{Z}$  with the shift. The group is nonabelian and its normal subgroup  $\mathcal{N}$  is abelian with cyclic quotient.  $\square$

**Remark 6.11** (All nonzero times [merge]). The same proof gives  $\langle A_s, B_u \rangle \cong \mathbb{Z} \wr \mathbb{Z}$  for all real  $s, u \neq 0$ : here  $B_u = N_{-ut^\omega}$ ,  $A_s N_h A_s^{-1} = N_{A_s(h)}$  and  $A_s(t^\omega) = t^\omega e^{st}$ , so with  $h_p = -ut^\omega p(e^{st})$  the map  $(p, m) \mapsto N_{h_p} A_s^m$  obeys the same law; the Vandermonde nodes  $ns$  are distinct, and  $A_s^m(P) = P e^{-mst}$  forces  $m = 0$ .

**Corollary 6.12** (Every set stabilizer is not nilpotent [04, 08]). *For every set  $P \subseteq \mathbf{No}$ , the pointwise stabilizer of  $P$  in  $\mathcal{U}_{\mathbf{Oz}}$  contains a subgroup isomorphic to  $\mathbb{Z} \wr \mathbb{Z}$ , metabelian and not nilpotent.*

*Proof.* Choose ordinals  $0 < \beta < \beta'$  above every inner exponent of every outer exponent of every member of  $P$  (lemma 7.1), and replace  $\omega, \omega^2$  by  $\epsilon = \omega^\beta, \omega^{\beta'}$ , using the functionals  $q_\beta, q_{\beta'}$  and the shifts  $1, \epsilon$ . Both functionals vanish on the exponents of  $P$  and satisfy the criterion, and  $q_\beta(\epsilon) = 1, q_\beta(1) = q_{\beta'}(\epsilon) = q_{\beta'}(1) = 0$ ; the proofs are unchanged.  $\square$

These examples live in rank three, consistent with theorem 6.3 (derived length at most two there) [08, 09].

**Example 6.13** (Binomial twists at three scales [10]). Source 10 shows noncommutativity with binomial twists at the scales of  $x = \omega, y = \omega^\omega$  and  $z = \omega^{\omega^2}$ . Let  $\sigma = \sigma_{1, -q_1}^{1+\xi}$  and  $\tau = \sigma_{\omega, -q_2}^{1+\xi}$ , that is  $\sigma(\omega^g) = \omega^g(1 + \omega^{-1})^{q_1(g)}$  and  $\tau(\omega^g) = \omega^g(1 + \omega^{-\omega})^{q_2(g)}$ ; both lie in  $\mathcal{U}_{\mathbf{Oz}}$  by theorem 4.18, since  $q_1(C_1) = 0$  and  $q_2(C_\omega) = 0$ . With  $u = 1 + x^{-1}$ ,

$$\sigma(x) = x, \sigma(y) = yu, \sigma(z) = z, \quad \tau(x) = x, \tau(y) = y, \tau(z) = z(1 + y^{-1}),$$

so  $\sigma\tau(z) - \tau\sigma(z) = -z/(xy(1 + x^{-1})) \neq 0$  and  $[\sigma, \tau](z) = z(1 + 1/(yu))/(1 + 1/y) \neq z$ , with  $[\sigma, \tau] = \sigma\tau\sigma^{-1}\tau^{-1}$ . The rational expressions denote the corresponding Hahn series. The example proves only that the integer-part stabilizer is not abelian and gives no presentation of it; the exponential shears at the same scales have the exact structure of theorems 6.8 and 6.10.

## 7 Fixed fields and arbitrarily large stabilizers

### 7.1 Exponent coefficient functionals

For  $\beta \in \mathbf{No}$  let  $q_\beta(\gamma) = [\omega^\beta]\gamma$  be the coefficient of  $\omega^\beta$  in the Conway normal form of the exponent  $\gamma$  (the *inner* normal form; the outer support of  $f = \sum c_\gamma t^\gamma$  consists of the exponents  $\gamma$ ). Normal-form addition makes  $q_\beta : \mathbf{No} \rightarrow \mathbb{R}$  additive; it need not commute with infinite sums inside the exponent group, and only finite additivity is used [04]. If  $0 < \delta \ll \omega^\beta$  then  $q_\beta(C_\delta) = 0$ : a nonzero  $\gamma$  with  $|\gamma| \leq n\delta$  has leading Conway exponent at most that of  $\delta$ , which is below  $\beta$  [04, 08, 09].

**Lemma 7.1** (Parameter avoidance [04, 08, 09, 10]). *Let  $P \subseteq \mathbf{No}(i)$  be a set,  $E$  the union of the outer supports of the real and imaginary parts of its members, and  $T$  the union of the inner supports of the members of  $E$ . Then  $E$  and  $T$  are sets, and for every ordinal  $\beta > 0$  above all members of  $T$ ,  $q_\beta$  vanishes on  $E$ .*

*Proof.* Replacement and union; every set of surreals has a strict upper bound, for example an ordinal above all their birthdays. This bounds a set of supports; it does not choose a bound for all exponents at once.  $\square$

**Theorem 7.2** (The common fixed field [04, 08, 09]).  *$\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$ , and already the elementary shears  $\sigma_{\delta, q_\beta}$  have common fixed field  $\mathbb{R}$ . The same holds for all field automorphisms of  $\mathbf{No}$  preserving  $\mathbf{Oz}$ . The common fixed elements in  $\mathbf{Oz}$  are  $\mathbb{Z}$ , and the common fixed subring of all ring automorphisms of  $\mathbf{Oz}$  is  $\mathbb{Z}$ .*

*Proof.* All these maps fix  $\mathbb{R}$  (corollary 5.2). If  $f \notin \mathbb{R}$ , some nonzero  $\gamma \in \text{supp } f$  has  $q_\beta(\gamma) \neq 0$  for some  $\beta$ , for instance its leading inner exponent [08]. Put  $\theta = q_\beta$  and  $\delta = \omega^{\beta-1}$ ; then  $\theta(C_\delta) = 0$  and  $\theta(\delta) = 0$ , and theorem 4.2 shows that  $\sigma_{\delta, \theta}$  moves  $f$ : its first displacement is  $a_\lambda \theta(\lambda) t^{\lambda+\delta}$ ,  $\lambda$  the least exponent of  $f$  with  $\theta(\lambda) \neq 0$ . Ring automorphisms of  $\mathbf{Oz}$  are restrictions of field automorphisms preserving  $\mathbf{Oz}$  (odg: def: thm: autreal).  $\square$

Source 04 reaches the same conclusion through lemma 7.3 with  $V = 0$ ; the explicit choice above is 08's and 09's.

## 7.2 Relative fixed fields and the Hahn hull

**Lemma 7.3** (Set-sized coefficient separation [04]). *Let  $V \subseteq \mathbf{No}$  be a set-sized  $\mathbb{Q}$ -subspace and  $g \in \mathbf{No} \setminus V$ . There are  $\delta > 0$  and an additive class map  $a : \mathbf{No} \rightarrow \mathbb{R}$  with  $a(V) = 0$ ,  $a(C_\delta) = 0$  and  $a(g) = 1$ .*

*Proof.* Let  $T = \bigcup_{x \in V \cup \{g\}} \text{supp}_\omega(x)$ , a nonempty set of inner exponents, and  $p_T(x) = ([\omega^\beta]x)_{\beta \in T} \in \mathbb{R}^T$ . It is injective on the set-sized space  $W = V + \mathbb{Q}g$ , so  $\ell_0(p_T(v + qg)) = q$  is a well-defined  $\mathbb{Q}$ -linear functional on  $p_T(W)$ ; extend it through a basis of the set  $\mathbb{R}^T$  to  $\ell$  and put  $a = \ell \circ p_T$ . Choose  $\beta_0$  below every member of  $T$  and  $\delta = \omega^{\beta_0}$ ; then  $p_T(C_\delta) = 0$ .  $\square$

For a set  $A \subseteq k((t^{\mathbf{No}}))$  ( $k = \mathbb{R}$  or  $\mathbb{C}$ ) let  $S_A$  be the union of the outer supports,  $V_A = \text{span}_{\mathbb{Q}}(S_A)$  and  $\mathcal{H}(A) = k((t^{V_A}))$ ; let  $\mathcal{E}_A$  be the group generated by the elementary shears  $\Phi_s$  (with real  $s$ ) that fix  $A$  pointwise, in the sense of [section 1.4](#).

**Theorem 7.4** (Exact elementary hull [04]). *For every set  $A \subseteq k((t^{\mathbf{No}}))$ ,  $\text{Fix}(\mathcal{E}_A) = \mathcal{H}(A) = k((t^{V_A}))$ .*

*Proof.* By [theorem 4.2](#) a nonidentity shear with functional  $a$  fixes all of  $A$  iff  $a$  vanishes on  $S_A$ , iff  $a(V_A) = 0$ ; such shears fix every series supported in  $V_A$ . If  $f \notin \mathcal{H}(A)$ , take  $g \in \text{supp } f \setminus V_A$  and apply [lemma 7.3](#).  $\square$

**Corollary 7.5** (Monomial parameters [04]). *For a set  $S \subseteq \mathbf{No}$  with  $V = \text{span}_{\mathbb{Q}}(S)$ , the pointwise stabilizer in  $\mathcal{U}_{\mathbf{Oz}}$  of  $\{t^g : g \in S\}$  has common fixed field exactly  $\mathbb{R}((t^V))$ .*

*Proof.* The elementary subgroup already has this fixed field. Conversely an ordered field automorphism fixing  $t^g$  fixes its positive rational powers, hence all  $t^v$ ,  $v \in V$ , and strong linearity fixes every series supported on  $V$ .  $\square$

Naming  $\omega = t^{-1}$  gives the fixed field  $\mathbb{R}((t^{\mathbb{Q}}))$ , much larger than  $\mathbb{R}(\omega)$ ; naming all monomials with real exponents gives  $\mathbb{R}((t^{\mathbb{R}}))$  [04]. On sets,  $A \mapsto \mathcal{H}(A)$  is extensive, monotone and idempotent, and  $\mathcal{H}(A)$  is the smallest Hahn subfield with divisible exponent group containing every monomial of the normal forms of  $A$  [04]. For nonmonomial parameters [theorem 7.4](#) computes the fixed field of  $\mathcal{E}_A$  only: a general strong automorphism fixing a series may do so by cancellation, and  $\mathcal{H}(A)$  is not asserted to be the model-theoretic definable closure [04] ([question 11.3](#)).

## 7.3 Product actions and nonsolvable stabilizers

**Theorem 7.6** (Faithful  $\mathbb{R}^\kappa$  actions [09]). *For every set  $P \subseteq \mathbf{No}$  and every set cardinal  $\kappa$  there is a faithful action of the full product  $\mathbb{R}^\kappa$  by strong 1-automorphisms preserving  $\mathbf{Oz}$  and fixing  $P$  pointwise.*

*Proof.* Let  $E, T$  be as in [lemma 7.1](#), and choose an ordinal  $\beta$  above  $\kappa$  and all of  $T$ . For  $j < \kappa$  put  $\delta_j = \omega^j$ ,  $\eta_j = \omega^{\beta+j}$  (ordinal addition in the index) and  $\theta_j = q_{\beta+j}$ . Then  $\theta_j|_E = 0$ ,  $\theta_j(C_{\delta_j}) = 0$ ,  $\theta_j(\delta_\ell) = 0$  and  $\theta_j(\eta_\ell)$  is the Kronecker delta. For  $\mathbf{a} \in \mathbb{R}^\kappa$  let  $D_{\mathbf{a}}(t^\gamma) = t^\gamma \sum_{j < \kappa} a_j \theta_j(\gamma) t^{\delta_j}$ ; the shifts form one well-ordered set, so [proposition 4.4](#) applies with no finiteness condition on  $\mathbf{a}$ .  $D_{\mathbf{a}}$  kills the exponents of  $P$ . The  $D_{\mathbf{a}}$  commute by (6.1) and add, so  $\mathbf{a} \mapsto \text{Exp}(D_{\mathbf{a}})$  is a homomorphism, and  $\text{Exp}(D_{\mathbf{a}})(t^{\eta_j}) = t^{\eta_j} \exp(a_j t^{\delta_j})$  detects every  $a_j$ . The result is one class relation encoding a set-indexed action.  $\square$

**Theorem 7.7** (Nonsolvable stabilizers [09]). *For every set  $P \subseteq \mathbf{No}$  the pointwise stabilizer of  $P$  in  $\mathcal{U}_{\mathbf{Oz}}$  is nonsolvable; for every  $r \geq 2$  it contains a subgroup whose derived length is at least  $r - 1$ , and exactly  $r - 1$  granted [cited theorem 1.1](#) for  $\Gamma = \mathbf{No}$ .*

*Proof.* Choose ordinals  $\beta \leq \eta_1 < \dots < \eta_r$  above  $T$ , and put  $e_j = \omega^{\eta_j}$ ,  $\theta_i = q_{\eta_i}$ . Then  $e_1 \ll \dots \ll e_r$ ,  $\theta_i(e_j)$  is the Kronecker delta, and each  $D_{e_j, \theta_i}$  ( $j < i$ ) satisfies the criterion and kills  $P$ . The balanced brackets of [lemma 6.4](#) are unchanged, so [lemma 6.5](#) shows that the group generated by all  $\text{Exp}(sD_{e_j, \theta_i})$ ,  $s \in \mathbb{R}$ , has derived length at least  $r - 1$ ; this already gives nonsolvability. The upper bound applies the filtration by the  $r - 1$  scales  $C_{e_1}, \dots, C_{e_{r-1}}$  as in [theorem 6.3](#); its group step uses BCH.  $\square$

#### 7.4 A definable-closure bound in set-sized Hahn fields [10]

For a set-sized  $\Gamma$  consider the structure  $K^\dagger = (K; 0, 1, +, -, \cdot, \mathcal{R}_o, (c)_{c \in k}, P_-, P_0, P_+)$ , with the Hahn order added when  $k$  is ordered. For a set  $A \subseteq K$  of parameters let  $S_A$  be the union of the supports of its members, as in [theorem 7.4](#), and put  $V_\delta(A) = \text{span}_{\mathbb{Q}}(S_A) + C_\delta$  for  $\delta > 0$  (10's  $S$ ,  $A_S$  and  $V_h(S)$ ).

**Proposition 7.8** (Definable-closure support bound [10]). *Every element of  $K$  definable over  $A$  in  $K^\dagger$  is supported in  $V_\delta(A)$  for every  $\delta > 0$ ; that is,  $\text{dcl}(A) \subseteq \bigcap_{\delta > 0} k((t^{V_\delta(A)}))$ .*

*Proof.* Let  $g \in \text{supp } f$  lie outside the  $\mathbb{Q}$ -subspace  $V_\delta(A)$ . A basis of the set-sized  $\mathbb{Q}$ -space  $\Gamma/V_\delta(A)$  gives a  $\mathbb{Q}$ -linear  $\theta : \Gamma \rightarrow \mathbb{Q} \subseteq k$  with  $\theta(V_\delta(A)) = 0$  and  $\theta(g) = 1$ . The twist  $\sigma_{\delta, \theta}^{1+\xi}$  lies in  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$  ([theorem 4.18](#)); it fixes  $k$ , preserves  $\mathcal{R}_o$ , commutes with  $P_-, P_0, P_+$  ([theorem 3.3](#)), preserves lc and hence the order, and fixes every monomial with exponent in  $S_A$ , hence every member of  $A$ . By [theorem 4.20](#) it moves  $f$ . An element definable over  $A$  is fixed by every automorphism of  $K^\dagger$  fixing  $A$ .  $\square$

The bound may be strict. It is not a quantifier-elimination result and does not describe the whole definable closure; it says only that support excluded by a separating exponent functional cannot be introduced by a definition [10]. For  $\Gamma = \mathbf{No}$  see [corollary 9.11](#).

## 8 What the pair $(\mathbf{No}, \mathbf{Oz})$ remembers

Sources 04, 08 and 09 each reconstruct the coefficient field from  $\mathbf{Oz}$ ; 08 and 09 add first-order definitions of the ordinary integers and of  $\Pi$ . All of this is now in the omnific Diophantine report, placed after the three pins, and is cited rather than reprinted:

- $\Pi$  is defined in  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  by  $\exists y(x^2 = 2y^2)$  (`odg:def:thm:ideal`); the first-order definitions of  $\Pi$  and of the graph of ct through ordinary integers are `odg:cor:definablect`, and a positive-existential definition of  $\mathbb{Z}$  is `odg:thm:guard`; Pell rigidity is `odg:cor:pell`;
- the multiplier identity  $(\Pi : \Pi) = \mathcal{A}_k$ ,  $k = \{0\} \cup \mathcal{A}_k^\times$  and its internal first-order form (`odg:def:thm:multiplier`, `odg:def:cor:internal`);  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  (`odg:thm:fractions`);
- the pure ring  $\mathbf{Oz}$  interprets  $\mathbf{No}$  with  $\mathbb{R}$ , the order, the finite ring, the infinitesimals and the standard part (`odg:def:thm:realrecovery`); the value group (`odg:def:cor:valuegroup`); every automorphism of  $\mathbf{Oz}$  extends uniquely to  $\mathbf{No}$  and fixes  $\mathbb{R}$  pointwise (`odg:def:thm:autreal`);

- for  $\mathbf{Oz}[i]$ :  $\mathbb{C}$  is definable (`odg:def:cor:internal`), every field automorphism of  $\mathbb{C}$  lifts coefficientwise and restriction to  $\mathbb{C}$  splits (`odg:def:prop:coeffaut`); a phase twist shows that neither  $\mathbf{No}$  nor conjugation is definable from  $\mathbf{Oz}[i]$  (`odg:def:lem:twist`, `odg:def:thm:norealaxis`).

In the sources: 04's intrinsic coefficient invariance, 08's characteristic reconstruction, its restriction correspondence and its split restriction to  $\text{Aut } \mathbb{C}$ , and 09's corollary on all pair automorphisms re-derive `odg:def:thm:multiplier`, `odg:def:thm:autreal` and `odg:def:prop:coeffaut`; source 10's coefficient-rigidity proposition re-derives the first two, with credit (section 9.2). 08's formula  $J(x) :\Leftrightarrow \forall n (\text{Int}(n) \wedge n \neq 0 \Rightarrow \exists y x = ny)$  is the definition `odg:cor:definablect` in structure (present at 08's pin); 09's order-free formula  $\Phi_J(a) :\Leftrightarrow a \in \mathbf{Oz} \wedge \forall u, v \in \mathbf{Oz} [u^2 - 2v^2 = 1 \Rightarrow \exists b \in \mathbf{Oz} a = ub]$  is a variant using all Pell first coordinates; both are dominated by the one-witness existential formula `odg:def:thm:ideal`, and 08's existential definition of  $\mathbb{Z}$  (six witnesses, two inequations) by `odg:thm:guard`. 09 observes that the order symbol can be omitted, positivity being definable by nonzero squares. The following extras are not in the Diophantine report.

**Lemma 8.1** (Order on  $\mathbf{Oz}$  through fractions [08]). *For  $z \in \mathbf{Oz}$ ,  $z \geq 0$  if and only if  $\exists a, b \in \mathbf{Oz} (b \neq 0 \wedge a^2 = zb^2)$ .*

*Proof.* If  $a^2 = zb^2$  with  $b \neq 0$  then  $z = (a/b)^2 \geq 0$ . Conversely a square root of  $z \geq 0$  exists in the real closed field  $\mathbf{No}$  and is a quotient  $a/b$  of omnific integers (`odg:thm:fractions`).  $\square$

The formula is existential with one inequation, consistent with the Diophantine report's result that the nonnegative cone is not positive-existentially definable (`odg:prop:notdiophantine`).

**Theorem 8.2** (Zero-cut projections and floor [08, 09]). *In  $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ , equivalently in the fraction field interpreted in the pure ring  $\mathbf{Oz}$ , the graphs of the projections  $P_-, P_0, P_+$  on all of  $\mathbf{No}$  (so  $P_0 = \text{ct}$  extended to  $\mathbf{No}$ ) and of the floor map  $x \mapsto \lfloor x \rfloor_{\mathbf{Oz}}$  are parameter-free definable. For every monomial  $m = t^\alpha$ , truncation below  $\alpha$ ,  $T_{<\alpha}(f) = m P_-(f/m)$ , is definable with the single parameter  $m$ . The same definitions work in  $(\mathbb{R}((t^\Gamma)), \mathbb{R}_\mathbb{Z})$  for every nonzero divisible ordered abelian  $\Gamma$ .*

*Proof.*  $\Pi$ ,  $\mathbb{R}$ ,  $\mathcal{O}$  and  $\mathfrak{m}$  are definable by the results cited above (in the  $t$ -convention  $\mathcal{O}$  is the finite ring and  $\mathfrak{m}$  the infinitesimals). For  $x \in \mathbf{No}$ ,  $P_-(x)$  is the unique  $j \in \Pi$  with  $x - j \in \mathcal{O}$ : existence by splitting the support at 0, uniqueness from  $\Pi \cap \mathcal{O} = 0$ . Then  $P_0(x)$  is the unique  $c \in \mathbb{R}$  with  $x - j - c \in \mathfrak{m}$ , since  $\mathbb{R} \cap \mathfrak{m} = 0$ , and  $P_+ = \text{id} - P_- - P_0$  [09]. The floor graph is  $y \in \mathbf{Oz} \wedge y \leq x < y + 1$ , unique because distinct omnific integers differ by at least one [08]; its value is `odg:thm:floor`. Dividing by  $m$  translates exponents by  $-\alpha$ ,  $P_-$  keeps those below 0, and multiplying by  $m$  restores them [09]. For the set-sized field the proof uses only real coefficients, the Hahn order (the field is real closed since  $\Gamma$  is divisible), and the definitions of  $\Pi$  and  $\mathbb{R}$ : the formula of `odg:def:thm:ideal` works for every  $\Gamma$ , and the multiplier identity `odg:def:thm:multiplier` holds inside the whole Hahn field by the same one-monomial proof [09].  $\square$

Every pair automorphism therefore preserves  $\Pi \oplus \mathbb{R} \oplus \mathfrak{m}$  summandwise, fixes the real summand, and satisfies  $\text{ct}(\sigma x) = \text{ct}(x)$  and  $\sigma(\lfloor x \rfloor) = \lfloor \sigma x \rfloor$  (corollary 5.2).

*Remark 8.3* (Zero truncation is not the normal form [08, 09]).  $P_0$  is definable on all of  $\mathbf{No}$  but is not multiplicative there. Definability of  $P_-$  does not define the set of monomials, and truncation at a named monomial is not a uniform definable choice of such a boundary at every valuation; the distinction matters in comparison with work on truncation [3]. Recovering every coefficient by  $[\omega^g]x = \text{ct}(\omega^{-g}x)$  would need the monomial map  $g \mapsto \omega^g$ , and section 9 shows that this is a genuine obstruction.

Source 08's Question 2, whether the reconstruction of  $\mathbb{C}$  from the pure ring  $\mathbf{Oz}[i]$  is first-order, is answered positively by `odg:def:cor:internal`, whose formula  $\exists y (x^2 = 2y^2)$  defines  $\Pi_{\mathbb{C}}$  in  $\mathbf{Oz}[i]$  and whose multiplier formula then defines  $\mathbb{C}$ ; 09 gives an independent route, its Pell-divisibility formula with the Gaussian Pell solutions (such as  $(i, i)$ ) handled, followed by the idealizer. The question is not repeated here.

## 9 What the pair forgets: monomials and representatives

**Theorem 9.1** (Set-parameter invisibility [04, 08, 09, 10]). *For every set  $P \subseteq \mathbf{No}$  there is a nonidentity strong real-linear 1-automorphism  $\sigma$  of  $\mathbf{No}$  with  $\sigma(\mathbf{Oz}) = \mathbf{Oz}$  that fixes  $P \cup \mathbb{R} \cup \{\omega\}$  pointwise and sends a positive Conway monomial in  $\mathbf{Oz}$  to a nonmonomial in  $\mathbf{Oz}$ . Explicitly, with  $\beta$  as in lemma 7.1 and  $\eta = \omega^\beta$ , one may take  $\sigma = \sigma_{1,q_\beta}$ , and then*

$$\sigma(t^{-\eta}) = t^{-\eta} \exp(-t) = \sum_{n \geq 0} \frac{(-1)^n}{n!} t^{-\eta+n}, \quad \text{that is} \quad \sigma(\omega^{\omega^\beta}) = \sum_{n \geq 0} \frac{(-1)^n}{n!} \omega^{\omega^\beta-n}. \quad (9.1)$$

Consequently, in  $\mathfrak{S} = (\mathbf{No}, 0, 1, +, \cdot, <, \mathbf{Oz})$  neither the monomial class  $\mathfrak{M} = \{\omega^x : x \in \mathbf{No}\}$  nor the graph of  $x \mapsto \omega^x$  is definable with parameters from any set, even after naming  $\mathbb{R}$  as a predicate, every ordinary real,  $\omega$ , and any further set of surreal constants.

*Proof.*  $q_\beta$  kills  $C_1$  since  $\beta > 0$ , so  $\sigma \in \mathcal{U}_{\mathbf{Oz}}$  by theorem 4.1. It fixes each monomial with exponent in  $E$ , hence every member of  $P$  by strongness, and it fixes  $\mathbb{R}$  and  $t^{-1} = \omega$  because  $q_\beta(-1) = 0$ . Since  $q_\beta(-\eta) = -1$ , (9.1) follows; all exponents  $-\eta + n$  are negative because  $\eta$  is infinite, and the terms at  $-\eta$  and  $-\eta + 1$  show that the image is not a monomial. A class definable from parameters fixed pointwise by an automorphism of  $\mathfrak{S}$  is invariant under it;  $\sigma$  preserves every symbol and moves an element of  $\mathfrak{M}$  out of  $\mathfrak{M}$ . A definable graph of the omega-map would make its range definable by an existential quantifier.  $\square$

The three sources give the same construction: 04 with  $a = [\omega^\beta]$  and explicit fixing of  $\mathbb{R} \cup \{\omega\}$ , 08 with the witness  $\omega^{\omega^\alpha} \mapsto \sum_n \omega^{\omega^\alpha-n}/n!$  (its sign convention), 09 with  $q_\eta$ ; 10 uses a binomial twist instead of the exponential shear, and its image has exactly two terms (theorem 9.8). An individual formula uses finitely many parameters; the theorem is the stronger statement that one automorphism fixes an entire prescribed set. Nothing is claimed after naming a proper class of parameters: naming every surreal would remove the symmetry [04, 08]. Expanding the structure by the already definable order, floor or zero-truncation maps cannot remove the obstruction [08].

**Corollary 9.2** ([04, 09, 10]). *The monomial class is not determined, as an automorphism-invariant subclass, by  $\mathbf{Oz}$ , its ordered fraction field and any set of parameters; the standard normal-form section of the natural valuation is not first-order recoverable from these data. The Conway monomials lying in  $\mathbf{Oz}$  are not definable in the pure ring  $\mathbf{Oz}$  with parameters from any set.*

*Proof.* The range of the section is  $\mathfrak{M}$ . For the pure ring, restrict the witness of theorem 9.1 to  $\mathbf{Oz}$ , with parameters in  $\mathbf{Oz}$ ; automorphisms of  $\mathbf{Oz}$  and field automorphisms of  $\mathbf{No}$  preserving  $\mathbf{Oz}$  correspond (`odg:def:thm:autreal`).  $\square$

### 9.1 No definable representative rule

Write  $x \sim_v y$  if  $v(x) = v(y)$  on  $\mathbf{No}^\times$ . A *valuation representative rule* is a function  $s : \mathbf{No}^\times \rightarrow \mathbf{No}^\times$  with  $v(s(x)) = v(x)$  and  $s(x) = s(y)$  whenever  $x \sim_v y$ . For a set-sized field this is a section

of the quotient by  $\mathcal{O}^\times$ ; the formulation also makes sense on the class **No** without treating proper-class equivalence classes as elements. No multiplicativity, positivity or monomial-image assumption is imposed [09].

**Lemma 9.3** (A whole fiber moves [09]). *For  $\sigma$  and  $\eta$  as in [theorem 9.1](#), every  $f$  with  $v(f) = -\eta$  satisfies  $v(\sigma f - f) = -\eta + 1$  and  $\text{lc}(\sigma f - f) = -\text{lc}(f)$ .*

*Proof.* This is [theorem 4.2](#) with  $\gamma_0 = -\eta$  and  $\theta(-\eta) = -1$ : later input terms contribute beyond  $-\eta + 1$ , higher iterates at  $-\eta + n$ ,  $n \geq 2$ .  $\square$

**Theorem 9.4** (No definable section [09]). *For every set  $P \subseteq \mathbf{No}$ , no  $P$ -definable function on  $\mathbf{No}^\times$  is a valuation representative rule.*

*Proof.* Let  $x = t^{-\eta}$ . Since  $\sigma$  fixes valuations,  $\sigma x \sim_v x$  and  $s(\sigma x) = s(x)$ , while  $P$ -definability gives  $s(\sigma x) = \sigma(s(x))$ . So  $s(x)$  would be a fixed point of  $\sigma$  with valuation  $-\eta$ , contradicting [lemma 9.3](#).  $\square$

This is the strongest nondefinability statement of the sources; it covers any representative rule, including one-term ones, and does not follow from character twists. The nondefinability is relative to the field-and-integer-part language; Conway's normal forms remain set-theoretically defined in the ambient foundation [09].

*Remark 9.5* (Character twists, one-term series and the RV sort [04, 02]). Bare nondefinability of the coefficient-one monomials already follows from character twists: for a nonzero additive  $a : \mathbf{No} \rightarrow \mathbb{R}$  with  $a(V_A) = 0$  ([lemma 7.3](#)),  $\sum c_g t^g \mapsto \sum c_g e^{a(g)} t^g$  preserves **Oz**, fixes  $A$  and sends some monomial to a non-unit multiple of itself [04]. These twists are the  $M_{\chi, \text{id}}$  of [theorem 5.1](#); they are 02's  $\tau_{\chi, t}$  ([remark 17.10](#)), the real form of the Diophantine report's phase twist (`odg:def:lem:twist`), and the collection's character twists (`saut:thm:decomp`); they are established and not claimed new. What the shears add is motion inside the leading-term kernel with more than one nonzero term. In particular the class of nonzero one-term series  $\{c\omega^g : c \in \mathbb{R}^\times, g \in \mathbf{No}\}$  is not definable from the ordered field, **Oz** and set parameters: the shear sends a one-term series to a multiterm one, which rescaling cannot do [04]. Nondefinability also survives adding the sort  $\text{RV}(K) = K^\times / (1 + \mathfrak{m})$  with its multiplication, the quotient map and every element of that sort named, since a 1-automorphism acts trivially on it (for the class **No** this is a metatheoretic expansion, read formula by formula). This must be distinguished from naming a representative map  $\text{RV}(K) \rightarrow K$ , which the Conway monomials partly constitute and which would destroy the symmetry by design:  $\Phi_s(t^{-\omega})$  has leading term  $t^{-\omega}$ , yet  $\Phi_s$  moves  $t^{-\omega}$  [04].

*Remark 9.6* (Contrasts in the collection). The single dilation  $S_2(\sum a_g t^g) = \sum a_g t^{2g}$  defines the canonical monomials, every coefficient and truncation in the language of fields with one automorphism (`dsup:thm:definability` in `docs/surcomplex/single-dilation-hahn-support`); (**No**, **Oz**) does not, even with set parameters. The two expansions differ exactly on the monomial section, and there is no conflict: the shears do not commute with  $S_2$ , in line with that report's centralizer theorem (`dsup:thm:centralizer`). The shears are not exponential: by `cor:question54` of the rigidity report `docs/surreal/exponential-automorphism-rigidity` (from its `thm:no`), every exponential 1-automorphism of **No** is the identity, which is stronger than [9, Proposition 5.2] (strongly linear exponential 1-automorphisms are trivial), the result cited by 09. The shears are compatible with that report's displacement lemma: their displacement  $-t^{-\eta+1}$  is infinite. Nor do they preserve the omega-map, whose compatibility questions [9, Questions 5.6, 5.7] are untouched here.

## 9.2 Two-term witnesses, simplicity and the exponential [10]

For a surreal  $b > 0$  and a surreal  $\delta$  with  $0 < \delta < 1$ , the functional  $q_b$  vanishes on  $C_\delta$ : every member of  $C_\delta$  is finite, and a finite surreal has no positive exponent in its normal form. Source 10's class twist is the binomial twist  $\Sigma_{b,\delta} = \sigma_{\delta,-q_b}^{1+\xi}$ ; in normal forms,

$$\Sigma_{b,\delta} \left( \sum_g a_g \omega^g \right) = \sum_g a_g \omega^g (1 + \omega^{-\delta})^{q_b(g)}, \quad (9.2)$$

with the real binomial series. Each image has a set-sized support inside  $\{g - n\delta\}$ , and the proof of [theorem 4.18](#) is elementwise, so  $\Sigma_{b,\delta}$  is a strong real-linear 1-automorphism of  $\mathbf{No}$  with inverse given by  $-q_b$ ; it fixes  $\mathbb{R}$ , preserves the order and  $\mathbf{Oz}$ , and commutes with  $P_-, P_0, P_+$  [10].

*Remark 9.7* (An exponent index is not a fixed element [10]).  $\Sigma_{b,\delta}$  fixes the monomial  $\omega^{-\delta}$  but need not fix the surreal  $\delta$ : the equality  $q_b(-\delta) = 0$  controls the monomial with exponent  $-\delta$ , not the exponents inside the normal form of  $\delta$ . (For  $\delta = \omega^{-\omega^b}$  one has  $\Sigma_{b,\delta}(\delta) \neq \delta$ .) This is the inner/outer distinction of [section 7](#), and it is why an automorphism can fix all valuations without commuting with the omega-map.

**Theorem 9.8** (Two-term witnesses [10]). *For every set  $S \subseteq \mathbf{No}$  there is an ordinal  $\beta > 0$  such that, for every surreal  $\delta$  with  $0 < \delta < 1$ ,  $\Sigma_{\beta,\delta}$  fixes  $S$  pointwise and*

$$\Sigma_{\beta,\delta}(\omega^{\omega^\beta}) = \omega^{\omega^\beta} + \omega^{\omega^\beta - \delta}, \quad (9.3)$$

*a two-term omnific integer and not a Conway monomial. These images are pairwise distinct as  $\delta$  ranges over  $(0, 1)$ , and they form a proper class.*

*Proof.* Take  $\beta$  as in [lemma 7.1](#), above the inner supports of the outer supports of the members of  $S$ . Then  $q_\beta$  vanishes on every exponent of every member of  $S$ , so (9.2) fixes their monomials and, by strongness, the members themselves. Since  $q_\beta(\omega^\beta) = 1$ , the binomial exponent at  $\omega^{\omega^\beta}$  is one, which gives (9.3) with no further terms. Both  $\omega^\beta$  and  $\omega^\beta - \delta$  are positive, since  $\omega^\beta$  is infinite, and they differ, so by uniqueness of normal forms the image is an omnific integer and not a monomial. Distinct  $\delta$  give distinct second exponents. The class  $(0, 1)$  is proper, since  $\alpha \mapsto \omega^{-(\alpha+1)}$  injects the ordinals into it, and  $\delta$  is recovered from the image, so the images form a proper class.  $\square$

The witness of [theorem 9.1](#) is an infinite series; here two terms suffice. The theorem gives, for each set of parameters, one uniform formula for individual automorphisms indexed by the set parameters  $\beta, \delta$  and a proper class of values at one input; it forms no proper class of automorphisms [10].

**Corollary 9.9** (Bounded birthdays [10]). *For every ordinal  $\alpha$  there is a nonidentity strong ordered-field automorphism of  $\mathbf{No}$  that preserves  $\mathbf{Oz}$  and fixes every surreal of birthday less than  $\alpha$ .*

*Proof.* The surreals of birthday less than  $\alpha$  form a set; apply [theorem 9.8](#).  $\square$

The automorphism is not asserted to preserve the birthdays of other elements: the statement is pointwise fixing of a stage, not preservation of the simplicity hierarchy [10], which by [theorem 9.10](#) no such automorphism preserves.

The proof of [theorem 4.20](#) uses only the set-sized support of one input, so the fixed class of a class twist is explicit [10]:

$$\text{Fix}(\Sigma_{b,\delta}) = \left\{ \sum_g a_g \omega^g : q_b(g) = 0 \text{ for every } g \in \text{supp} \right\}, \quad (9.4)$$

and for a nonfixed  $f$ , with  $g_*$  its greatest exponent having  $q_b(g_*) \neq 0$ , the leading term of  $\Sigma_{b,\delta}(f) - f$  is  $a_{g_*} q_b(g_*) \omega^{g_* - \delta}$ . The fixed class field is relatively algebraically closed in  $\mathbf{No}$ , in the finite-polynomial sense, by the proof of [corollary 4.21](#). If  $(r_j)_{j \in J}$  is a  $\mathbb{Q}$ -linearly independent family of reals with  $|J| = 2^{\aleph_0}$ , the monomials  $\omega^{r_j \omega^b}$  are algebraically independent over (9.4) by [proposition 4.22](#). No exact transcendence degree of the class extension is claimed [10].

Let  $\mathfrak{S}^\dagger = (\mathbf{No}; 0, 1, +, -, \cdot, <, \mathbf{Oz}, (r)_{r \in \mathbb{R}}, P_-, P_0, P_+)$ , the structure  $\mathfrak{S}$  of [theorem 9.1](#) with every real and the three zero-cut projections named; naming all reals uses a set-sized language. Definability has the usual finite-formula meaning, formula by formula in the class structure; no truth predicate for classes is assumed. Simplicity is the relation of being a proper initial segment of a sign sequence, not comparison of birthdays [10].

**Theorem 9.10** (What remains undefinable [10]). *For every set  $S \subseteq \mathbf{No}$  none of the following is first-order definable over  $S$  in  $\mathfrak{S}^\dagger$ : (i) the monomial class  $\mathfrak{M}$ ; (ii) the graph of  $x \mapsto \omega^x$ ; (iii) the simplicity relation; (iv) the graph of the Gonshor exponential. The class  $\mathfrak{M} \cap \mathbf{Oz}$  is not definable over any set of parameters in the ordered ring  $(\mathbf{Oz}; 0, 1, +, -, \cdot, <)$ .*

*Proof.* Take  $\Sigma = \Sigma_{\beta,\delta}$  from [theorem 9.8](#). It fixes  $S$  and every named real and preserves every symbol of  $\mathfrak{S}^\dagger$ . Items (i) and (ii) follow as in [theorem 9.1](#), since  $\Sigma$  moves the monomial  $Z = \omega^{\omega^\beta}$  out of  $\mathfrak{M}$ .

(iii) A bijection of  $\mathbf{No}$  preserving order and simplicity is the identity [9, Proposition 5.1]: the simplicity tree has one root, each node has two immediate successors distinguished by the order, and each limit sign sequence has a unique continuation, so induction on length fixes every node. A definition of simplicity would make  $\Sigma$  such a bijection, but  $\Sigma \neq \text{id}$ .

(iv) Put  $d = \Sigma(Z) - Z = \omega^{\omega^\beta - \delta}$ , which is positive and infinite. The Gonshor exponential is strictly increasing, extends the real exponential and satisfies  $\exp(x + y) = \exp(x) \exp(y)$ , so  $\exp(d)$  is positive infinite and  $v(\exp d) \neq 0$ . As  $\Sigma$  fixes valuations,  $v(\Sigma(\exp Z)) = v(\exp Z)$ , whereas  $v(\exp(\Sigma Z)) = v(\exp Z) + v(\exp d) \neq v(\exp Z)$ . So  $\Sigma$  does not commute with the exponential, while the graph of a definable function is invariant.

For the ordered ring take  $S \subseteq \mathbf{Oz}$ : the restriction of  $\Sigma$  is an automorphism of  $(\mathbf{Oz}; 0, 1, +, -, \cdot, <)$  fixing  $S$ , and  $Z$  and  $\Sigma(Z)$  both lie in  $\mathbf{Oz}$ , but only  $Z$  lies in  $\mathfrak{M}$ .  $\square$

Items (i), (ii) and the ordered-ring clause are [theorem 9.1](#) and [corollary 9.2](#), the order of  $\mathbf{Oz}$  being definable in its pure ring ([lemma 8.1](#)), and the projections being definable in the pair ([theorem 8.2](#)); 10 re-derives them. Items (iii) and (iv) are new here. They need only some nonidentity 1-automorphism of  $\mathfrak{S}^\dagger$  fixing  $S$ : the merge notes that for (iv) the rigidity report's `cor:question54` (every exponential 1-automorphism of  $\mathbf{No}$  is the identity, from `thm:no`, whose proof is the same valuation comparison) would equally serve with the witness of [theorem 9.1](#). The formal `exp` of principal units is not the Gonshor exponential, and the class twists are not claimed to preserve the latter; they provably do not. The displacement mechanism is related to the exponential rigidity arguments of [9] and of that report; what is added is that the witness preserves  $\mathbf{Oz}$  and fixes an arbitrary set of parameters [10].

**Corollary 9.11** (Support bounds for definable elements [10]; outer bound [merge]). *Let  $A \subseteq \mathbf{No}$  be a set and  $f$  definable over  $A$  in  $\mathfrak{S}^\dagger$ . Then  $f \in \mathbb{R}((t^{V_A}))$ , with  $V_A = \text{span}_{\mathbb{Q}}(S_A)$  as in [theorem 7.4](#). In particular  $B_+(\{f\}) \subseteq B_+(A)$ , where  $B_+(A) = \bigcup_{s \in A} \bigcup_{g \in \text{supp } s} (\text{supp } g \cap \mathbf{No}^{>0})$  is the positive second-level support: definability creates no new positive inner support position.*

*Proof.* Every elementary shear fixing  $A$  is an automorphism of  $\mathfrak{S}^\dagger$  fixing  $A$ , so  $f \in \text{Fix}(\mathcal{E}_A) = \mathbb{R}((t^{V_A}))$  by [theorem 7.4](#). Each exponent of  $f$  is then a finite rational combination of exponents of members of  $A$ , and the normal-form support of a finite sum lies in the union of the supports. Source 10's direct proof of the inner bound: if  $b > 0$  is in the inner support of an exponent of  $f$  but not in  $B_+(A)$ , then  $\Sigma_{b,\delta}$  fixes  $A$  and moves  $f$  by (9.4);  $b$  need not be an ordinal.  $\square$

Both are upper bounds only. They do not say that every element obeying them is definable, nor that every coefficient is recoverable from the named projections [10]; compare [question 11.3](#).

**Floor, fractions and coefficient rigidity [10].** Source 10 also proves that every ordered-field automorphism stabilizing  $\mathbf{Oz}$  commutes with the floor map, that  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  makes any automorphism fixing  $\mathbf{Oz}$  pointwise the identity, and that every ring automorphism of  $\mathbf{Oz}$  extends uniquely to  $\mathbf{No}$  and fixes  $\mathbb{R}$  pointwise, through  $\Pi = \bigcap_n n\mathbf{Oz}$ , the multiplier ring  $\mathcal{A}_k$  of  $\Pi$  and its units  $\mathbb{R}^\times$ , with no strongness assumption. These are [corollary 5.2](#), `odg:thm:fractions`, `odg:def:thm:multiplier` and `odg:def:thm:autreal`, which 10 credits as prior repository results; they are not reprinted. There is no conflict between  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  and the moving of monomials: a ring determines its fraction field up to isomorphism, not a distinguished monomial section inside it; the automorphisms preserve  $\mathbf{Oz}$  setwise and move some of its elements [10].

## 10 Surcomplex and Gaussian consequences

Identify  $\mathbf{No}(i)$  with  $\mathbb{C}((t^{\mathbf{No}}))$  by taking coefficientwise real and imaginary parts; the union of the two supports is well ordered. Then  $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}} = \mathbf{Oz} + i\mathbf{Oz}$ , which is not an ordered ring; conjugation is coefficientwise [04, 08, 09].

**Theorem 10.1** (Gaussian stabilizer and conjugation-compatible witnesses [04, 09, 10]). *Theorem 3.3 applies to  $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$  with the same condition. The real-coefficient shears and actions of [sections 7](#) and [9](#) extend  $\mathbb{C}$ -linearly, commute with conjugation, preserve  $\mathbf{No}$  and  $\mathbf{Oz}[i]$ ; for every set of surcomplex parameters they fix the parameters and give nondefinability of the monomials and of any valuation representative rule, even with the real axis, conjugation and every ordinary complex constant named. Their common fixed field is  $\mathbb{C}$ , and for every set  $A \subseteq \mathbf{No}(i)$  the real elementary shears fixing  $A$  have common fixed field  $\mathbb{C}((t^{V_A}))$ .*

*Proof.* The criterion was proved for every characteristic-zero  $k$  and unital  $\mathfrak{o}$ , and its proof uses no order on coefficients. The functionals  $q_\beta$  are real, so the derivations and their exponentials commute with conjugation and restrict to the real automorphisms on  $\mathbf{No}$ . Use the union of the real and imaginary supports in [lemma 7.1](#); the moved monomial is real, and [lemma 9.3](#) holds with any nonzero complex leading coefficient. For the fixed field, the first displacement of [theorem 4.2](#) is a product of a nonzero real functional value and a nonzero complex coefficient. The hull statement is [theorem 7.4](#) with  $k = \mathbb{C}$  [04].  $\square$

For these real-axis-preserving maps the surreal modulus is equivariant,  $|\sigma(z)| = \sigma(|z|)$ , since  $|z|$  is the nonnegative square root of  $z\bar{z}$ ; this is not a statement that moduli are fixed [09].

**Theorem 10.2** (The real axis inside the leading-term kernel [04]). *For every set  $A \subseteq \mathbf{No}(i)$  there is a strong  $\mathbb{C}$ -linear 1-automorphism of  $\mathbf{No}(i)$  that preserves  $\mathbf{Oz}[i]$ , fixes  $A$  pointwise and does not preserve  $\mathbf{No}$ . Hence  $\mathbf{No}$  is not definable in  $(\mathbf{No}(i), 0, 1, +, \cdot, \mathbf{Oz}[i])$  with parameters from any set, even with all ordinary complex constants named and the Hahn valuation ring as a predicate.*

*Proof.* Take  $\beta$  above the inner exponents of the outer supports of  $A$ , and the flow  $\Phi_s$  of [theorem 4.1](#) with  $\theta = q_\beta$ ,  $\delta = 1$  and the imaginary time  $s = i$ . It fixes  $A$  and is strong,  $\mathbb{C}$ -linear, leading-term preserving and  $\mathbf{Oz}[i]$ -preserving. With  $\eta = \omega^\beta$ ,  $\Phi_i(t^{-\eta}) = t^{-\eta} \exp(-it) = t^{-\eta} - it^{-\eta+1} - \frac{1}{2}t^{-\eta+2} + \dots$  has a nonreal coefficient, although  $t^{-\eta}$  is real; all its exponents are negative. A 1-automorphism preserves every valuation, hence the valuation ring.  $\square$

Phase twists preserving  $\mathbf{Oz}[i]$  and moving the real axis are already in the collection (`saut:thm:phase`; `odg:def:lem:twist`, `odg:def:thm:norealaxis`); the new points are arbitrary set-parameter fixing *inside the complete leading-term kernel* and multiterm images. The real shears commute with conjugation and the imaginary-time shear does not; the two conclusions must not be combined into a claim that one automorphism commutes with conjugation and moves its fixed field [04].

**Proposition 10.3** (A wild coefficient lift [08]). *There is a strong automorphism of  $\mathbf{No}(i)$  that preserves  $\mathbf{Oz}[i]$ , fixes  $\mathbb{Z}[i]$  and every Conway monomial, and sends an ordinary real number outside  $\mathbb{R}$ . So neither  $\mathbb{R}$  nor  $\mathbf{No}$  is invariant under all field automorphisms preserving  $\mathbf{Oz}[i]$ .*

*Proof.* Choose a real  $r$  transcendental over  $\mathbb{Q}$  and a transcendence basis of  $\mathbb{C}$  over  $\overline{\mathbb{Q}}$  containing it. On the rational function field it generates, fix  $\overline{\mathbb{Q}}$  and the other basis elements and send  $r \mapsto r + i$  (inverse  $r \mapsto r - i$ ), and extend to an automorphism  $\rho$  of  $\mathbb{C}$  by the set-sized algebraic-closure extension theorem; the extension is onto, since its image is algebraically closed and  $\mathbb{C}$  is algebraic over the base. The coefficientwise lift  $\hat{\rho}$  (`odg:def:prop:coeffaut`) preserves  $\mathbf{Oz}[i]$  and monomials and sends  $r$  to  $r + i$ .  $\square$

**Corollary 10.4** (Gaussian fixed field [08, 09]). *The common fixed elements in  $\mathbf{No}(i)$  of all field automorphisms preserving  $\mathbf{Oz}[i]$  are exactly  $\mathbb{Q}$ ; in  $\mathbf{Oz}[i]$  they are  $\mathbb{Z}$ .*

*Proof.* The shears of [theorem 10.1](#) reduce a common fixed point to  $\mathbb{C}$ . Every field automorphism of  $\mathbb{C}$  lifts coefficientwise and preserves  $\mathbf{Oz}[i]$  (`odg:def:prop:coeffaut`). A nonrational algebraic number is moved by a conjugate embedding extended to  $\mathbb{C}$ ; a transcendental one by a translation of a transcendence basis element; rationals are fixed. Finally  $\mathbb{Q} \cap \mathbf{Oz}[i] = \mathbb{Z}$ .  $\square$

This agrees with `saut:prop:fixedfields` of the surcomplex automorphism report,  $\text{Fix}(\text{Aut } \mathbf{No}(i)) = \mathbb{Q}$ , for the larger collection of all field automorphisms. The real and complex answers differ because a real automorphism preserving  $\mathbf{Oz}$  fixes every ordinary real,  $\mathbb{R}$  being parameter-free definable in the pair ([corollary 5.2](#), `odg:def:thm:autreal`), whereas a complex one preserving  $\mathbf{Oz}[i]$  can move reals unless real structure is required; the explicit witnesses of [theorem 10.1](#) fix all of  $\mathbb{C}$  [09]. *Correction* (batch 30): this sentence formerly gave as the reason that a real automorphism “preserves order and cannot move an ordinary real”. Read literally that is false: the Taylor automorphism of `saut:thm:coeffflow` is a strong order-preserving automorphism of  $\mathbf{No}$  sending a real  $b$  to  $b + t^\delta$ . It does not preserve  $\mathbf{Oz}$  ([example 4.12](#)); preservation of  $\mathbf{Oz}$ , not of the order, is what fixes  $\mathbb{R}$ . The comparison and the fixed fields above are unchanged, so this corrects the stated reason and retracts no result. Three mechanisms move the real axis while preserving  $\mathbf{Oz}[i]$ : the phase twist of the Diophantine report (fixes  $\mathbb{C}$ , preserves supports), the imaginary-time shear of [theorem 10.2](#) (fixes  $\mathbb{C}$  and all leading terms), and the wild lift of [proposition 10.3](#) (fixes monomials, moves  $\mathbb{R}$ ). The real shears of [theorem 10.1](#) preserve the real axis.

**Proposition 10.5** (Two-term Gaussian witnesses [10]). *For every set  $S \subseteq \mathbf{No}(i)$  there is an ordinal  $\beta > 0$  such that for every surreal  $0 < \delta < 1$  the  $\mathbb{C}$ -linear extension  $\tilde{\Sigma}_{\beta,\delta}(x + iy) = \Sigma_{\beta,\delta}(x) + i \Sigma_{\beta,\delta}(y)$  is a strong  $\mathbb{C}$ -linear 1-automorphism of  $\mathbf{No}(i)$  which (i) fixes  $S$  and  $\mathbb{C}$  pointwise; (ii) preserves  $\mathbf{No}$  and  $\mathbf{Oz}[i]$  and commutes with conjugation; (iii) preserves the*

order of  $\mathbf{No}$  and commutes with  $P_-, P_0, P_+$ ; (iv) sends  $\omega^{\omega^\beta}$  to the two-term element (9.3). Consequently, with parameters from any set, neither the monomial class nor, on the real axis, the graph of the omega-map, the simplicity relation or the graph of the Gonshor exponential is definable in  $\mathbf{No}(i)$  expanded by conjugation, the predicate  $\mathbf{No}$ , the order of  $\mathbf{No}$ , the predicate  $\mathbf{Oz}[i]$ , every complex constant and  $P_-, P_0, P_+$ .

*Proof.* Apply theorem 9.8 to the set of real and imaginary parts of the members of  $S$ . The binomial coefficients in (9.2) are real, so  $\tilde{\Sigma}_{\beta,\delta}$  is the same formula on complex-coefficient normal forms; it commutes with conjugation because it acts by one real map on both components and fixes  $i$ . Theorem 3.3 with  $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$  gives (ii) and (iii) for  $\mathbf{Oz}[i]$  and the projections. The witness is real. A definition of the monomial class would be preserved, contradicting (iv); for the real-axis statements restrict a putative definition to  $\mathbf{No}$  and argue as in theorem 9.10. No global surcomplex exponential is introduced.  $\square$

Its fixed class field is the complex-coefficient analogue of (9.4), with the same first displacement, no nontrivial finite orbits and relative algebraic closedness [10]. Beyond theorem 10.1, the proposition adds the two-term witness, the order of the real axis, the projections, and simplicity and the exponential. It differs from the phase twists of the Diophantine report, which move the real axis (`odg:def:lem:twist`): here the real axis and conjugation are preserved, and the obstruction lies inside the fixed real axis [10].

## 11 Scope, answered questions, and further questions

### 11.1 What is settled, and status in other reports

For a strong infinitesimal automorphism, preservation of the integer part is exactly a convex-scale annihilation condition; the size of the residual symmetry is measured by a sharp solvability bound in finite rank and by arbitrarily large parameter-fixing actions on  $\mathbf{No}$ ; and the omnific predicate recovers the coefficient fields and zero-cut truncation but no definable valuation cross-section [09]. An attempt to reconstruct Conway monomials by a first-order formula in the pair cannot succeed with any set of constants; a classification of strong  $\mathbf{Oz}$ -automorphisms by coefficient and exponent actions alone misses the higher-rank factor  $\mathcal{U}_{\mathbf{Oz}}$ ; and an alleged rank-one nontrivial 1-automorphism preserving a divisible-exponent integer part must violate a support hypothesis, as example 4.12 shows [09].

The results bear on four questions of other reports. They are recorded here; the other reports are not edited by this merge.

- `osq:q:invisible` of docs/surreal/set-sized-quotients-of-omnific-integers asks, in its second clause, how much an automorphism of  $\mathbf{Oz}$  can do inside  $\Pi$  while fixing  $\text{ct}$ . *Partly answered:* every automorphism of  $\mathbf{Oz}$  fixes  $\mathbb{R}$  and  $\text{ct}$  (`odg:def:thm:autreal`, corollary 5.2); the *strong* ones are exactly  $\{M_{\chi,\tau}\} \ltimes \mathcal{U}_{\mathbf{Oz}}$ , with  $\mathcal{U}_{\mathbf{Oz}}$  classified by theorem 3.3; they move monomials to infinite series, fix any set of parameters, carry  $\mathbb{R}^\kappa$  and have nonsolvable stabilizers (theorems 7.6, 7.7 and 9.1). Nonstrong automorphisms, and the first clause of that question, remain open. *Status* (batch 30): there are no nonstrong automorphisms of  $\mathbf{Oz}$  (theorem 23.2), so the second clause is *answered* (corollary 24.4 and section 32.1); the first clause remains open.
- `odg:def:q:realform` of the Diophantine report asks which natural structure on  $\mathbf{Oz}[i]$  rules out the coefficient-fixing twists and determines the real omnific subring. *Negative information, not an answer:* naming the valuation ring, every complex constant, the RV

sort, or requiring leading-term fixing does not suffice ([theorem 10.2](#)); and in  $(\mathbf{No}(i), \mathbf{Oz}[i])$  even  $\mathbb{R} \subseteq \mathbb{C}$  is not invariant ([proposition 10.3](#)).

- The surcomplex automorphism report’s research direction “effective descriptions inside the leading-term kernel” (its section `saut:sec:questions`) is *not* answered. Its  $\mathbf{Oz}$ -stabilizing part is now described by coefficient functionals ([theorem 3.3](#) and [corollary 3.5](#)), with finite support certificates on the common-positive-shift class ([theorem 4.8](#)); no exhaustion or generation theorem is proved ([question 11.4](#)). Its omega-map compatibility questions are untouched.
- `dsn:q:languages` of `docs/foundations-and-computation/definable-surreals-and-omnific-integers` asks about languages between the ordered field with  $\mathbf{Oz}$  and set theory. *Further information, not an answer* [10]: in  $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ , even with every real, the zero-cut projections and any set of parameters named, neither the simplicity relation nor the graph of the Gonshor exponential is definable ([theorem 9.10](#)), whereas order and simplicity together already define  $\omega$  (`dsn:prop:simplicity-omega`); definable elements obey the support bound of [corollary 9.11](#).

The novelty statements of the batch-26 sources were checked against the tree at their merge: no report contains a classification of  $\mathbf{Oz}$ -preserving automorphisms, the rank dichotomy or set-parameter nondefinability of monomials, and `osq:q:invisible` was recorded as open. Their reconstruction and definability claims were stale ([section 8](#)). Source 10’s statement that none of the report guides it inspected states the support-cut criterion, and its proposal of the criterion, the rank boundary, the fixed fields and the parameter witnesses as its contribution, were accurate at its pin `58cd8e1`, which does not contain this report. They are stale at the merge: this report was placed in `f4c9504` and written in `9b80a30`, and those results are [theorems 3.3](#), [4.2](#), [4.13](#) and [9.1](#). 10 is recorded as an independent re-derivation of them, and its new material is the list in [section 1.2](#). Its other repository statements (the fraction field and the multiplier and coefficient results of the Diophantine report, the phase twists of its reconstruction audit) are accurate.

## 11.2 Further questions

These questions arise from the proofs; they are not presented as named historical problems [04, 08, 09, 10].

**Question 11.1** (Strongness [04, 09, 10]). Which ring automorphisms of  $\mathbf{Oz}$  extend to strong field automorphisms of  $\mathbf{No}$ ? Does preservation of  $\mathbf{Oz}$  force strongness, or is there a nonstrong  $\mathbf{Oz}$ -preserving example? Equivalently for 09: characterize the nonstrong automorphisms of  $(\mathbf{No}, \mathbf{Oz})$  that commute with all set-indexed normal-form sums.

[Corollary 5.2](#) removes coefficient motion from this question but does not settle preservation of infinite sums; nothing is inferred from the existence of nonstrong automorphisms of the ambient field [9]. Source 10 asks what characterizes the nonstrong  $\mathbf{Oz}$ -preserving 1-automorphisms, and whether every one is strong. *Partial information* [10]: the necessity half of the criterion holds without strongness on every monomial ([remark 3.9](#)); the passage to arbitrary series is what remains. The question stays open. *Status* (batch 30): *answered* by sources 12 and 13. Every ring automorphism of  $\mathbf{Oz}$  extends uniquely to a strongly  $\mathbb{R}$ -linear automorphism of  $\mathbf{No}$  with strong inverse ([theorem 23.2](#)): all of them extend to strong automorphisms, preservation of  $\mathbf{Oz}$  forces strongness, there is no nonstrong example, 09’s characterization is vacuous, and 10’s question has the answer yes. The analogous question for  $\mathbf{Oz}[i]$  has a positive answer under conjugation or valuation compatibility and is open otherwise ([theorems 30.1](#) and [30.2](#) and [question 32.1](#)); for general  $(k, \mathfrak{o})$  see [question 32.13](#).

**Question 11.2** (Admissible logarithmic characters [04]). Give testable necessary and sufficient conditions on an additive  $h : \Gamma \rightarrow \mathfrak{m}$  for  $t^g \mapsto t^g \exp h(g)$  to define a strong bijective field map. Within that class the integer-part condition is exactly  $a_\delta(C_\delta) = 0$ .

Common positive support is one sufficient regime; the convex condition alone is not asserted to solve the global problem. This is the problem the surcomplex report describes under “what monomial prescriptions do not automatically give”. Sources 12 and 13 ask it again; it stays open (section 32.1).

**Question 11.3** (Full relative fixed fields [04]). For a set  $A$  of parameters, determine  $\text{Fix}((\mathcal{U}_{\mathbf{Oz}})_A)$ . It is contained in  $\mathbb{R}((t^{V_A}))$ , with equality for monomial parameters (corollary 7.5); which conditions on  $A$  force equality?

Source 10 asks the definable-closure analogue: when is the bound of proposition 7.8 sharp, and which support and coefficient conditions characterize definable closure in  $\mathfrak{S}^\dagger$ ? Its upper bounds (proposition 7.8 and corollary 9.11) are obstructions, not converse criteria; a full answer would have to separate algebraic support generation from the information contributed by the order and by  $\mathbf{Oz}$  [10].

**Question 11.4** (Generation and exhaustion [04, 08, 10]). How much of  $\mathcal{U}_{\mathbf{Oz}}$  is generated by elementary shears and their support-controlled limits, and can an intrinsic summation convention make this precise without one common shift set for all inputs? Can all strong leading-term-fixing  $\mathbf{Oz}$ -preserving automorphisms be described by a necessary and sufficient uniform support condition on their monomial phases?

No density or generation theorem is claimed; fine-topology convergence is not a substitute for a strong summation rule; the finite-dimensional phase condition of theorem 4.17 is sufficient only. Source 10 asks the same for set-sized divisible  $\Gamma$  with its twists as generators, and for compositions indexed by a well-ordered set: example 6.13 shows that one additive family is not enough, and a transfinite composition is not justified merely because each factor is strong; the accumulated changes must be controlled and a genuine limit map with an inverse proved [10].

**Question 11.5** (Weaker normal-form data [08]). Which normal-form data weaker than the omega-map remove the monomial nondefinability? Set-sized collections of constants do not (theorem 9.1).

Nor do every real constant, the zero-cut projections and the floor map, and simplicity and the Gonshor exponential are not definable either (theorem 9.10) [10].

**Question 11.6** (Non-divisible exponent groups [merge]). For non-divisible  $\Gamma$  and  $\mathfrak{o} = \mathbb{Z}$ , is a strong  $k$ -linear 1-automorphism preserving  $\mathcal{R}_{\mathbb{Z}}$  characterized by a convex-annihilator condition, as single flows are (theorem 4.10)?

This records the scope correction of remark 4.11. Source 10 assumes divisibility throughout, and its example with  $\Gamma = \mathbb{Z}$  is stated for  $\mathfrak{o} = k$  (remark 4.23); the question stays open.

**Question 11.7** (Other integer parts [10]). Which parts of the support-cut criterion survive for noncanonical integer parts of a real closed Hahn field, or for subrings whose positive-support condition is replaced by a more general support family?

The proof uses closure under every positive rational power of a monomial and the exact boundary at exponent zero. An arbitrary integer part need not present that boundary termwise, and the criterion should not be transferred to it without a replacement for the least-forbidden-shift argument [10]. Source 10 also records that its construction does not answer the question of

[9] on nontrivial omega-map-preserving automorphisms, since its witnesses do not preserve the omega-map, nor factorization problems for omnific integers, nor the order-unit question of the holonomic-rigidity report ([docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions](#)). Source 12 asks the analogous question for automatic strongness ([question 32.13](#)).

## Part II

# Algebraic-parameter rigidity of the omnific integers

This part is manuscript 02. Its question is different from that of [part I](#): not which automorphisms preserve a support condition, but whether **Oz** admits nonconstant *algebraic* families of embeddings. It does not: the decisive property is that every nonzero element divides an element with roots of every order. It does admit formal families of all orders. The relation to [part I](#) is recorded in [section 19](#).

## 12 Conventions and the root-covered property

### 12.1 Conventions of this part

Ring homomorphisms are unital, domains are commutative integral domains, and  $\mathbb{N} = \{0, 1, 2, \dots\}$ ; degrees of polynomials are ordinary integers. This part keeps its source's *large-monomial* convention, which is that of the other omnific reports: for a characteristic-zero field  $k$ , a unital subring  $\mathfrak{o} \subseteq k$  and a nonzero divisible ordered abelian group  $\Gamma$ ,

$$k((X^\Gamma)) = \left\{ \sum_{g \in S} a_g X^g : S \subseteq \Gamma \text{ reverse well ordered} \right\}, \quad X^g = t^{-g},$$

so the leading exponent  $\ell(x) = \max \text{supp } x$  is the largest, and  $\ell(xy) = \ell(x) + \ell(y)$ . The ideal  $\Pi_k(\Gamma)$  consists of the series supported in  $\Gamma^{>0}$  (the same set as  $\Pi$  in [part I](#)), and  $\mathcal{R}_{\mathfrak{o}}(k, \Gamma) = \mathfrak{o} + \Pi_k(\Gamma)$  is the ring  $\mathcal{R}_{\mathfrak{o}}$  of [part I](#), the Diophantine report's  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . For surreals  $X = \omega$ , so **Oz** =  $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \mathbf{No})$  and **Oz**[ $i$ ] =  $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \mathbf{No})$ . Renamings from the source: its  $R_\Gamma(D, k)$  and  $\mathcal{P}_\Gamma(k)$  are  $\mathcal{R}_{\mathfrak{o}}(k, \Gamma)$  and  $\Pi_k(\Gamma)$ ; its principal convex subgroup  $H(a)$  is  $C(a)$ , the transcendence report's notation ( $C_\delta$  in [part I](#); the quotient report's  $H_a$  means the opposite, the elements infinitesimal relative to  $a$ ); its covering element  $c$  is  $z$ ; its bounded-scale field  $K_b$  is  $K_{\text{bd}}$  and its stages  $H_j$  are  $\Gamma_{\leq j}$ . Elsewhere in this part the letters  $R, B, A, K, F, L$  denote rings and fields local to each statement.

The general theorems are proved for sets. Their conclusions for **Oz**, **Oz**[ $i$ ], **No** and **No**( $i$ ) use class functions and set-supported normal forms. A statement about one polynomial contains finitely many coefficients and variables, so the degree arguments apply to a class-sized coefficient domain unchanged. No proper-class scheme  $\text{Spec}(\mathbf{Oz})$  and no class compactness or Noetherian theorem is used; the finite-type theorem is a theorem about sets, applied to the set-sized models of [section 18](#). Expressions involving  $\text{Aut}(\mathbf{Oz})$  or ML of a class ring abbreviate statements about each automorphism or derivation;  $(b, x) \mapsto D_b(x)$  is one definable class operation. For cancellation against a class-sized  $B$ , the transcendence-degree arguments involve only finite tuples of indeterminates inside one rational-function field [02].

**Definition 12.1** ([02]). A domain  $R$  is *root-covered* if for every  $0 \neq a \in R$  there are  $b \in R$  and  $z = ab \neq 0$  such that for every  $n \geq 1$  some  $z_n \in R$  has  $z_n^n = z$ . (No compatibility between the  $z_n$  is required; the terminology is local to this report.)

Fields are root-covered. The interesting examples have few units:  $\mathbf{Oz}^\times = \{\pm 1\}$  and  $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$ , and in general  $\mathcal{R}_o(k, \Gamma)^\times = \mathfrak{o}^\times$  by additivity of leading exponents (the Diophantine report’s `odg:prop:units`, `odg:def:lem:units`).

**Lemma 12.2** (Localization [02]). *If  $R$  is root-covered and  $S \subseteq R \setminus \{0\}$  is multiplicative, then  $S^{-1}R$  is root-covered.*

*Proof.*  $(a/s)(sb) = z$ , and the roots  $z_n$  remain.  $\square$

Root-covered is sufficient for the conclusions below and is not asserted necessary; failure of the monomial criterion is not used to claim failure of cancellation or stable rigidity [02].

### 13 The monomial-divisibility criterion

For  $a > 0$  in  $\Gamma$  let  $C(a) = \{g : |g| \leq na \text{ for some } n \geq 1\}$ ;  $a$  is an *order unit* if  $C(a) = \Gamma$ .

**Theorem 13.1** (Monomial divisibility [02]; after `bst:thm:fieldcriterion`). *For  $\Gamma \neq 0$  divisible and  $R = \mathcal{R}_o(k, \Gamma)$  the following are equivalent: (i)  $\Gamma$  has no order unit; (ii) for every  $0 \neq x \in R$  some monomial  $X^c$  with  $c > 0$  is divisible by  $x$  in  $R$ . Then the quotient can be chosen in  $\Pi_k(\Gamma)$ , and  $R$  is root-covered.*

*Proof.* Assume (i). The support of  $x$  lies in  $[0, a]$  for some  $a > 0$ . The inverse of  $x$  in the Hahn field has support in the subgroup generated by  $\text{supp } x$ , hence in  $C(a)$  (write  $x = \alpha X^h(1 + \varepsilon)$  and expand the geometric series). Choose  $c > na$  for all  $n$ ; then  $X^c x^{-1}$  has strictly positive support, lies in  $\Pi_k(\Gamma)$ , and  $x \cdot X^c x^{-1} = X^c$ . Divisibility of  $\Gamma$  gives  $(X^{c/n})^n = X^c$ . Conversely, if  $a$  is an order unit and  $x = 1 + X^a$ , the unique Hahn quotient  $X^c/(1 + X^a) = \sum_{n \geq 1} (-1)^{n-1} X^{c-na}$  has a nonzero coefficient at a negative exponent, so it is not in  $R$ .  $\square$

This is the transcendence report’s field criterion `bst:thm:fieldcriterion` (`docs/surreal/transcendence-over-bounded-support`: the bounded-support ring is a field if and only if  $\Gamma$  has no order unit, by the same two arguments, the inverse support in a proper convex subgroup and the geometric series of  $1 - t^u$ ), restricted to positive-support elements; its “order unit” is defined identically. Source 02’s audit read only the openings of the omnific reports and did not see it. The positive-support phrasing is what the rigidity theorems use.

**Example 13.2** ([02]; [11]). In  $\mathbf{No}$ ,  $\omega > n$  for all ordinary  $n$ , so  $\omega^\omega/(1 + \omega) = \omega^{\omega-1} - \omega^{\omega-2} + \omega^{\omega-3} - \dots$  is an omnific integer:  $1 + \omega$  divides  $\omega^\omega$ . With exponent group  $\mathbb{Q}$  no exponent can absorb the tail. The example is L’Innocente and Mantova’s; the distinction is a larger Archimedean scale, not convergence.

**Corollary 13.3** (Omnific root coverage [02]).  *$\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  are root-covered: every nonzero element divides a positive real monomial  $\omega^c$ , with quotient of strictly positive support.*

*Proof.* This is immediate from the global monomial clearing `odg:thm:fractions` of the Diophantine report, applied to  $x^{-1}$ : some  $\omega^c$  makes  $\omega^c x^{-1}$  purely infinite. Source 02 gives the same argument through [theorem 13.1](#), using a strict upper bound of the set  $\{na : n \geq 1\}$ ; the coefficients’ order is not used.  $\square$

## 14 Finite-type domains carry no varying embedding

The results of this section concern set-sized fields and domains.

**Lemma 14.1** (Unbounded roots [02]). *Let  $F/K$  be finitely generated and  $u \in F^\times$ . If  $u$  has  $n$ th roots in  $F$  for an unbounded set of  $n$ , then  $u$  is algebraic over  $K$ .*

*Proof.* Otherwise extend  $u$  to a transcendence basis  $u, t_2, \dots, t_r$ , with  $[F : E] = d < \infty$  for  $E = K(u, t_2, \dots, t_r)$ .  $Z^n - u$  is Eisenstein at the prime  $u$  of  $K(t_2, \dots, t_r)[u]$ , hence irreducible over  $E$  by Gauss's lemma, so an  $n$ th root has degree  $n \leq d$ .  $\square$

**Lemma 14.2** (Algebraic constants [02]). *If  $F/K$  is finitely generated, the relative algebraic closure  $K'$  of  $K$  in  $F$  is finite over  $K$ ; if  $A$  is an integral  $K$ -algebra with  $\text{Frac } A = F$ , then  $C_A = A \cap K'$  is a field finite over  $K$ .*

*Proof.* With a transcendence basis  $\mathbf{t}$  and  $[F : K(\mathbf{t})] = d$ , every finitely generated  $L \subseteq K'$  has  $[L : K] = [L(\mathbf{t}) : K(\mathbf{t})] \leq d$  (adjoining algebraically independent indeterminates preserves linear independence;  $\mathbf{t}$  stays independent over  $L$ ). An  $L$  of maximal degree contains every element of  $K'$ . For  $0 \neq a \in C_A$  the minimal polynomial has nonzero constant term, which expresses  $a^{-1}$  as a polynomial in  $a$  over  $K$ .  $\square$

**Lemma 14.3** (Finitely many unit translates [02]). *Let  $A$  be a finitely generated integral  $K$ -algebra and  $f \in A$  transcendental over  $K$ . Only finitely many  $\lambda \in K$  make  $f - \lambda$  a unit of  $A$ .*

*Proof.* Extend  $f$  to a transcendence basis  $f, t_2, \dots, t_r$  of  $\text{Frac } A$ , and put  $B_0 = K[f, t_2, \dots, t_r]$ . Generators of  $A$  are algebraic over  $\text{Frac } B_0$ ; clearing denominators gives  $0 \neq d \in B_0$  with  $A$  integral over  $B = B_0[1/d]$  in the common overring. If  $(f - \lambda)^{-1} \in A$ , it lies in  $\text{Frac } B_0$  and is integral over the integrally closed  $B$ , so  $f - \lambda$  divides a power of  $d$  in the UFD  $B_0$ , hence divides  $d$ ; the primes  $f - \lambda$  are pairwise nonassociate. The argument does not assume  $B \subseteq A$ .  $\square$

**Theorem 14.4** (Finite-type embedding rigidity [02]). *Let  $R$  be a characteristic-zero root-covered domain,  $K$  a characteristic-zero field, and  $A$  a finitely generated integral  $K$ -algebra. Every injective homomorphism  $\phi : R \rightarrow A$  has image in the field  $C_A = \{x \in A : x \text{ algebraic over } K\}$ , finite over  $K$ , and extends uniquely to  $\text{Frac } R \rightarrow C_A$ . If  $K$  is relatively algebraically closed in  $\text{Frac } A$ , then  $C_A = K$ . Neither  $K$ -linearity of  $\phi$  nor a map  $K \rightarrow R$  is assumed.*

*Proof.* For  $a \neq 0$  take  $ab = z$  and  $z^n = z$ . Then  $\phi(z) \neq 0$  has roots of all orders, so it is algebraic (lemma 14.1) and a unit of  $A$  (lemma 14.2); hence  $\phi(a)$  is a unit. If  $f = \phi(a)$  were transcendental, every  $f - m = \phi(a - m)$ ,  $m \in \mathbb{Z}$ , would be a unit (as  $a - m \neq 0$ ), contradicting lemma 14.3. So  $\phi(R) \subseteq C_A$ , and nonzero images are invertible there.  $\square$

The theorem is nontrivial only for set-sized  $R$ , such as the Hahn models of section 18; it is not a disguised statement that a proper class cannot inject into a set. One must not replace  $C_A$  by the whole relative algebraic closure:  $A = \mathbb{Q}[x, \sqrt{2}x]$  has  $\text{Frac } A = \mathbb{Q}(\sqrt{2})(x)$  but  $C_A = \mathbb{Q}$ , since elements of  $A$  have rational constant term; no normality of  $A$  is assumed [02]. If  $K$  is relatively algebraically closed in  $\text{Frac } A$ , every embedding  $R \rightarrow A$  has image in  $K$ , and none exists if  $\text{Frac } R$  does not embed in  $K$ .

**Theorem 14.5** (Pointed parameter spaces [02]). *Let  $R$  be a set-sized characteristic-zero root-covered domain,  $F = \text{Frac } R$ ,  $A$  a finite-type integral  $F$ -algebra with an  $F$ -algebra map  $\varepsilon : A \rightarrow F$ . If a ring homomorphism  $\psi : R \rightarrow A$  has  $\varepsilon\psi$  equal to the inclusion, then  $\psi(x) = x$  for all  $x$ .*

*Proof.*  $\psi$  is injective, so  $\psi(R) \subseteq C_A$ . The restriction of  $\varepsilon$  to the field  $C_A$  is injective and fixes  $F$ ; for  $a \in C_A$ ,  $a - \varepsilon(a) \in C_A$  has image 0, so  $C_A = F$ .  $\square$

Examples are  $A = F[s_1, \dots, s_r, 1/h]$  with an  $F$ -point where  $h \neq 0$ ; relations, singularities and localizations are allowed, and relative algebraic closedness is not required. The hypothesis is one common finite-type algebra containing the images of all of  $R$ , not separate formulas in unrelated algebras. For a finite extension  $L/F$ ,  $L[s]$  contains the root-covered  $L$  as constants and has no  $F$ -valued augmentation when  $L \neq F$  [02].

## 15 Polynomial and Laurent coefficient rigidity

These proofs work over arbitrary coefficient domains, need no finite generation and assume no preservation of infinite sums [02].

**Theorem 15.1** (Polynomial coefficient rigidity [02]). *If  $R$  is root-covered and  $B$  a domain, every injective  $\phi : R \rightarrow B[T_1, \dots, T_n]$  has image in  $B$ .*

*Proof.*  $\deg \phi(z) = n \deg \phi(z_n)$  for all  $n$  forces  $\deg \phi(z) = 0$ , and  $\deg \phi(a) + \deg \phi(b) = 0$ .  $\square$

**Lemma 15.2** (Laurent units [02]). *For a domain  $B$ , the units of  $B[U_1^{\pm 1}, \dots, U_q^{\pm 1}]$  are the  $b\mathbf{U}^{\mathbf{e}}$  with  $b \in B^\times$  and  $\mathbf{e} \in \mathbb{Z}^q$ .*

*Proof.* The widths  $M_j - m_j$  of the  $U_j$ -exponents are nonnegative and additive, so a unit has a single exponent vector.  $\square$

**Theorem 15.3** (Laurent coefficient rigidity [02]). *If  $R$  is root-covered and  $B$  a domain, every injective  $\phi : R \rightarrow B[\mathbf{U}^{\pm 1}]$  has image in  $B$ ; hence so does every injective  $\phi : R \rightarrow B[\mathbf{T}, \mathbf{U}^{\pm 1}]$ .*

*Proof.* The largest and smallest  $U_j$ -exponents of  $\phi(z)$  are divisible by every  $n$ , so  $\phi(z)$  is a nonzero constant, a unit over  $\text{Frac } B$ , and  $\phi(a)$  is a Laurent monomial by lemma 15.2. If  $a \neq -1$  the same holds for  $\phi(a) + 1$ , which forces  $\phi(a)$  to be constant. For mixed targets apply theorem 15.1 over  $B[\mathbf{U}^{\pm 1}]$  first.  $\square$

Injectivity and a domain target are needed:  $x \mapsto x + sD(x)$  into  $R[s]/(s^2)$  can be nonconstant with identity specialization (section 17). No noninjective analogue for all targets is claimed [02].

**Theorem 15.4** (Strong coefficient invariance [02]). *Let  $R$  be root-covered,  $B$  any commutative unital ring, and  $\Phi : R[X_1, \dots, X_p, U_1^{\pm 1}, \dots, U_q^{\pm 1}] \rightarrow B[Y_1, \dots, Y_{p'}, V_1^{\pm 1}, \dots, V_{q'}^{\pm 1}]$  an isomorphism with  $p + q = p' + q' < \infty$ . Then  $\Phi(R) = B$ ,  $q = q'$  and  $p = p'$ . In particular, for  $R = \mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , every isomorphism  $R[\mathbf{X}_n] \cong B[\mathbf{Y}_n]$  carries  $R$  onto  $B$ :  $R$  is strongly invariant and cancels against an arbitrary coefficient ring with the same number of variables. If  $B$  is also root-covered, the equality of total variable counts can be dropped.*

*Proof.* The target is a domain, so  $B$  is, and theorem 15.3 gives  $\Phi(R) \subseteq B$ . With  $F = \text{Frac } \Phi(R) \subseteq L = \text{Frac } B \subseteq E$ ,  $\text{trdeg}_F E = p + q = p' + q' = \text{trdeg}_L E$ , so  $L/F$  is algebraic.  $E$  is a rational function field over  $F$ , which has no new algebraic constants (a nonconstant rational function is transcendental, one variable at a time), so  $B \subseteq F$ , and comparing coefficients gives  $B \subseteq \Phi(R)$ . Polynomial variables add no units, and the quotient unit groups give  $\mathbb{Z}^q \cong \mathbb{Z}^{q'}$ . If both rings are root-covered, apply the first step to  $\Phi$  and  $\Phi^{-1}$ .  $\square$

Equality of total variable counts cannot be dropped for arbitrary  $B$ :  $R[X_1, X_2] = B[X_2]$  with  $B = R[X_1]$ . The general statements do not solve the unrestricted cancellation problem for finitely generated algebras; the strong-invariance terminology and history are older [8] [02].

**Theorem 15.5** (Automorphisms of polynomial and torus rings [02]). *For a root-covered  $R$ , every automorphism of  $R[\mathbf{T}]$  or  $R[\mathbf{U}^{\pm 1}]$  preserves  $R$ ; for polynomial extensions, restriction to  $R$  is split surjective onto  $\text{Aut } R$ , with kernel the  $R$ -algebra automorphisms. Every automorphism of  $R[T]$  is  $T \mapsto uT + b$  with  $u \in R^\times$ ,  $b \in R$ , over a unique  $\sigma \in \text{Aut } R$ ; every automorphism of  $R[\mathbf{U}^{\pm 1}]$  is  $U_i \mapsto u_i \prod_j U_j^{m_{ji}}$  over  $\sigma$ , with  $u_i \in R^\times$  and  $(m_{ji}) \in \text{GL}_q(\mathbb{Z})$ . For  $\mathbf{Oz}$ ,  $u \in \{\pm 1\}$ ; for  $\mathbf{Oz}[i]$ ,  $u \in \{\pm 1, \pm i\}$ .*

*Proof.* Apply theorem 15.3 to an automorphism and its inverse; coefficientwise extension is the section. In one variable,  $\deg g \deg f = 1$ ; units give the torus formula via lemma 15.2.  $\square$

The affine and torus formulas are standard once coefficients are preserved; the new omnific conclusion is that *every* ring automorphism preserves  $\mathbf{Oz}$ . No classification of multivariable polynomial automorphisms over  $R$  and no claim about the Jacobian conjecture follows [02].

## 16 Locally nilpotent derivations, coactions and gradings

A derivation  $D$  of a domain is *locally nilpotent* if every element is killed by some iterate, and  $\text{ML}(A) = \bigcap_{D \in \text{LND}(A)} \ker D$ . Derivations are ring derivations, hence  $\mathbb{Z}$ -linear;  $A$  need not contain  $\mathbb{Q}$ .

**Lemma 16.1** ([02]). *For a locally nilpotent  $D$  of a characteristic-zero domain,  $\deg_D(a) = \max\{j : D^j a \neq 0\}$  satisfies  $\deg_D(ab) = \deg_D a + \deg_D b$ .*

*Proof.*  $D^{r+s}(ab) = \binom{r+s}{r} D^r a D^s b \neq 0$ , and higher iterates vanish.  $\square$

**Theorem 16.2** (Stable locally nilpotent rigidity [02]). *Let  $R$  be a characteristic-zero root-covered domain. Every locally nilpotent derivation of any domain containing  $R$  kills  $R$ , and  $\text{ML}(R[T_1, \dots, T_n]) = R$  for every finite  $n$ . In particular  $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$ ,  $\text{ML}(\mathbf{Oz}[i][\mathbf{T}]) = \mathbf{Oz}[i]$ , and  $\mathbf{Oz}$  has no nonzero locally nilpotent derivation.*

*Proof.*  $\deg_D(z) = m \deg_D(z_m)$  for all  $m$  forces  $\deg_D z = 0$ , hence  $\deg_D a = 0$ . The partial derivatives  $\partial/\partial T_j$  are locally nilpotent with common kernel  $R$ .  $\square$

**Corollary 16.3** (No coactions or gradings [02]). *If  $R$  is root-covered, a homomorphism  $\psi : R \rightarrow R[s_1, \dots, s_r]$  with identity specialization at  $\mathbf{s} = 0$ , or  $\theta : R \rightarrow R[u_1^{\pm 1}, \dots, u_r^{\pm 1}]$  with identity specialization at  $\mathbf{u} = 1$ , is the constant inclusion; so there are no nontrivial additive-group or split-torus coactions of finite rank, and every  $\mathbb{Z}^r$ -grading of  $R$  is concentrated in degree zero.*

*Proof.* Identity specialization makes the map injective; apply theorems 15.1 and 15.3. A grading gives  $\sum x_{\mathbf{m}} \mapsto \sum x_{\mathbf{m}} \mathbf{u}^{\mathbf{m}}$ .  $\square$

A Hahn expansion  $\sum a_g X^g$  is not a grading: it has infinitely many terms, and the no-grading result concerns finite homogeneous decompositions. Theorem 16.2 does not say that  $R[\mathbf{T}]$  has no locally nilpotent derivation, and it is not a solution of a general stable-rigidity conjecture [4] [02].

## 17 Euler derivations and formal flows

The parameter  $s$  in this section is a new formal indeterminate, not a surreal and not a Hahn monomial.

**Proposition 17.1** (Support-preserving Euler derivations [02]). *For additive  $\chi : \Gamma \rightarrow (k, +)$ ,  $D_\chi(\sum a_g X^g) = \sum \chi(g) a_g X^g$  is a  $k$ -linear derivation of  $k((X^\Gamma))$  that preserves  $\mathcal{R}_0(k, \Gamma)$ , kills  $\mathfrak{o}$  and maps  $\mathcal{R}_0(k, \Gamma)$  into  $\Pi_k(\Gamma)$ ; and  $D_\chi D_\psi = D_\psi D_\chi$  for additive  $\chi, \psi$ .*

*Proof.* The coefficient of  $X^h$  in  $xy$  is a finite sum of  $a_g b_{h-g}$ , and  $\chi(h) = \chi(g) + \chi(h-g)$  splits it;  $\chi(0) = 0$ .  $\square$

In the notation of [part I](#),  $D_\chi$  is the shift-zero derivation  $t^\gamma \mapsto \chi(-\gamma)t^\gamma$ . Division by  $n!$  is allowed on  $\Pi_k(\Gamma)$ , and  $D_\chi^n(x) \in \Pi_k(\Gamma)$  for  $n \geq 1$ , so

$$\mathcal{E}_{\chi,s}(x) = \sum_{n \geq 0} \frac{D_\chi^n(x)}{n!} s^n \quad (17.1)$$

has coefficients in  $\mathcal{R}_0(k, \Gamma)$ ; no division of an integer constant occurs.

**Theorem 17.2** (Compatible formal flows [02]).  *$\mathcal{E}_{\chi,s}$  is a unital ring homomorphism  $\mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[[s]]$  with identity specialization, satisfying  $\mathcal{E}_{\chi,s} \mathcal{E}_{\chi,t} = \mathcal{E}_{\chi,s+t}$ ; its coefficientwise extension fixing  $s$  is an automorphism of  $\mathcal{R}_0(k, \Gamma)[[s]]$  with inverse from  $-s$ , and it induces compatible automorphisms of  $\mathcal{R}_0(k, \Gamma)[s]/(s^{N+1})$  for every  $N$ .*

*Proof.*  $D_\chi^n(xy)/n! = \sum_r D_\chi^r(x)/r! \cdot D_\chi^{n-r}(y)/(n-r)!$ , first in the Hahn field; the coefficient of  $s^r t^m$  in both sides of the flow law is  $D_\chi^{r+m}(x)/(r!m!)$ ; each coefficient uses finitely many inputs. Substituting  $t = -s$  is valid since both parameters have positive order.  $\square$

**Theorem 17.3** (Nonalgebraization at the first jet [02]). *If  $\Gamma$  has no order unit and  $\chi \neq 0$ , the reduction of  $\mathcal{E}_{\chi,s}$  modulo  $s^2$  is nonconstant and is induced by no homomorphism  $\psi : \mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[s]$  with  $\text{ev}_{s=0} \psi = \text{id}$ , and  $D_\chi$  is not locally nilpotent.*

*Proof.* Some  $g > 0$  has  $\chi(g) \neq 0$ ; then  $D_\chi^n(X^g) = \chi(g)^n X^g \neq 0$ . By [theorem 13.1](#) the ring is root-covered, and by [corollary 16.3](#) every such  $\psi$  is constant.  $\square$

Formal automorphisms exist to *all* orders; the obstruction appears only when a single polynomial family must realize even the first-order part. A formal power series ring is not of finite type, and  $R[s]/(s^2)$  is not a domain, so there is no conflict with [theorem 14.4](#) [02]. The flows involve no analytic convergence, no fine-topology limit and no identification of  $s$  with a surreal infinitesimal, and they are not an exponentiation of a derivation *inside* the Hahn field [02].

### 17.1 The monomial orbit field

Let  $R = \mathcal{R}_0(k, \Gamma)$  be set sized,  $F = \text{Frac } R$  (which contains  $k$ :  $a = aX^g/X^g$ ) and  $W = \chi(\Gamma)$ , a  $\mathbb{Q}$ -subspace of  $k$ . In  $F((s))$  put

$$L_\chi = F\left(\frac{\mathcal{E}_{\chi,s}(X^g)}{X^g} : g > 0\right) = F(e^{ws} : w \in W), \quad e^{ws} = \sum_n w^n s^n / n!. \quad (17.2)$$

It uses only transformed monomials; it is not claimed to contain the image of every Hahn series [02].

**Lemma 17.4** (Formal exponential independence [02]). *For distinct  $\lambda_j \in F$  the  $e^{\lambda_j s}$  are  $F$ -linearly independent; for  $\mathbb{Q}$ -independent  $w_1, \dots, w_r$  the  $e^{w_j s}$  are algebraically independent over  $F$ .*

*Proof.* Comparing coefficients of  $s^n$ ,  $n < m$ , gives a Vandermonde system. A polynomial relation is a linear relation among  $e^{(\alpha \cdot \mathbf{w})s}$  with distinct exponents.  $\square$

**Theorem 17.5** (Monomial orbit field [02]).  $\text{trdeg}_F L_\chi = \dim_{\mathbb{Q}} W$ . *If  $\chi \neq 0$ ,  $L_\chi$  lies in no finitely generated extension of  $F$  inside  $F((s))$ , so  $\mathcal{E}_{\chi,s} : R \rightarrow F[[s]]$  factors through no finite-type  $F$ -algebra with an  $F$ -algebra second map.*

*Proof.* A  $\mathbb{Q}$ -basis  $(w_j)$  of  $W$  gives algebraically independent  $e^{w_j s}$ , and each  $e^{w s}$  has a power equal to a monomial in finitely many of them, so  $L_\chi$  is algebraic over  $F(e^{w_j s})$ . For  $0 \neq w \in W$ ,  $e^{w s}$  is transcendental and has roots  $e^{(w/n)s}$  in  $L_\chi$ , so lemma 14.1 excludes a finitely generated overfield, even when  $\dim_{\mathbb{Q}} W = 1$ . A factorization would put the image of  $R$  in a finite-type domain  $B \subseteq F[[s]]$  with  $\text{Frac } B \supseteq L_\chi$ .  $\square$

## 17.2 An omnific family of commuting directions

**Theorem 17.6** (Commuting omnific derivations [02]; after `osq:prop:classder`). *For  $b \in \mathbf{No}$ ,  $D_b(\sum a_g \omega^g) = \sum q_b(g) a_g \omega^g$  is an  $\mathbb{R}$ -linear derivation of  $\mathbf{No}$  and, with complex coefficients, a  $\mathbb{C}$ -linear derivation of  $\mathbf{No}(i)$  commuting with conjugation; it preserves  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ . The  $D_b$  commute, and every finite family with distinct indices is linearly independent over  $\mathbf{Oz}$ , respectively  $\mathbf{Oz}[i]$ .*

*Proof.* Proposition 17.1 with  $\chi = q_b$ . For  $m_c = \omega^{\omega^c} \in \mathbf{Oz}$ ,  $D_b(m_c) = m_c$  if  $b = c$  and 0 otherwise; applying  $\sum_j a_j D_{b_j} = 0$  to  $m_{b_\ell}$  gives  $a_\ell m_{b_\ell} = 0$  in a domain.  $\square$

These derivations, with the test elements  $m_c$ , are the quotient report's class-valued derivations `osq:prop:classder` (docs/surreal/set-sized-quotients-of-omnific-integers), which combine that report's sources 06, 08 and 10 and state independence over  $\mathbb{R}$ ; the same evaluation gives independence over the omnific ring, as 02 states, and 02 adds their commutation and flows. In the  $t$ -convention of part I,  $D_b(t^\gamma) = -q_b(\gamma)t^\gamma$ .

**Corollary 17.7** (Multi-parameter flows [02]). *For distinct  $b_1, \dots, b_r$  and  $R = \mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , the map*

$$x \mapsto \sum_{\alpha \in \mathbb{N}^r} \frac{D_{b_1}^{\alpha_1} \dots D_{b_r}^{\alpha_r}(x)}{\alpha_1! \dots \alpha_r!} s_1^{\alpha_1} \dots s_r^{\alpha_r} \quad \text{into } R[[s_1, \dots, s_r]]$$

*obeys the additive formal group law; its coordinate directions are independent, and no nonzero coordinate direction is induced by a polynomial family on  $R$ .*

*Proof.* Compose one-parameter flows, which commute; independence is theorem 17.6, and the last claim is theorem 17.3 in one parameter.  $\square$

**Example 17.8** ([02]). For  $b = 0$ ,  $q_0(g)$  is the real constant coefficient of the exponent  $g$ :  $D_0(\omega^r) = r\omega^r$  for real  $r$ ,  $D_0(\omega^\omega) = 0$ , and  $\mathcal{E}_{0,s}(\omega) = \omega + \omega s + \frac{\omega}{2}s^2 + \frac{\omega}{6}s^3 + \dots$ , every coefficient in  $\mathbf{Oz}$  although  $\frac{1}{2}, \frac{1}{6} \notin \mathbf{Oz}$ ; “ $\omega e^s$ ” abbreviates this series and is not the surreal exponential. Also  $D_0(\omega^\omega/(1+\omega)) = -\omega^{\omega+1}/(1+\omega)^2$ , again omnific, supported at  $\omega - n$  for positive ordinary  $n$  (checked coefficientwise to 29 terms by the placement dossier).

**Corollary 17.9** (A continuum of independent orbit elements [02]). *For  $R = \mathbf{Oz}$ ,  $F = \mathbf{No}$  (or  $R = \mathbf{Oz}[i]$ ,  $F = \mathbf{No}(i)$ ), the monomial orbit field  $F(e^{rs} : r \in \mathbb{R})$  of  $D_0$  has transcendence degree  $2^{\aleph_0}$  over  $F$ , in the sense of a set-sized transcendence basis with finite algebraic relations.*

*Proof.*  $q_0(r) = r$ , and  $e^{rs} = \mathcal{E}_{0,s}(\omega^r)/\omega^r$  for  $r > 0$ . A  $\mathbb{Q}$ -basis of  $\mathbb{R}$ , made positive, has  $2^{\aleph_0}$  elements; lemma 17.4 holds over the class field  $F$  since each relation uses finitely many coefficients.  $\square$

Only a lower bound is asserted for the field generated by the formal images of all Hahn series [02].

*Remark 17.10* (Actual automorphisms remain abundant [02]). For additive  $\chi : \mathbf{No} \rightarrow \mathbb{R}$  and ordinary  $t \in \mathbb{R}$ ,  $\tau_{\chi,t}(\sum a_g \omega^g) = \sum e^{t\chi(g)} a_g \omega^g$  is a support-preserving automorphism of  $\mathbf{No}$  and  $\mathbf{No}(i)$  preserving  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , with  $\tau_{\chi,t}\tau_{\chi,u} = \tau_{\chi,t+u}$  and positive multipliers. These character twists are established [9] and not claimed new (remark 9.5); they prevent reading the rigidity theorems as  $\text{Aut}(\mathbf{Oz}) = 1$ . What fails is realization of a family by a homomorphism into a polynomial or finite-type parameter algebra: with  $\chi = q_0$  the twists send  $\omega^r$  to  $e^{tr}\omega^r$  for all real  $r$  at once, which no single polynomial parameter algebra can carry.

## 18 Set-sized workspaces and a fraction-field warning

Let  $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}e_j$ , ordered by the sign of the coefficient at the largest index, so  $e_{j+1} > ne_j$  for all  $n$ :  $\Gamma$  is divisible with no order unit. For  $(\mathfrak{o}, k) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $R = \mathcal{R}_{\mathfrak{o}}(k, \Gamma)$  is a root-covered set (theorem 13.1), so all the results of this part apply to it in ordinary set theory. The order embedding  $e_j \mapsto \omega^j$  of exponent groups induces  $X^{e_j} \mapsto \omega^{\omega^j}$  and embeds  $R$  in  $\mathbf{Oz}$ , respectively  $\mathbf{Oz}[i]$ ; the coordinate projections give commuting derivations already here, and replacing  $\mathbb{Q}e_j$  by  $\mathbb{R}e_j$  gives a set-sized realization of corollary 17.9 [02].

Let  $\Gamma_{\leq j} = \bigoplus_{\ell \leq j} \mathbb{Q}e_\ell$  (convex) and  $K_{\text{bd}} = \bigcup_j k((X^{\Gamma_{\leq j}}))$ .

**Proposition 18.1** (Bounded-scale fraction field [02]; after `bst:prop:localdensity`). *For this  $R$ ,  $\text{Frac } R = K_{\text{bd}} \subsetneq k((X^\Gamma))$ .*

*Proof.* An element of  $R$  with positive leading exponent in  $\Gamma_{\leq j}$  has all its exponents in  $\Gamma_{\leq j}$  by convexity, so quotients lie in  $K_{\text{bd}}$ . Conversely  $y \in k((X^{\Gamma_{\leq j}}))$  gives  $X^{e_{j+1}}y \in R$ . Finally  $\sum_{j \geq 0} X^{-e_j}$  is a Hahn series in no  $k((X^{\Gamma_{\leq j}}))$ .  $\square$

This is the transcendence report's localization proposition `bst:prop:localdensity` together with its field criterion (condition (2) of `bst:thm:fieldcriterion`) for this group. For the full class ring,  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  and  $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$  (`odg:thm:fractions`); that fact does not justify replacing  $K_{\text{bd}}$  by the full Hahn field, since a fixed set-sized group may lack an upper bound for a set of its scales even without an order unit [02].

## 19 Relation to Part I and the questions of this part

The two parts are complementary. Part I concerns derivations with *positive* shifts and their exponentials inside the Hahn field; this part concerns shift-zero (support-preserving) derivations and needs an external formal parameter.

- 02's question on formal directions below is partly answered by part I: the positive-shift derivations preserving  $\mathbf{Oz}$  are classified exactly, by  $c_\delta(C_\delta) = 0$  (theorem 3.4) and, on common positive shifts, by  $a_\delta(C_\delta) = 0$  (theorem 4.7). Those with a common well-ordered set of shifts integrate *inside*  $\mathbf{No}$  to automorphisms of  $\mathbf{Oz}$  (theorem 4.8), so they are integrable by an honest automorphism rather than only by a formal flow; for an arbitrary contracting strong one this holds granted cited theorem 1.1 for  $\Gamma = \mathbf{No}$  (remark 1.2).

The shift-zero derivations, the Hasse–Schmidt extensions of other derivations of  $\mathbf{Oz}$ , and derivations with no common well-ordered shift set are not covered.

- There is no conflict with [theorem 16.2](#): the shift derivations  $D_{\delta,\theta}$  restrict to derivations of  $\mathbf{Oz}$  but are not locally nilpotent, since  $D_{\delta,\theta}^n(t^\gamma) = \theta(\gamma)^n t^{\gamma+n\delta} \neq 0$  when  $\theta(\gamma) \neq 0$ ; and the flows  $\Phi_s$  are not polynomial in  $s$ , since  $\Phi_s(t^{-\eta})$  involves every power of  $s$ .
- The character twists of [remark 17.10](#) are the factor  $M_{\chi,\text{id}}$  of [theorem 5.1](#).

[Theorem 15.5](#) concerns  $\text{Aut}(\mathbf{Oz}[T])$ , not  $\text{Aut}(\mathbf{Oz})$ , and does not bear on `osq:q:invisible`. The rational self-maps of  $\mathbf{Oz}$  studied in the quotient report are maps of the ring given by rational functions, not ring automorphisms of  $\mathbf{Oz}[T]$ ; the objects differ. Derivations into set-sized modules vanish (`osq:thm:derivations`), which does not conflict with [theorem 17.6](#), whose values lie in the class module  $\Pi$ .

**Question 19.1** (Beyond the scale criterion [02]). For a nonzero divisible  $\Gamma$  *with* an order unit, which rings  $\mathcal{R}_o(k, \Gamma)$  remain strongly invariant or stably rigid?

[Theorem 13.1](#) shows that their elements need not divide monomials; it does not show that cancellation or locally nilpotent rigidity fails.

**Question 19.2** (Formal directions [02]; re-scoped). Which derivations of  $\mathbf{Oz}$  preserve its purely infinite ideal, and which admit an iterative integral Hasse–Schmidt extension  $\mathbf{Oz} \rightarrow \mathbf{Oz}[[s]]$ ? How much larger is this class than the Euler family of [theorem 17.6](#)?

*Status.* Partly answered for positive shifts by [part I](#), as recorded above; open otherwise. The factorial argument of [theorem 17.2](#) is a sufficient condition for the displayed family and is not asserted to classify all integrable derivations or all formal automorphisms tangent to the identity.

**Question 19.3** (Reduced parameter algebras [02]). How far does coefficient rigidity extend to reduced parameter rings with several irreducible components when a prescribed specialization stays injective on the source?

The present proofs apply to domain quotients on which the map stays injective; a component can kill root-cover elements, and the dual-number example shows that nilpotent parameter rings need a different statement.

## Part III

# Automatic strongness and proper embeddings of the omnific pair

This part is built from three manuscripts of batch 30, numbered 11, 12 and 13 here. It has two themes. The first is automatic strongness: every ring automorphism of  $\mathbf{Oz}$  extends to a strong automorphism of  $\mathbf{No}$ , which answers [question 11.1](#) and extends [theorem 5.1](#) and [corollary 5.2](#) to all automorphisms. The second is embeddings: proper self-embeddings of  $\mathbf{No}$  can preserve and reflect  $\mathbf{Oz}$  and still move a real number, and the strong exact-monomial ones are classified by a convex subgroup of the target value group. Its labels carry the sub-prefix `opa:as:`.

## 20 Sources, results and conventions of this part

### 20.1 Three sources and where the merge chose

The three manuscripts were placed together in commit 21375f8 as items 03, 04 and 05 of batch 30. They are numbered 11, 12 and 13 here, after their file prefixes:

- **11** *Coefficient Gaps and Proper Self-Embeddings of the Surreal Field: omnific preservation, topological dichotomies, and nonconjugate copies*, pinned to cf56b89; files prefixed 11-coefficient-gaps-.
- **12** *Automatic Strongness and Proper Embeddings of the Omnific Integers: constant-term duality, full stabilizers, and the surreal-surcomplex boundary*, pinned to a6c68ac; files prefixed 12-automatic-strongness-.
- **13** *Automatic Hahn Linearity of Omnific Isomorphisms: residue duality, full stabilizers, and coefficient-drifting self-embeddings*, pinned to 0865f04; files prefixed 13-omnific-isomorphisms-.

The three pins are commits of one line that forked at c6359e4, the placement of source 10. None contains b3fa9e2, the batch-28 write of source 10 into this report. So no author saw [section 3.3](#), [remark 3.9](#) or the paragraph of partial information after [question 11.1](#); their comparisons were with the report of batch 26, and they cite its strong factorization ([theorem 5.1](#)) and its strongness-free [corollary 5.2](#) accurately.

Sources 12 and 13 prove the same main theorems, in nearly the same order: scalar detection of summability, automatic strongness of constant-term-covariant isomorphisms, automatic strongness of  $\text{Aut}(\mathbf{Oz})$ , the factorization of the whole stabilizer, scalar adjoints, a proper-class obstruction, a proper embedding moving a real, and surcomplex versions. They are printed once, with both sources named. The merge made these choices:

- *Base of the automatic-strongness sections: 12.* Its abstract theorem assumes only a coefficient isomorphism  $\alpha$  and derives  $F|_k = \alpha$ , where 13 assumes  $\sigma(k) = \ell$  ([theorem 22.1](#)); it also proves the factorization for set-sized groups, the Archimedean corollary, the regular-cardinal case and retractions for arbitrary strong embeddings. 13's genuinely different material is added with its proofs: the countable binary witness and its finite-row lemma, kept as a second route to detection ([lemma 21.3](#) and [theorem 21.4](#)); isomorphisms between different, possibly non-divisible groups ([proposition 23.1](#) and [corollary 23.4](#)); the coefficient-matrix formula ([proposition 22.3](#)); the example that constant-term preservation does not give preservation of  $\mathbf{Oz}$  ([example 23.6](#)); the small-integer-part fraction example ([example 31.2](#)); and the failure of homogeneity ([corollary 27.5](#)).
- *Base of the embedding sections: 11.* Its bottom-gap classification ([theorem 26.3](#)) contains the embeddings of 12 and 13 as instances ([theorem 27.3](#)). The three explicit maps are the same construction with different order isomorphisms  $\mathbf{No} \rightarrow \mathbf{No}_{>0}$  or  $\mathbf{No} \rightarrow (1, \infty)$ ; they are printed once. 12's image intersection and annihilator and 13's non-homogeneity are kept.
- *Printed once, not reprinted.* The reconstruction ingredients that all three reprove (the ideal  $\Pi = \bigcap_n n\mathbf{Oz}$ , the existential test, the multiplier ring, its units,  $\text{Frac } \mathbf{Oz} = \mathbf{No}$ , fixing of  $\mathbb{R}$ ) are cited from the Diophantine report and [section 8](#); the convex-support criterion, which 12 and 13 reprove, is [theorems 3.3](#) and [3.8](#); the Taylor coefficient automorphism, which 11 reproves, is `saut:thm:coeffflow` of the surcomplex automorphism report.

- *Credit added.* 12 and 13 do not credit `saut:thm:coeffflow`, which already moves a real  $b$  to  $b + t^\delta$  by a strong automorphism of **No**; 11 does. The Taylor motion in every embedding below is that theorem's (section 26.1).

No mathematical error was found in the three manuscripts. They agree with one another and with the rest of the report: automorphisms preserving **Oz** fix  $\mathbb{R}$  (corollary 5.2), embeddings need not. Two statements in their delivered audits are wrong and are recorded in section A.3; neither affects the mathematics.

## 20.2 What this part settles

- Question 11.1 is *answered* in all its clauses (theorem 23.2): every ring automorphism of **Oz** extends uniquely to a strongly  $\mathbb{R}$ -linear automorphism of **No** whose inverse is strong, so preservation of **Oz** forces strongness. There is no nonstrong **Oz**-preserving automorphism, so the clause of 09 (characterize the nonstrong automorphisms of **(No, Oz)** that commute with normal-form sums) is vacuous, and 10's variant (is every **Oz**-preserving 1-automorphism strong?) has the answer yes.
- Theorem 5.1 and corollary 5.2 hold for all automorphisms of **(No, Oz)** and of the full real Hahn pairs with nonzero divisible set-sized exponent group (theorem 24.1). For Archimedean groups every stabilizer element is a monomial map (corollary 24.3), which upgrades theorem 4.13 from 1-automorphisms to the whole stabilizer.
- The second clause of `osq:q:invisible` (how much can an automorphism of **Oz** do inside  $\Pi$  while fixing  $\text{ct}$ ?) is answered (corollary 24.4); its first clause is untouched.
- Scope. The answers concern  $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$ : **Oz** and the full real Hahn pairs. For  $(\mathbb{C}, \mathbb{Z}[i])$  they hold when conjugation or the natural valuation is preserved (section 30); whether every automorphism of **Oz** $[i]$  is strong stays open (question 32.1). For a general characteristic-zero  $k$  and unital  $\mathfrak{o} \subseteq k$ , as in theorem 3.3, nothing here removes the strongness hypothesis (question 32.13).
- Embeddings: a strong field embedding of the full real Hahn pairs sending monomials to monomials and preserving the integer parts is classified by a coefficient section into the Hahn field on the *bottom gap*; real numbers can move exactly when the positive part of the value-group image is not coinitial (theorem 26.3 and corollary 26.7). Whether every embedding of **(No, Oz)** reflecting **Oz** is strong has a negative answer if there is a measurable cardinal (question 32.2).

## 20.3 Conventions and notation of this part

All three sources use the small- $t$  convention (1.1) of part I,  $t^\gamma = \omega^{-\gamma}$ , so the purely infinite series have *negative*  $t$ -exponents. For a field  $k$  and an ordered abelian group  $\Gamma$  put  $K_\Gamma = k((t^\Gamma))$ ,  $\Pi_\Gamma$  for the series supported in  $\Gamma^{<0}$  and  $\mathfrak{m}_\Gamma$  for those supported in  $\Gamma^{>0}$ . For  $k = \mathbb{R}$  put

$$\mathcal{R}_\Gamma = \mathbb{Z} \oplus \Pi_\Gamma, \quad \mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi_\Gamma,$$

as in theorem 4.10; for  $\Gamma = \mathbf{No}$  these are **Oz** and  $\mathcal{A}_k$ . For  $k = \mathbb{C}$  write  $\Pi_\Gamma^\mathbb{C} = \mathbb{Z}[i] \oplus \Pi_\Gamma^\mathbb{C}$  and  $K_\Gamma^\mathbb{C} = \mathbb{C}((t^\Gamma))$ ; for  $\Gamma = \mathbf{No}$  these are  $\Pi_\mathbb{C}$ , **Oz** $[i]$  and **No** $(i)$ . The constant term  $\text{ct}(x) = [t^0]x$  is the projection  $P_0$  on all of  $K_\Gamma$ ; it is  $k$ -linear and not multiplicative, since  $\text{ct}(t^\gamma) \text{ct}(t^{-\gamma}) = 0 \neq \text{ct}(1)$  for  $\gamma \neq 0$ . The *constant-term pairing* is

$$\langle x, y \rangle_0 = \text{ct}(xy), \quad [t^\gamma]x = \langle x, t^{-\gamma} \rangle_0. \quad (20.1)$$

It is symmetric,  $k$ -bilinear and nondegenerate, and not positive definite: it is not an inner product, a trace or a residue integral [12, 13].

*Strong maps.* An additive map is *strongly additive* if it sends every Hahn summable family to a summable family and commutes with its sum; this is one-directional. A strongly additive map need not be  $\mathbb{R}$ -linear: the Taylor maps below move real constants. As in [section 1.5](#), a *strong automorphism* is one whose inverse is also strongly additive; every theorem below states both directions where it asserts them. For scalar families in  $k$  “summable” means “only finitely many nonzero terms”; this is not convergence in  $\mathbb{R}$  [13].

*Exponent maps.* Here  $\tau$  is an *ordered additive embedding* of exponent groups; when it is onto it is the order automorphism  $\tau$  of (5.1). The monomial map  $M_{\chi,\tau}$  of (5.1) is defined by the same formula for an embedding  $\tau$  and then is a strong field embedding. A field embedding is *exact-monomial* if it sends every  $t^\gamma$  to a monomial  $t^{\tau(\gamma)}$  with coefficient one (11 says “monomial-preserving”). This does *not* mean commuting with the omega-map  $x \mapsto \omega^x$ , and it excludes characters and principal-unit corrections [11].

| Notation                                  | Meaning in <a href="#">part III</a> , and the sources’ symbols                                                                                                                                         |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{R}_\Gamma, \mathcal{A}_\Gamma$  | $\mathbb{Z} \oplus \Pi_\Gamma$ and $\mathbb{R} \oplus \Pi_\Gamma$ . Sources: $I_\Gamma, A_\Gamma$ (11); $A_G, B_\mathbb{R}$ (12); $\mathcal{J}_\Gamma, \mathcal{A}$ (13).                              |
| $\langle x, y \rangle_0$                  | $\text{ct}(xy)$ . Sources: $\langle x, y \rangle_0$ (12, 13); 11 writes $[x]_0$ for $\text{ct } x$ .                                                                                                   |
| $h_\sigma(g), a_{\sigma,\delta}$          | The logarithmic character (3.1). Sources: $\ell_g, a_\delta$ (12); $L_\gamma, a_\delta$ (13).                                                                                                          |
| $G_\tau$                                  | The <i>bottom gap</i> of $\tau : \Gamma \hookrightarrow \Delta$ ( <a href="#">definition 26.1</a> ). Source: $C_\tau$ (11).                                                                            |
| $\rho$                                    | An ordered field embedding $\mathbb{R} \rightarrow \mathbb{R}((t^{G_\tau}))$ , a <i>coefficient section</i> (11). 13’s $\rho = \sigma _k$ is $\alpha$ here, 12’s letter.                               |
| $J_{\tau,\rho}$                           | The classified embedding (26.1). Source: $\Phi_{\tau,\rho}$ (11).                                                                                                                                      |
| $d$                                       | A $\mathbb{Q}$ -derivation of $\mathbb{R}$ . Sources: $d$ (11 and the surcomplex report), $D$ (12, 13); $D$ stays reserved for derivations of Hahn fields.                                             |
| $\rho_{d,\delta}, \mathcal{T}_{d,\delta}$ | The Taylor section (26.3) and the ambient Taylor automorphism (26.4), which is $\Phi_d$ of <code>saut:thm:coeffflow</code> . Sources: $\rho_{d,\delta}, E_{d,\delta}$ (11); $\rho_D$ (12); $E_D$ (13). |
| $\iota_\mu$                               | The inner lift (27.1) of an order embedding $\mu : \mathbf{No} \hookrightarrow \mathbf{No}$ . Sources: $T_h$ (11); 12’s $\theta$ is $\iota_{\mu_{12}}$ and 13’s second $\tau$ is $\iota_{\mu_{13}}$ .  |
| $\mu_{11}, \mu_{12}, \mu_{13}$            | The order isomorphisms (27.2). Sources: $p$ (11), $\psi$ (12), $p$ (13).                                                                                                                               |
| $J_{\mu,d}$                               | The explicit embedding (27.3). Sources: $\Phi$ (11), $J_D$ (12), $F_D$ (13).                                                                                                                           |
| $T^\dagger, e_\phi, \text{ret}_\phi$      | Constant-term adjoint, $\phi^\dagger(1)$ and the linear retraction of a strong embedding $\phi$ ( <a href="#">section 25</a> ). Sources: $T^\dagger, d = j^\dagger(1), E_j$ (12); $T^*, A$ (13).       |
| $\lambda_{\mathbf{On}}$                   | The class functional (25.2). Sources: $\Lambda$ (12), $\lambda$ (13).                                                                                                                                  |
| $\mathcal{V}_\lambda$                     | The valuation ball $\{x : v(x) > \lambda\}$ . Source: $\mathcal{B}_\lambda$ (11).                                                                                                                      |

The collisions are resolved in the new material only, and the tempting false readings are wrong:

- 11's  $C_\tau$  is *not* a principal convex subgroup  $C_\delta$ . It is the largest convex subgroup of the target group meeting  $\tau(\Gamma)$  only in 0; in the notation of [definition 2.4](#),  $G_\tau = \bigcap_{h \in \tau(\Gamma) > 0} H_h$ . For the explicit maps below it happens to be a principal convex subgroup,  $C_1$  or  $C_\omega$  ([lemma 27.2](#)). 11's  $H = \tau(\Gamma)$  is written  $\tau(\Gamma)$ , leaving  $H_g$  to [definition 2.4](#); 11's class  $\mathcal{F}$  of finite surreals is  $C_1$ .
- 12's  $\theta$  is an *exponent embedding*, not an additive profile as in  $D_{\delta, \theta}$ ; it is written  $\iota_{\mu_{12}}$ . 13 uses  $\tau$  both for the order automorphism of  $M_{\chi, \tau}$  and for its exponent embedding; the second is  $\iota_{\mu_{13}}$ .
- 11's and 13's order isomorphisms are both called  $p$ , and 11's  $\rho$  and 13's  $\rho$  are different maps; the renamings above separate them.
- 13 writes the factorization as  $\sigma = U \circ M$ , 12 and [theorem 5.1](#) as  $\sigma = M \circ u$ ; the two agree with  $U = MuM^{-1}$ , since the monomial maps normalize  $\mathcal{U}_{\mathbf{Oz}}$ .
- 11's target group  $\Lambda$  is  $\Delta$  here, 13's letter, so that  $\lambda$  and  $\Lambda$  stay free; 11's Gaussian ring  $J_\Gamma$  and field  $L_\Gamma$  are  $\mathcal{R}_\Gamma^\mathbb{C}$  and  $K_\Gamma^\mathbb{C}$ .
- The large-cardinal report's sign-sequence embedding, called  $J$  there and in its dossier, is written  $\hat{j}$  here, to keep  $J$  for the strong embeddings of this part.

## 21 Constant-term tests detect Hahn summability

This section is 12's Section 3 and 13's Section 3, printed once. Let  $k$  be any field and  $K = k((t^\Gamma))$  a full Hahn field;  $\Gamma$  may be a set or  $\mathbf{No}$ , with set-sized supports. The point is that one scalar functional, ct, together with multiplication, recognizes Hahn summability.

**Lemma 21.1** (Finite intersection [12, 13]). *If  $S, T \subseteq \Gamma$  are well ordered, then  $S \cap (-T)$  is finite.*

*Proof.* An infinite subset of the well-ordered  $S$  contains a strictly increasing sequence  $s_0 < s_1 < \dots$ , and then  $-s_0 > -s_1 > \dots$  would be a descending sequence in  $T$ .  $\square$

**Lemma 21.2** (Triangular detector [12]). *Let  $(x_j)_{j \in I}$  be a set-indexed family in  $K$  in which each exponent occurs in only finitely many supports. If  $\bigcup_j \text{supp } x_j$  is not well ordered, there is  $y \in K$  with countable support and all coefficients in  $\{0, 1\}$  such that  $\langle x_j, y \rangle_0 \neq 0$  for infinitely many  $j$ .*

*Proof.* Choose a strictly decreasing sequence in the union of the supports. Each support meets it in finitely many places, since an infinite subsequence would descend inside a well order. Select recursively a decreasing subsequence  $g_0 > g_1 > \dots$  and distinct indices  $j_0, j_1, \dots$  with

$$[t^{g_n}]x_{j_n} \neq 0, \quad [t^{g_m}]x_{j_n} = 0 \quad (m > n) : \quad (21.1)$$

at each stage discard the finitely many places lying in the previously selected supports, and pick a later place and an index whose support contains it. Choose  $c_n \in \{0, 1\}$  recursively with  $\sum_{m \leq n} c_m [t^{g_m}]x_{j_m} \neq 0$ ; the new variable  $c_n$  has the nonzero coefficient  $[t^{g_n}]x_{j_n}$ , so at most one of the two choices is forbidden, over every field. Put  $y = \sum_n c_n t^{-g_n}$ , a Hahn series because the exponents  $-g_n$  increase. By (21.1),  $\langle x_{j_n}, y \rangle_0$  is exactly that finite sum, nonzero for every  $n$ .  $\square$

**Lemma 21.3** (Binary finite-row lemma [13]). *Let  $(L_n)_{n \in \mathbb{N}}$  be nonzero linear forms on  $k^J$ , each using only finitely many coordinates. There is  $a \in \{0, 1\}^J$  with  $L_n(a) \neq 0$  for infinitely many  $n$ .*

*Proof.* Let  $E_n \neq \emptyset$  be the finite support of  $L_n$ . If  $E = \bigcup_n E_n$  is finite, some coordinate  $j$  has a nonzero coefficient in infinitely many rows, and  $a = e_j$  works. If  $E$  is infinite, select rows  $n_0 < n_1 < \dots$  inductively. At stage  $r$  finitely many coordinates have been assigned, and the rows after  $n_{r-1}$  still use infinitely many coordinates; choose  $n_r$  whose support contains an unassigned coordinate  $j_r$ , set the other new coordinates of that row to 0, and choose  $a_{j_r} \in \{0, 1\}$  so that  $L_{n_r}(a) \neq 0$  (the two choices change  $L_{n_r}(a)$  by a nonzero field element). Mark all of  $E_{n_r}$  as assigned; later choices do not change this row. Finally set the remaining coordinates to 0.  $\square$

The lemma holds in every characteristic, including over  $\mathbb{F}_2$ , and uses no uncountability, genericity or analytic argument. Asking for *every* row to be nonzero would be false: over  $\mathbb{F}_2$  the forms  $x, y, x + y$  cannot all be nonzero [13].

**Theorem 21.4** (Scalar detection of Hahn summability [12, 13]). *For a set-indexed family  $(x_j)_{j \in I}$  in the full Hahn field  $K$  the following are equivalent:*

- (i)  $(x_j)$  is Hahn summable;
- (ii) for every  $y \in K$ , only finitely many  $j$  have  $\langle x_j, y \rangle_0 \neq 0$ ;
- (iii) the same holds for every  $y$  with countable support and coefficients in  $\{0, 1\}$ .

If they hold, then for every  $y$

$$\left\langle \sum_j x_j, y \right\rangle_0 = \sum_j \langle x_j, y \rangle_0, \quad (21.2)$$

a finite scalar sum, and  $\sum_j x_j$  is the unique element with this property.

*Proof.* (i) $\Rightarrow$ (ii): with  $S = \bigcup_j \text{supp } x_j$  and  $T = \text{supp } y$ , only exponents in the finite set  $S \cap (-T)$  contribute to a constant term (lemma 21.1), and each occurs in finitely many  $x_j$ ; all rearrangements in (21.2) are finite. (ii) $\Rightarrow$ (iii) is trivial. (iii) $\Rightarrow$ (i): the tests  $y = t^{-g}$  give pointwise finiteness, and if the union of the supports were not well ordered, lemma 21.2 would give a countable binary  $y$  violating (iii). Uniqueness follows from the tests  $t^{-g}$  and nondegeneracy.  $\square$

*Second route to (iii) $\Rightarrow$ (i) [13].* Assume (iii); monomial tests give pointwise finiteness. If the union  $S$  is not well ordered, choose  $\gamma_0 > \gamma_1 > \dots$  in  $S$ . Each support meets  $\{\gamma_n\}$  finitely, so infinitely many distinct indices  $j_0, j_1, \dots$  meet it. The forms  $L_r(a) = \sum_{n: \gamma_n \in \text{supp } x_{j_r}} a_n [t^{\gamma_n}] x_{j_r}$  on  $k^{\mathbb{N}}$  are nonzero and use finitely many coordinates, so lemma 21.3 gives binary  $a_n$  with infinitely many  $L_r(a) \neq 0$ . For  $y = \sum_n a_n t^{-\gamma_n}$ , a Hahn series, one has  $\text{ct}(yx_{j_r}) = L_r(a)$ , contradicting (iii).  $\square$

The two routes differ in how cancellation is prevented: 12 first makes the family triangular and then chooses coefficients; 13 keeps the family and chooses coefficients row by row. 12's witness is also countable and binary, although 12 does not state it; 13 does. Every witness has countable set support, so the theorem holds for  $\Gamma = \mathbf{No}$  [12, 13].

**Corollary 21.5** (Countable certificates [13]). *Every nonsummable set-indexed family has a countable subfamily and a countably supported binary  $y$  for which infinitely many products  $x_j y$  have nonzero constant term.*

*Proof.* An exponent occurring in infinitely many supports gives a monomial witness; otherwise use lemma 21.2.  $\square$

This does not say that summable families are countable: a Hahn sum may have an arbitrarily large set-sized support. It says that a failure of summability has a countable certificate.

**Example 21.6** (Monomial tests are not enough [13]). The family  $x_n = t^{-n}$ ,  $n \geq 1$ , has each exponent in one member, but its supports descend. Each monomial test detects at most one member, whereas the Hahn series  $y = \sum_{n \geq 1} t^n$  has  $\text{ct}(yx_n) = 1$  for every  $n$ .

*Remark 21.7* (Finiteness spaces [12, 13]). **Lemma 21.1** is finite-intersection orthogonality, and its linearization is the finiteness-space duality of Ehrhard, applied to generalized power series by Blute, Cockett, Jacqmin and Scott [2]. Neither support duality nor the notion of a strong linear map is new; what the sources prove directly is the scalar criterion with its cancellation argument and set supports over the class **No**. It is an external characterization of summation, not first-order definability of summation, and not convergence in the valuation topology: a Hahn-summable family need not converge in that topology [13].

## 22 Constant-term-covariant isomorphisms are strong

**Theorem 22.1** (Automatic strongness from constant-term covariance [12, 13]). *Let  $K = k((t^\Gamma))$  and  $L = l((t^\Delta))$  be full Hahn fields, the exponent groups sets or **No**, with set supports. Let  $F : K \rightarrow L$  be a field isomorphism and  $\alpha : k \rightarrow l$  a field isomorphism with*

$$\text{ct}(Fx) = \alpha(\text{ct } x) \quad (x \in K). \quad (22.1)$$

*Then  $F(r) = \alpha(r)$  for all  $r \in k$ ,  $F$  is  $\alpha$ -semilinear, and  $F$  and  $F^{-1}$  are strongly additive.*

*Proof.* For  $r \in k$  and  $x \in K$ , multiplicativity and (22.1) give  $\text{ct}((F(r) - \alpha(r))F(x)) = \alpha(\text{ct}(rx)) - \alpha(r)\alpha(\text{ct } x) = 0$ . As  $F$  is onto,  $F(r) - \alpha(r)$  pairs to zero with all of  $L$ , so  $F(r) = \alpha(r)$ ; semilinearity follows. Let  $(x_j)$  be summable with sum  $s$ . For  $y \in L$  write  $y = F(z)$ ; then  $\langle F(x_j), y \rangle_0 = \alpha(\langle x_j, z \rangle_0)$  is nonzero for finitely many  $j$  (theorem 21.4 in  $K$ ), so  $(F(x_j))$  is summable (theorem 21.4 in  $L$ ). With  $w = \sum_j F(x_j)$ ,  $\langle F(s), y \rangle_0 = \alpha(\sum_j \langle x_j, z \rangle_0) = \sum_j \langle F(x_j), y \rangle_0 = \langle w, y \rangle_0$  for all  $y$ , so  $F(s) = w$ . The inverse satisfies the hypotheses with  $\alpha^{-1}$ .  $\square$

13 proves this with the extra hypothesis  $F(k) = l$  and  $\alpha = F|_k$ ; 12 derives it. The proof detects summability of the image family before any target sum is used, and it does not represent arbitrary strong functionals, which would fail for **No** (theorem 25.6). Surjectivity enters twice: it supplies a preimage of every target test  $y$ , and it identifies the image of  $k$ . An embedding supplies only the tests in its image; nondegeneracy of the pairing on the image does not detect all target support obstructions. This is a real boundary, not a removable word: the embeddings of section 27 satisfy (22.1) with  $\alpha = \text{id}$  and move real coefficients; they are strongly additive, so they contradict coefficient rigidity without surjectivity, not strongness [12, 13].

**Corollary 22.2** (The summation operation is recovered [12]). *An isomorphism of full Hahn fields that carries the coefficient field onto the coefficient field and commutes with the constant term up to that coefficient isomorphism respects the whole set-indexed Hahn summation. It need not preserve the monomial sections.*

**Proposition 22.3** (Coefficient matrix [13]). *If in theorem 22.1  $k = l$  and  $\alpha = \text{id}$ , then for all  $f \in K$  and  $\delta \in \Delta$*

$$[t^\delta]F(f) = \text{ct}(F^{-1}(t^{-\delta})f) = \sum_{\gamma \in \text{supp } f} [t^\gamma]f [t^{-\gamma}]F^{-1}(t^{-\delta}), \quad [t^\delta]F(t^\gamma) = [t^{-\gamma}]F^{-1}(t^{-\delta}), \quad (22.2)$$

*with finite sums. So the coefficient matrix of  $F$  is the inverse transpose of that of  $F^{-1}$ , with exponent reversal from the pairing.*

*Proof.*  $[t^\delta]F(f) = \text{ct}(F(f)t^{-\delta}) = \text{ct}(F(fF^{-1}(t^{-\delta}))) = \text{ct}(fF^{-1}(t^{-\delta}))$  by (22.1); expand the last constant term, which has finitely many contributions by lemma 21.1.  $\square$

The formula presupposes the global well-ordering of the image, which theorem 22.1 proves; a term-by-term formal argument alone would leave both the coefficient and the well-ordering gaps open [13].

## 23 Automatic strongness of the omnific stabilizer

### 23.1 What the integer part reconstructs

All three sources reprove the reconstruction of the coefficients from the integer part:  $\Pi = \bigcap_n n\mathbf{Oz}$  (lemma 2.3), the existential test  $\exists y (x^2 = 2y^2)$  (odg: def: thm: ideal), the multiplier ring  $(\Pi : \Pi) = \mathcal{A}_k$  with units  $\mathbb{R}^\times$  (odg: def: thm: multiplier),  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  (odg: thm: fractions; also [11, Proposition 2.4.5], which records it as [5, Theorem 32]), and the fixing of  $\mathbb{R}$  by every automorphism (odg: def: thm: autreal, corollary 5.2). They credit these as prior work, and they are not reprinted. The one form needed below that is not already stated in the report is 13's version for isomorphisms between different, possibly non-divisible groups.

**Proposition 23.1** (Pair isomorphisms preserve the constant term [12, 13]). *Let  $\Gamma, \Delta$  be nonzero ordered abelian groups, sets or  $\mathbf{No}$ , and let  $F : K_\Gamma \rightarrow K_\Delta$  be an ordered field isomorphism of the real Hahn fields with  $F(\mathcal{R}_\Gamma) = \mathcal{R}_\Delta$ . Then  $F$  fixes  $\mathbb{R}$  pointwise, maps  $\Pi_\Gamma, \mathbb{R}$  and  $\mathfrak{m}_\Gamma$  onto  $\Pi_\Delta, \mathbb{R}$  and  $\mathfrak{m}_\Delta$ , and  $\text{ct}(Fx) = \text{ct}(x)$  for all  $x$ . If  $\Gamma$  and  $\Delta$  are divisible, “ordered” may be omitted.*

*Proof.* By lemma 2.3,  $\Pi_\Gamma = \bigcap_n n\mathcal{R}_\Gamma$  for every ordered abelian group, so  $F(\Pi_\Gamma) = \Pi_\Delta$ . The multiplier description  $\{x : x\Pi_\Gamma \subseteq \Pi_\Gamma\} = \mathcal{A}_\Gamma$  holds for every nonzero  $\Gamma$ : a nonzero coefficient of  $x$  at  $\delta > 0$  gives  $\text{ct}(xt^{-\delta}) \neq 0$ . As  $F$  is onto,  $F(\mathcal{A}_\Gamma) = \mathcal{A}_\Delta$ , and the units  $\mathbb{R}^\times$  correspond, so  $F$  maps  $\mathbb{R}$  onto  $\mathbb{R}$ ; a field isomorphism  $\mathbb{R} \rightarrow \mathbb{R}$  is the identity. The order is preserved by hypothesis, or because divisible groups give real closed Hahn fields whose order is defined by squares. The infinitesimals  $\{x : |x| < 1/n \text{ for all } n\}$  are  $\mathfrak{m}_\Gamma$ , so  $F(\mathfrak{m}_\Gamma) = \mathfrak{m}_\Delta$ , and the decomposition  $K = \Pi \oplus \mathbb{R} \oplus \mathfrak{m}$  gives the rest.  $\square$

For  $\Gamma = \Delta$  divisible this is also a consequence of the parameter-free definitions of theorem 8.2; for  $\Gamma = \Delta = \mathbf{No}$  it is part of corollary 5.2. The coefficient field is reconstructed as a set and then fixed pointwise; an ordered embedding  $\mathbb{R} \hookrightarrow \mathbf{No}$  need not have image  $\mathbb{R}$ , and section 27 gives embeddings that move reals by infinitesimals [13].

### 23.2 The main theorem

**Theorem 23.2** (Automatic strongness of omnific automorphisms [12, 13]). *Every unital ring automorphism  $\sigma$  of  $\mathbf{Oz}$  extends uniquely to a field automorphism  $\tilde{\sigma}$  of  $\mathbf{No}$ . The extension preserves the order and fixes  $\mathbb{R}$  pointwise, and  $\tilde{\sigma}$  and  $\tilde{\sigma}^{-1}$  are strongly  $\mathbb{R}$ -linear: for every Hahn-summable family  $(f_j)$  the family  $(\tilde{\sigma}(f_j))$  is summable and  $\tilde{\sigma}(\sum_j f_j) = \sum_j \tilde{\sigma}(f_j)$ . Consequently*

$$\text{Aut}(\mathbf{Oz}) \cong \text{Aut}(\mathbf{No}; \mathbf{Oz}) = \text{Aut}_{\mathbb{R}}^{\text{str}}(\mathbf{No}; \mathbf{Oz}), \quad (23.1)$$

*by restriction, with extension to fractions as inverse; here  $\text{Aut}(\mathbf{No}; \mathbf{Oz})$  denotes the field automorphisms of  $\mathbf{No}$  preserving  $\mathbf{Oz}$  setwise and the right-hand side those among them that are strongly  $\mathbb{R}$ -linear with strong inverse.*

*Proof.* Extension, uniqueness, order and fixing of  $\mathbb{R}$  are `odg:def:thm:autreal`. By [corollary 5.2](#) (or [proposition 23.1](#)) the extension commutes with  $P_0 = \text{ct}$ , without any strongness assumption. [Theorem 22.1](#) with  $k = l = \mathbb{R}$  and  $\alpha = \text{id}$  gives strongness of  $\tilde{\sigma}$  and its inverse.  $\square$

The dependencies are not circular [13]:

$$\mathbf{Oz} \longrightarrow \Pi \longrightarrow \mathcal{A}_k \longrightarrow \mathbb{R} \longrightarrow (\Pi, \mathbb{R}, \mathbf{m}) \longrightarrow \text{ct} \longrightarrow \text{Hahn summation}.$$

The first five steps use only algebra, the order and finite decompositions; the residue criterion is used only after the constant term has been recovered. The group notation in (23.1) is schematic, in the sense of [section 1.4](#): it concerns individual class maps and their compositions, and forms no class of all of them.

*Remark 23.3* (What is and is not asserted [12, 13]). The theorem does not say that every automorphism of the field  $\mathbf{No}$  is strong: Kaplan, Krapp and Serra construct automorphisms of  $\mathbf{No}$  that are not strongly additive [9, Example 4.4], and the theorem shows that none of them preserves  $\mathbf{Oz}$ . Nor does it say that the automorphisms of  $(\mathbf{No}, \mathbf{Oz})$  are few: [part I](#) constructs large families of them, and they are strong by construction. The theorem is a regularity statement. It identifies the summation operation as preserved by these maps; invariance under automorphisms is not first-order definability of summation. It also gives an intrinsic certificate for a transported sum: a proposed value  $g$  of  $\tilde{\sigma}(\sum_j f_j)$  is correct exactly when  $\text{ct}(yg) = \sum_j \text{ct}(y\tilde{\sigma}(f_j))$  for every  $y \in \mathbf{No}$ , each a finite equation [13]. With [theorems 3.3](#) and [5.1](#) the theorem answers [question 11.1](#); see [section 32.1](#).

**Corollary 23.4** (Full real Hahn pairs [12, 13]). *Every isomorphism  $F$  in [proposition 23.1](#) is strongly  $\mathbb{R}$ -linear with strongly  $\mathbb{R}$ -linear inverse. In particular, for a nonzero divisible set-sized  $\Gamma$ , every field automorphism of  $K_\Gamma$  stabilizing  $\mathcal{R}_\Gamma$  is strongly  $\mathbb{R}$ -linear.*

*Proof.* [Proposition 23.1](#) and [theorem 22.1](#) with  $\alpha = \text{id}$ .  $\square$

The corollary assumes an isomorphism of the ambient Hahn fields. It does not assert that an abstract isomorphism  $\mathcal{R}_\Gamma \cong \mathcal{R}_\Delta$ , or an automorphism of  $\mathcal{R}_\Gamma$ , extends to the Hahn fields when  $\text{Frac } \mathcal{R}_\Gamma$  is smaller than  $K_\Gamma$ , which can happen ([example 31.2](#), [11, Proposition 2.4.5]); 12 states the automorphism case, 13 the isomorphism case [12, 13].

**Corollary 23.5** (Determination by monomials [12, 13]). *Two automorphisms of  $\mathbf{Oz}$  whose extensions agree on every Conway monomial are equal. An  $\mathbf{Oz}$ -preserving field automorphism of  $\mathbf{No}$  fixing every monomial is the identity.*

*Proof.* Both extensions fix  $\mathbb{R}$  and commute with the normal-form sum of every element.  $\square$

**Example 23.6** (Constant-term preservation does not give preservation of  $\mathbf{Oz}$  [13]). Choose a  $\mathbb{Q}$ -linear  $q : \mathbb{R} \rightarrow \mathbb{R}$  with  $q(1) = 0$  and  $q(\sqrt{2}) = 1$ , by extending  $\{1, \sqrt{2}\}$  to a Hamel basis, and put  $\theta(\gamma) = q(\text{ct } \gamma)$ , the constant term of the exponent's own normal form. Then  $\theta$  is additive,  $\theta(1) = 0$  and  $\theta(\sqrt{2}) = 1$ , and  $\Psi(t^\gamma) = t^\gamma \exp(\theta(\gamma)t)$  defines a strong  $\mathbb{R}$ -linear 1-automorphism of  $\mathbf{No}$  with inverse given by  $-\theta$ : it is the flow  $T^1$  of [proposition 4.9](#) with  $\delta = 1$ , which needs only  $\theta(1) = 0$ , not  $\theta(C_1) = 0$ . A term at exponent 0 in  $\Psi(t^\gamma)$  requires  $\gamma = -n$  with  $n \in \mathbb{N}$ , and  $\theta(-n) = 0$ ; so  $\text{ct} \circ \Psi = \text{ct}$  on monomials and, by strongness, everywhere. But

$$\Psi(t^{-\sqrt{2}}) = t^{-\sqrt{2}} \exp(-t)$$

has coefficient  $\frac{1}{2}$  at the positive exponent  $2 - \sqrt{2}$ , so  $\Psi(\mathbf{Oz}) \not\subseteq \mathbf{Oz}$ . Constant-term preservation suffices for automatic strongness of isomorphisms, but not for preservation of  $\mathbf{Oz}$ . The merge notes the agreement with [theorem 4.1](#): here  $\theta(C_1) \neq 0$ , since  $\sqrt{2} \in C_1$ .

## 24 The whole stabilizer

**Theorem 24.1** (Factorization of every pair automorphism [12, 13]). *Let  $\Gamma = \mathbf{No}$ , or a nonzero divisible set-sized ordered abelian group. Every field automorphism  $\sigma$  of  $K_\Gamma = \mathbb{R}((t^\Gamma))$  with  $\sigma(\mathcal{R}_\Gamma) = \mathcal{R}_\Gamma$  factors uniquely as*

$$\sigma = M_{\chi, \tau} \circ u, \quad u \in \mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma),$$

where  $\tau(\gamma) = v(\sigma(t^\gamma))$  is an order automorphism of  $\Gamma$  and  $\chi(\gamma) = \text{lc}(\sigma(t^\gamma))$  is a positive character;  $u$  satisfies the equivalent conditions of [theorem 3.3](#), in particular  $a_{u, \delta}(C_\delta) = 0$  for every  $\delta > 0$ . Conversely every such composition preserves  $\mathcal{R}_\Gamma$ . So the stabilizer is  $\{M_{\chi, \tau}\} \ltimes \mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma)$ , and  $\text{Aut}(\mathbf{Oz}) \cong \mathcal{U}_{\mathbf{Oz}} \rtimes \{M_{\chi, \tau}\}$ .

*Proof.* By [theorem 23.2](#) or [corollary 23.4](#),  $\sigma$  is strong and fixes  $\mathbb{R}$ . It preserves the order, hence Archimedean equivalence, which is expressed with ordinary integers; so it maps the valuation ring onto itself and is valued in the sense of [section 5](#). Every character into  $\mathbb{R}^\times$  of a divisible group is positive. [Theorem 5.1](#) now applies, with  $k = \mathbb{R}$  and  $\mathfrak{o} = \mathbb{Z}$ .  $\square$

This is [theorem 5.1](#) and the second sentence of [corollary 5.2](#) without the hypothesis of strongness. 12 proves the factorization for  $\mathbf{No}$  and set-sized groups, 13 for  $\mathbf{No}$ , in the order  $\sigma = U \circ M_{\chi, \tau}$  ([section 20.3](#)). Both reprove the convex-support criterion: 12 by the least forbidden shift, which is source 10's route ([theorem 3.8](#)), and 13 by a polynomial in the rational parameter  $q$  with infinitely many zeros, which is 04's route ([theorem 3.3](#)); those proofs are printed there. 13's nontrivial example, the shear  $t^\gamma \mapsto t^\gamma \exp(q_1(\gamma)t)$  with  $\omega^\omega \mapsto \omega^\omega \exp(-\omega^{-1})$ , is [example 4.3](#).

*Remark 24.2* (Admissibility is retained [12, 13]). [Theorem 24.1](#) classifies *existing* automorphisms. An arbitrary array of additive  $a_\delta$  with  $a_\delta(C_\delta) = 0$  is not thereby shown to define one: the monomial substitution must be summable on every well-ordered support, multiplicative and invertible with the same admissibility. Local convex vanishing is not proved sufficient, and no class-wide exponential–logarithm correspondence is used or established ([question 11.2](#)). The theorem removes a hypothesis on actual automorphisms; it licenses no unchecked infinite substitution.

**Corollary 24.3** (Archimedean groups [12]). *For a nonzero divisible Archimedean set-sized  $\Gamma$ , every field automorphism of  $\mathbb{R}((t^\Gamma))$  stabilizing  $\mathcal{R}_\Gamma$  is a monomial map  $M_{\chi, \tau}$ .*

*Proof.* In [theorem 24.1](#),  $u \in \mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma) = \{\text{id}\}$  by [theorem 4.13](#).  $\square$

[Theorem 4.13](#) gave this among strong maps; automatic strongness removes the qualification. No claim is made that omnific-preserving automorphisms are trivial: the monomial factor, and in higher rank the factor  $\mathcal{U}$ , can be large [12].

**Corollary 24.4** (What an automorphism of  $\mathbf{Oz}$  does inside  $\Pi$  [merge]). *Every ring automorphism of  $\mathbf{Oz}$  fixes  $\text{ct}$  and  $\mathbb{R}$ , and its extension is  $M_{\chi, \tau} \circ u$  with  $\tau$  an order automorphism of  $\mathbf{No}$ ,  $\chi$  a positive character and  $u \in \mathcal{U}_{\mathbf{Oz}}$ . So on  $\Pi$  it acts by an exponent automorphism and a positive rescaling of monomials, followed by a strong leading-term-fixing automorphism whose logarithmic character has, for every  $g \neq 0$ , support in  $H_g$ , the elements infinitesimal relative to  $g$  ([theorem 3.3](#), condition (v)); every such composition occurs, and every monomial image stays in the signed Archimedean block of its exponent up to the action of  $\tau$  ([proposition 3.10](#)).*

*Proof.* [Theorem 24.1](#) with  $\Gamma = \mathbf{No}$ , and [theorem 3.3](#) and [proposition 3.10](#) for  $u$ .  $\square$

With [theorems 7.6, 7.7](#) and [9.1](#) this describes the automorphisms of **Oz** inside  $\Pi$  as fully as [part I](#) describes the strong ones; the admissibility of logarithmic data remains [question 11.2](#). Source 10's [remark 3.9](#) is subsumed: its monomial condition now holds for every automorphism because every automorphism is strong.

## 25 Constant-term adjoints and a proper-class obstruction

In this section  $\Gamma, \Delta$  are *sets*,  $K = k((t^\Gamma))$  and  $L = k((t^\Delta))$  share the coefficient field  $k$ , and linearity is over  $k$ , not over  $K$ . This is the scalar  $k$ -linear dual of a full Hahn field; the three-duals report (`docs/surcomplex/three-duals-of-hahn-vector-spaces, duals:`) studies  $K$ -linear maps on vector-valued Hahn spaces and topological duals, a different convention, and no identification with its functionals is asserted [12, 13]. A  $k$ -linear functional  $\lambda : K \rightarrow k$  is *strong* if it sends every summable family to a finitely supported scalar family and commutes with the finite sum.  $T$  denotes a  $k$ -linear map  $K \rightarrow L$  in this section only.

**Proposition 25.1** (Scalar representability at set size [12, 13]). *Every strong  $k$ -linear functional  $\lambda$  on  $K$  has a unique representation  $\lambda(x) = \text{ct}(ax)$  with  $a \in K$ ; conversely every such functional is strong.*

*Proof.* Put  $a_g = \lambda(t^{-g})$ , a function on the set  $\Gamma$ . If its support contained  $g_0 > g_1 > \dots$ , the monomials  $t^{-g_n}$ , with increasing exponents, would form a summable family with all images nonzero; so  $a = \sum a_g t^g \in K$ . Strongness on the normal-form family of  $x$  gives  $\lambda(x) = \sum_g [t^g]x a_{-g} = \text{ct}(ax)$ , a finite sum; uniqueness is nondegeneracy, and the converse is (21.2). 13 argues on the set  $\{\gamma : \lambda(t^\gamma) \neq 0\}$ , which meets every well-ordered set finitely and is therefore reverse well ordered, by dependent choice.  $\square$

**Theorem 25.2** (Adjoint characterization [12, 13]). *A  $k$ -linear map  $T : K \rightarrow L$  is strongly  $k$ -linear if and only if it has a constant-term adjoint  $T^\dagger : L \rightarrow K$ ,*

$$\langle Tx, y \rangle_0 = \langle x, T^\dagger y \rangle_0 \quad (x \in K, y \in L). \quad (25.1)$$

*The adjoint is unique,  $k$ -linear and strong;  $T^{\dagger\dagger} = T$ ,  $(ST)^\dagger = T^\dagger S^\dagger$  and  $(T + T')^\dagger = T^\dagger + T'^\dagger$ . Multiplication by a fixed series is self-adjoint, and a coefficient-fixing, constant-term-preserving field isomorphism has adjoint its inverse.*

*Proof.* If  $T$  is strong,  $x \mapsto \text{ct}(T(x)y)$  is a strong functional, represented by a unique  $T^\dagger y$  ([proposition 25.1](#)); uniqueness gives linearity. Conversely, given  $T^\dagger$  and a summable  $(x_j)$ , the scalars  $\langle Tx_j, y \rangle_0 = \langle x_j, T^\dagger y \rangle_0$  vanish for almost all  $j$ , so  $(Tx_j)$  is summable by [theorem 21.4](#), and (21.2) on both sides of (25.1) gives  $T(\sum x_j) = \sum Tx_j$ . Symmetry of the pairing makes  $T$  the adjoint of  $T^\dagger$ , so  $T^\dagger$  is strong. The identities are substitutions; for an isomorphism  $F$  with  $\text{ct} \circ F = \text{ct}$ ,  $\text{ct}(F(x)y) = \text{ct}(xF^{-1}(y))$ .  $\square$

The direction “adjoint implies strong” uses only [theorem 21.4](#) and holds for **No** as well; the converse genuinely needs set-sized groups ([theorem 25.6](#)) [13]. The theorem can be read as a concrete transpose statement for the linearized finiteness space; transpose duality in finiteness spaces is not claimed new [12].

**Theorem 25.3** (A canonical linear retraction [12]). *Let  $\phi : K \hookrightarrow L$  be a strong  $k$ -linear field embedding and  $e_\phi = \phi^\dagger(1)$ . Then  $e_\phi \neq 0$  and  $\phi^\dagger(\phi(a)y) = a\phi^\dagger(y)$  for  $a \in K, y \in L$ . The map*

$$\text{ret}_\phi(y) = e_\phi^{-1}\phi^\dagger(y)$$

*is a strong  $K$ -linear retraction ( $L$  a  $K$ -space through  $\phi$ ), with  $\text{ret}_\phi \circ \phi = \text{id}_K$ . If  $\text{ct} \circ \phi = \text{ct}$  then  $e_\phi = 1$  and  $\text{ret}_\phi = \phi^\dagger$ . If  $\phi$  is not onto,  $\text{ret}_\phi$  is not multiplicative.*

*Proof.* For  $b \in K$ ,  $\text{ct}(b\phi^\dagger(\phi(a)y)) = \text{ct}(\phi(ba)y) = \text{ct}(ba\phi^\dagger(y))$ , which gives the projection formula by nondegeneracy;  $\text{ct}(e_\phi) = \text{ct}(\phi(1)) = 1$ . With  $y = 1$ ,  $\phi^\dagger(\phi(a)) = ae_\phi$ . Multiplication by a fixed series is strong, so  $\text{ret}_\phi$  is strong and  $K$ -linear. If  $\text{ct} \circ \phi = \text{ct}$ , then  $\text{ct}(xe_\phi) = \text{ct}(x)$  for all  $x$  and  $e_\phi = 1$ . A multiplicative unital retraction would be an injective field homomorphism with  $\text{ret}_\phi(y - \phi(\text{ret}_\phi y)) = 0$ , forcing  $y = \phi(\text{ret}_\phi y)$  for all  $y$ .  $\square$

13 proves the case  $\text{ct} \circ \phi = \text{ct}$  with  $\phi$  coefficient-fixing (its residue splitting); it adds that  $\ker \phi^\dagger$  is the annihilator of  $\phi(K)$  for the pairing, and that  $L = \phi(K) \oplus \ker \phi^\dagger$  as  $\phi(K)$ -spaces. More generally a proper field extension admits no unital ring retraction fixing the subfield [13].

**Corollary 25.4** (Tower law and complement [12]). *For composable constant-term-preserving strong  $k$ -linear field embeddings  $K \xrightarrow{\phi} L \xrightarrow{\psi} P$ ,  $\text{ret}_{\psi\phi} = \text{ret}_\phi \circ \text{ret}_\psi$ . For one such embedding  $L = \phi(K) \oplus \ker \text{ret}_\phi$  as  $K$ -spaces, and  $\phi \circ \text{ret}_\phi$  is a strong projection onto  $\phi(K)$ .*

*Proof.* Adjoints reverse composition and all normalizing elements are 1; the complement follows from  $\text{ret}_\phi \circ \phi = \text{id}$ .  $\square$

$\text{ret}_\phi$  is not a field retraction and not a positive conditional expectation; neither multiplicativity nor positivity is part of its conclusion, and it is canonical only relative to the two pairings and to  $\phi$  [12].

**Example 25.5** (Monomial embeddings [12, 13]). For an ordered group embedding  $\tau : \Gamma \hookrightarrow \Delta$  and a character  $\chi : \Gamma \rightarrow k^\times$ , the strong embedding  $M_{\chi,\tau}$  preserves constant terms and

$$M_{\chi,\tau}^\dagger \left( \sum_\eta y_\eta t^\eta \right) = \sum_{g \in \Gamma} \chi(g)^{-1} y_{\tau(g)} t^g,$$

supported in the preimage under an order embedding of a well-ordered set; compare constant coefficients, using  $\chi(-g) = \chi(g)^{-1}$ . The same formula works for class groups with set supports, by replacement on the target support [13]. If  $\eta \notin \tau(\Gamma)$ , then  $M_{\chi,\tau}^\dagger(t^\eta) = M_{\chi,\tau}^\dagger(t^{-\eta}) = 0$  while  $M_{\chi,\tau}^\dagger(1) = 1$ , so the retraction is not multiplicative on two monomials. For instance  $\Gamma = \Delta = \mathbb{Z}$ ,  $\tau(g) = 2g$ ,  $\chi(g) = 2^g$  over  $k = \mathbb{Q}$  give a weighted even-coefficient extraction [12].

For a constant-term-compatible embedding  $\phi$  the source tests show only that  $\text{ct}(\phi(h)\phi(x_j))$  is finitely supported in  $j$  for each  $h \in K$ ; detection in  $L$  needs all  $y \in L$ . An adjoint supplies a source representative of every target test, so the question is whether all target residue functionals can be pulled back; this locates the remaining embedding questions without answering them [13] ([questions 32.2](#) and [32.3](#)).

**Theorem 25.6** (A strong functional with no representing surreal [12, 13]). *For  $x \in \mathbf{No}$  put*

$$\lambda_{\mathbf{On}}(x) = \sum_{\alpha \in \mathbf{On}} [t^{-\alpha}]x. \quad (25.2)$$

*For each  $x$  only finitely many terms are nonzero.  $\lambda_{\mathbf{On}}$  is a strongly  $\mathbb{R}$ -linear functional, but no  $a \in \mathbf{No}$  satisfies  $\lambda_{\mathbf{On}}(x) = \text{ct}(ax)$  for all  $x$ . Hence the strong  $\mathbb{R}$ -linear map  $x \mapsto \lambda_{\mathbf{On}}(x) \cdot 1$  of  $\mathbf{No}$  has no constant-term adjoint.*

*Proof.* A well-ordered set of exponents meets  $-\mathbf{On}$  finitely, since an infinite intersection would give a strictly decreasing sequence of ordinals. The same applies to the total support of a summable family, and only finitely many members contribute at each of those exponents; this gives strongness. As  $\lambda_{\mathbf{On}}(t^{-\alpha}) = 1$  for every ordinal  $\alpha$ , a representing  $a$  would have  $[t^\alpha]a = 1$  for all  $\alpha$ , a proper-class support. An adjoint evaluated at 1 would be such an  $a$ .  $\square$

The definition is not a proper-class Hahn sum: for each input one first finds the finite subset of its support lying in  $-\mathbf{On}$  and adds finitely many reals; the class only selects coefficients [12]. There is no conflict with [theorem 22.1](#): there the inverse of an isomorphism supplies each needed test element, and no arbitrary strong functional is represented. So the isomorphism theorem extends to  $\mathbf{No}$  while the self-duality of [proposition 25.1](#) and [theorem 25.2](#) does not [12, 13].

## 26 The bottom-gap classification of exact-monomial embeddings

This section and the next three are source 11's, with the embeddings of 12 and 13 as instances. Automorphisms preserving  $\mathbf{Oz}$  fix  $\mathbb{R}$  because  $\mathbb{R}$  is parameter-free definable in the pair (`odg:def:cor:internal`, [theorem 8.2](#)); the definition uses a universally quantified multiplier condition, which a proper embedding need not preserve [11, 12], as the Diophantine report already cautions after `odg:def:thm:autreal`. The existential part survives: by `odg:def:thm:ideal` and `odg:def:thm:homct`, every unital ring homomorphism  $\mathcal{R}_\Gamma \rightarrow \mathcal{R}_\Delta$  maps  $\Pi_\Gamma$  into  $\Pi_\Delta$  and preserves the constant term [11]. Throughout,  $\Gamma$  and  $\Delta$  are nonzero divisible ordered abelian groups (sets, or  $\mathbf{No}$  with set supports), so  $K_\Gamma$  and  $K_\Delta$  are real closed, and  $\tau : \Gamma \hookrightarrow \Delta$  is an ordered additive embedding.

**Definition 26.1** (Bottom gap [11]). The *bottom gap* of  $\tau$  is

$$\mathbf{G}_\tau = \{c \in \Delta : n|c| < h \text{ for every } h \in \tau(\Gamma)^{>0} \text{ and every } n \geq 1\}.$$

Since  $\tau(\Gamma)$  is divisible, this is  $\{c : |c| < h \text{ for every } h \in \tau(\Gamma)^{>0}\}$ .

The word “gap” refers to a convex subgroup of the value group, not to an omitted cut of the field [11].

**Lemma 26.2** (Intrinsic description [11]).  $\mathbf{G}_\tau$  is the largest convex subgroup of  $\Delta$  meeting  $\tau(\Gamma)$  only in 0; it is divisible, and  $\mathbf{G}_\tau = 0$  if and only if  $\tau(\Gamma)^{>0}$  is coinitial in  $\Delta^{>0}$ . The sum  $\tau(\Gamma) + \mathbf{G}_\tau$  is direct and ordered lexicographically with  $\tau(\Gamma)$  outer:  $h < h'$  in  $\tau(\Gamma)$  and  $c, c' \in \mathbf{G}_\tau$  give  $h + c < h' + c'$ . Hence a well-ordered family of well-ordered blocks lying in distinct cosets  $h + \mathbf{G}_\tau$  has well-ordered union.

*Proof.* For  $c, c' \in \mathbf{G}_\tau$  and  $0 < h \in \tau(\Gamma)$ ,  $|c + c'| < h/2 + h/2$ ; symmetry and convexity are clear, and  $0 < c \in \tau(\Gamma)$  fails the condition at  $h = c$ . If  $D$  is convex with  $D \cap \tau(\Gamma) = 0$ , then  $h \leq n|c|$  with  $c \in D$  would put  $h \in D$ , so  $D \subseteq \mathbf{G}_\tau$ . Convex subgroups of divisible groups are divisible. If  $\tau(\Gamma)^{>0}$  is coinitial no  $c > 0$  lies in  $\mathbf{G}_\tau$ ; otherwise some  $c > 0$  lies below every  $h \in \tau(\Gamma)^{>0}$ , hence below every  $h/n$ . For the blocks,  $h' - h > |c - c'|$ ; a nonempty subset of a union of blocks has a least outer coordinate and a least element in that block.  $\square$

**Theorem 26.3** (Bottom-gap classification [11]). *There is a bijection between*

- (i) *strongly additive field embeddings  $J : K_\Gamma \hookrightarrow K_\Delta$  with  $J(t^\gamma) = t^{\tau(\gamma)}$  for all  $\gamma$  and  $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$ ; and*
- (ii) *ordered field embeddings  $\rho : \mathbb{R} \hookrightarrow \mathbb{R}((t^{\mathbf{G}_\tau}))$ ,*

*given by  $\rho = J|_{\mathbb{R}}$  and*

$$J_{\tau, \rho} \left( \sum_{\gamma} r_{\gamma} t^{\gamma} \right) = \sum_{\gamma} \rho(r_{\gamma}) t^{\tau(\gamma)}. \quad (26.1)$$

Every  $J = J_{\tau, \rho}$  satisfies

$$\begin{aligned} v(Jx) &= \tau(v(x)), & \text{lc}(Jx) &= \text{lc}(x), & \text{ct}(Jx) &= \text{ct}(x), \\ J^{-1}(\mathcal{R}_\Delta) &= \mathcal{R}_\Gamma, & J^{-1}(\Pi_\Delta) &= \Pi_\Gamma. \end{aligned} \quad (26.2)$$

Without strongness, every field embedding with  $J(t^\gamma) = t^{\tau(\gamma)}$  and  $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$  still has  $J(\mathbb{R}) \subseteq \mathbb{R}((t^{\mathbb{G}_\tau}))$ .

Here  $\text{ct}(Jx) = \text{ct}(x)$  is an equality of coefficient extractions, not multiplicativity of  $\text{ct}$ .

*Proof. Step 1: residues.* An ordered embedding of  $\mathbb{R}$  into a real Hahn field fixes  $\mathbb{Q}$ ; a rational bound on  $|r|$  bounds  $|J(r)|$ , so  $J(r)$  has nonnegative support, and its constant term lies between all rational bounds of  $r$ . So  $J(r) = r + (\text{a series with positive support})$ , with first exponent 0 and first coefficient  $r$  when  $r \neq 0$ . (An embedding of a real closed field into an ordered field preserves positivity, positive elements being squares; so every field embedding here is ordered.)

*Step 2: the gap is necessary.* For  $\gamma > 0$  and  $r \in \mathbb{R}$ ,  $rt^{-\gamma} \in \Pi_\Gamma$ , so  $J(r)t^{-\tau(\gamma)} \in \Pi_\Delta$  by the existential test. Hence every  $c \in \text{supp } J(r)$  has  $c < \tau(\gamma)$  for all  $\gamma > 0$ ; with  $c \geq 0$  and divisibility,  $c \in \mathbb{G}_\tau$ . No strongness is used. If  $J$  is strong, expanding  $x$  into its summable family of terms gives (26.1), hence uniqueness.

*Step 3: every  $\rho$  is admissible.* For a support  $S$ , the blocks  $\tau(\gamma) + \text{supp } \rho(r_\gamma)$ ,  $\gamma \in S$ , lie in distinct cosets of  $\mathbb{G}_\tau$  in the order of  $S$  (lemma 26.2), so (26.1) is a Hahn series. For a summable family  $(x_j)$ , the outer exponents form a well-ordered set, finitely many  $j$  contribute at each outer exponent, and each block is a finite union of well-ordered supports; so the images are summable, and additivity of  $\rho$  gives strong additivity.

*Step 4: field operations.*  $J(1) = 1$ . Products of terms map to  $\rho(r)t^{\tau(\gamma)}\rho(s)t^{\tau(\eta)} = \rho(rs)t^{\tau(\gamma+\eta)}$ , and the product family is summable (the support principle of section 2); regrouping by Step 3 gives multiplicativity. The first block of  $J(x)$  starts at  $\tau(v(x))$  with coefficient  $\text{lc}(x)$ , which gives injectivity, the order and the valuation formula.

*Step 5: preservation and reflection.* For  $\gamma < 0$  the block at  $\tau(\gamma)$  is negative, for  $\gamma > 0$  positive; the zero block is  $\rho(r_0)$  with constant term  $r_0$ ; blocks cannot cancel. So  $\text{ct}(Jx) = \text{ct}(x)$ . If  $x \in \mathcal{R}_\Gamma$  its zero block is the integer  $r_0$  and all other blocks are negative. Conversely, if  $J(x) \in \mathcal{R}_\Delta$ , every positive block is absent and  $r_0 \in \mathbb{Z}$ , so  $x \in \mathcal{R}_\Gamma$ ; for  $\Pi$  also  $r_0 = 0$ .  $\square$

So these embeddings automatically *reflect* the integer part and  $\Pi$ . The classification is inside the exact-monomial category. It does not cover maps  $t^\gamma \mapsto c_\gamma t^{\tau(\gamma)}(1 + u_\gamma)$  with characters or unit corrections, which already occur among the automorphisms of part I, and it does not assert that omnific-preserving embeddings are strong; beyond the category it gives only the necessary coefficient restriction of Step 2 [11] (questions 32.2 and 32.5).

**Corollary 26.4** (Surjectivity [11]).  *$J_{\tau, \rho}$  is onto if and only if  $\tau$  is onto. Then  $\mathbb{G}_\tau = 0$ ,  $\rho = \text{id}_\mathbb{R}$  and  $J = M_{1, \tau}$ .*

*Proof.* The valuation formula forces  $\tau$  onto. Then  $\mathbb{G}_\tau = 0$  by lemma 26.2, and an ordered embedding  $\mathbb{R} \rightarrow \mathbb{R}$  is the identity; reindexing by  $\tau^{-1}$  inverts  $M_{1, \tau}$ .  $\square$

This agrees with theorem 24.1: an automorphism of the pair that is exact-monomial is  $M_{1, \tau}$ .

## 26.1 Taylor sections and the rigidity boundary

A  $\mathbb{Q}$ -derivation  $d : \mathbb{R} \rightarrow \mathbb{R}$  is additive with  $d(rs) = r d(s) + s d(r)$ ; it need not be continuous. Given a transcendence basis  $B$  of  $\mathbb{R}/\mathbb{Q}$ , arbitrary values  $d(b)$ ,  $b \in B$ , extend uniquely to  $\mathbb{Q}(B)$  and then through the algebraic extension  $\mathbb{R}/\mathbb{Q}(B)$ : if  $p(X) = \sum_j c_j X^j$  is the minimal polynomial of  $a$ , then  $d(a) = -\sum_j d(c_j) a^j / p'(a)$ , with  $p'(a) \neq 0$  in characteristic zero, and the extensions agree on finite subextensions. So for a transcendental  $b$  there is  $d$  with  $d(b) = 1$ , and every such  $d$  vanishes on the real algebraic numbers; the choice uses a transcendence basis of the set  $\mathbb{R}/\mathbb{Q}$  and is not computable [11, 12, 13]. This is also the derivation used in `saut:thm:coeffflow`.

**Lemma 26.5** (Taylor section [11, 12, 13]). *For  $\delta > 0$  in an ordered group and a  $\mathbb{Q}$ -derivation  $d$  of  $\mathbb{R}$ ,*

$$\rho_{d,\delta}(r) = \sum_{n \geq 0} \frac{d^n(r)}{n!} t^{n\delta} \quad (26.3)$$

*is an ordered field embedding of  $\mathbb{R}$  into  $\mathbb{R}((t^{\mathbb{Z}\delta}))$ , with constant term  $r$ ; if  $d(b) = 1$  then  $\rho_{d,\delta}(b) = b + t^\delta$ .*

*Proof.* The iterated Leibniz rule  $d^n(rs) = \sum_j \binom{n}{j} d^j(r) d^{n-j}(s)$  gives multiplicativity coefficient-wise; the constant term  $r$  gives injectivity and the sign;  $d^2(b) = d(1) = 0$ .  $\square$

The ambient version is `saut:thm:coeffflow` of the surcomplex automorphism report, restricted to real coefficients (a derivation of  $\mathbb{R}$  extends to  $\mathbb{C}$  with  $d(i) = 0$ , and then the map preserves **No**):

$$\mathcal{T}_{d,\delta} \left( \sum_{\lambda} r_{\lambda} t^{\lambda} \right) = \sum_{\lambda, n \geq 0} \frac{d^n(r_{\lambda})}{n!} t^{\lambda + n\delta} \quad (26.4)$$

is a strong automorphism of  $K_{\Delta}$  fixing every monomial, preserving valuations and leading coefficients, with inverse  $\mathcal{T}_{-d,\delta}$  and  $v(\mathcal{T}_{d,\delta}(x) - x) \geq v(x) + \delta$ . 11 reproves it and credits the surcomplex report; 12 and 13 use its Taylor ingredient without that credit. On its own it never preserves **Oz** (the remark after [example 4.12](#)): with  $\delta = 1$ ,  $\Delta = \mathbf{No}$  and  $d(b) = 1$  it sends  $\frac{b}{2}\omega = \frac{b}{2}t^{-1}$  to  $\frac{b}{2}\omega + \frac{1}{2} \notin \mathbf{Oz}$  [11]. What changes below is its interaction with the image of a proper exponent embedding.

**Theorem 26.6** (Exact Taylor admissibility [11]). *Let  $\delta > 0$  and let  $d \neq 0$  be a derivation of  $\mathbb{R}$ . The strong exact-monomial embedding  $\mathcal{T}_{d,\delta} \circ M_{1,\tau}$  preserves  $\mathcal{R}_{\Gamma}$  if and only if  $\delta \in \mathbf{G}_{\tau}$ , that is  $n\delta < \tau(\gamma)$  for all  $\gamma > 0$  and  $n \geq 1$ . It is then  $J_{\tau,\rho_{d,\delta}}$  and also reflects  $\mathcal{R}_{\Gamma}$ .*

*Proof.*  $\mathcal{T}_{d,\delta} \circ M_{1,\tau}$  is  $\sum r_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} \rho_{d,\delta}(r_{\gamma}) t^{\tau(\gamma)}$ . If  $\delta \in \mathbf{G}_{\tau}$ , then  $\rho_{d,\delta}$  takes values in  $\mathbb{R}((t^{\mathbf{G}_{\tau}}))$  and [theorem 26.3](#) applies. Conversely, for  $r$  with  $d(r) \neq 0$  the coefficient of  $\rho_{d,\delta}(r)$  at  $\delta$  is nonzero, and Step 2 of that theorem forces  $\delta \in \mathbf{G}_{\tau}$ .  $\square$

**Corollary 26.7** (Coinitiality is exactly coefficient rigidity [11]). *For fixed  $\tau$ , every embedding in [theorem 26.3](#) fixes  $\mathbb{R}$  pointwise if and only if  $\tau(\Gamma)^{>0}$  is coinitial in  $\Delta^{>0}$ . Otherwise some such embedding sends a chosen transcendental real  $b$  to  $b + t^{\delta}$  with  $\delta > 0$ .*

*Proof.* Coinitial means  $\mathbf{G}_{\tau} = 0$ , and an ordered embedding  $\mathbb{R} \rightarrow \mathbb{R}$  is the identity. Otherwise take  $0 < \delta \in \mathbf{G}_{\tau}$  and  $d(b) = 1$  in [theorem 26.6](#).  $\square$

**Example 26.8** (A set-sized model [11]). Let  $\Gamma = \mathbb{Q}$ ,  $\Delta = \mathbb{Q} \oplus \mathbb{Q}$  ordered lexicographically with the first coordinate dominant, and  $\tau(q) = (q, 0)$ . Then  $\mathbf{G}_{\tau} = \{0\} \oplus \mathbb{Q}$  and  $\delta = (0, 1)$  is admissible; for  $q > 0$ ,  $\rho_{d,\delta}(r) t^{(-q,0)} = \sum_n \frac{d^n(r)}{n!} t^{(-q,n)}$  has only negative exponents. No proper class is involved.

## 27 Explicit proper embeddings of $(\mathbf{No}, \mathbf{Oz})$

### 27.1 Two levels of normal-form substitution

At the *inner* level write an exponent  $g \in \mathbf{No}$  in its large- $\omega$  Conway normal form  $g = \sum_{a \in S} r_a \omega^a$ ,  $S$  reverse well ordered. For an order embedding  $\mu : \mathbf{No} \hookrightarrow \mathbf{No}$  put

$$\iota_\mu(g) = \sum_{a \in S} r_a \omega^{\mu(a)}. \quad (27.1)$$

**Lemma 27.1** (Inner lift [11]).  *$\iota_\mu$  is an  $\mathbb{R}$ -linear ordered embedding of  $(\mathbf{No}, +)$  into itself, with the same leading coefficient; its image is the class of normal forms supported in  $\mu(\mathbf{No})$ , and  $\iota_\mu \iota_{\mu'} = \iota_{\mu\mu'}$ .*

*Proof.* An order embedding preserves reverse well-ordering and identifies no exponents; addition is coefficientwise. Reindex a normal form supported in  $\mu(\mathbf{No})$  by  $\mu^{-1}$  for the image.  $\square$

At the *outer* level  $\iota_\mu$  is an exponent embedding, and  $M_{1, \iota_\mu}$  is a strong exact-monomial field embedding fixing  $\mathbb{R}$  (theorem 26.3 with  $\rho = \text{id}$ ).  $\iota_\mu$  is only additive: it is not a field map, and  $\mu$  need not be additive. The three sources use three order isomorphisms:

$$\mu_{11}(a) = \begin{cases} (1-a)^{-1}, & a < 0, \\ a+1, & a \geq 0, \end{cases} \quad \mu_{13}(a) = a + \sqrt{a^2 + 1}, \quad \mu_{12}(a) = 1 + a + \sqrt{a^2 + 1}. \quad (27.2)$$

$\mu_{11}$  maps  $\mathbf{No}$  onto  $\mathbf{No}^{>0}$ , its branches covering  $(0, 1)$  and  $[1, \infty)$ .  $\mu_{13}$  maps  $\mathbf{No}$  onto  $\mathbf{No}^{>0}$  with inverse  $y \mapsto (y - y^{-1})/2$ , since  $(\sqrt{a^2 + 1} + a)(\sqrt{a^2 + 1} - a) = 1$  and  $y \mapsto (y - y^{-1})/2$  is strictly increasing. So  $\mu_{12} = 1 + \mu_{13}$  maps  $\mathbf{No}$  onto  $(1, \infty)$ . They use only the ordered-field operations of  $\mathbf{No}$  [11, 12, 13].

**Lemma 27.2** (The three value images and their gaps). *The images  $\iota_{\mu_{11}}(\mathbf{No}) = \iota_{\mu_{13}}(\mathbf{No})$  are the additive group of purely infinite surreals, the normal forms with all exponents positive;  $\iota_{\mu_{12}}(\mathbf{No})$  consists of the normal forms with all exponents greater than 1. The gaps are*

$$\mathbf{G}_{\iota_{\mu_{11}}} = \mathbf{G}_{\iota_{\mu_{13}}} = C_1 \text{ [11, 13]}, \quad \mathbf{G}_{\iota_{\mu_{12}}} = C_\omega \text{ [merge]},$$

*the finite surreals and the surreals bounded by an ordinary multiple of  $\omega$ . All three images are cofinal in  $\mathbf{No}$  and none of their positive parts is coinital.*

*Proof.* The images follow from lemma 27.1. A positive purely infinite surreal exceeds every real, hence every ordinary multiple of a finite surreal; so  $C_1 \subseteq \mathbf{G}$ . An infinite  $c$  has leading exponent  $a > 0$ , and  $\omega^{a/2}$  is a positive image element below  $|c|$ ; so  $c \notin \mathbf{G}$ . For  $\mu_{12}$ : a positive image element has leading term  $r\omega^a$  with  $r > 0$ ,  $a > 1$ , so it exceeds  $n\omega$  for every  $n$ , and  $C_\omega \subseteq \mathbf{G}$ . If  $c \notin C_\omega$ , its leading exponent  $a$  exceeds 1, and  $\omega^{(1+a)/2}$  is an image element below  $|c|$ . Arbitrarily large exponents give cofinality; no positive image element is at most 1.  $\square$

The purely infinite surreals here form a subgroup of the *value group*; it is the same set as the ideal  $\Pi$  of  $\mathbf{No}$ , but it is used additively and is not a convex subgroup [11, 13].

### 27.2 One map, three instances

**Theorem 27.3** (Proper embeddings moving a real [11, 12, 13]). *Let  $b$  be a transcendental real,  $d$  a derivation of  $\mathbb{R}$  with  $d(b) = 1$ , and  $\mu \in \{\mu_{11}, \mu_{12}, \mu_{13}\}$ . Then*

$$J_{\mu, d}\left(\sum_{\gamma} r_{\gamma} t^{\gamma}\right) = \sum_{\gamma} \sum_{n \geq 0} \frac{d^n(r_{\gamma})}{n!} t^{\iota_{\mu}(\gamma) + n} \quad (27.3)$$

is  $J_{\iota_\mu, \rho_{d,1}} = \mathcal{T}_{d,1} \circ M_{1, \iota_\mu}$ . It is a proper, strongly additive, exact-monomial ordered field embedding  $\mathbf{No} \hookrightarrow \mathbf{No}$  with

$$v(J_{\mu,d}x) = \iota_\mu(v(x)), \quad \text{ct} \circ J_{\mu,d} = \text{ct}, \quad J_{\mu,d}^{-1}(\mathbf{Oz}) = \mathbf{Oz}, \quad J_{\mu,d}(b) = b + \omega^{-1},$$

and it fixes every real algebraic number. Moreover  $J_{\mu,d}(\omega) = \omega^\omega$  for  $\mu = \mu_{11}, \mu_{13}$  and  $J_{\mu_{12},d}(\omega) = \omega^{\omega^2}$ . Its restriction is a proper ring embedding  $\mathbf{Oz} \hookrightarrow \mathbf{Oz}$ ; it is not  $\mathbb{R}$ -linear, and no automorphism of  $(\mathbf{No}, \mathbf{Oz})$  agrees with it on  $b$ .

*Proof.*  $1 \in \mathbf{G}_{\iota_\mu}$  by lemma 27.2, so theorems 26.3 and 26.6 give the embedding, strongness, valuation, constant term and reflection.  $\rho_{d,1}(b) = b + t$  and  $t = \omega^{-1}$ ;  $d$  kills the real algebraic numbers.  $\iota_\mu(1) = \omega^{\mu(0)}$  with  $\mu_{11}(0) = \mu_{13}(0) = 1$  and  $\mu_{12}(0) = 2$ , and  $J(\omega) = J(t^{-1}) = t^{-\iota_\mu(1)}$ . The value  $1 = v(t)$  is not in  $\iota_\mu(\mathbf{No})$ , so  $t \notin J(\mathbf{No})$ ; if the restriction to  $\mathbf{Oz}$  were onto, extension to fractions (odg:thm:fractions) would make  $J$  onto. Every  $\mathbf{Oz}$ -preserving automorphism fixes  $b$  (corollary 5.2), and the restriction to  $\mathbf{Oz}$  determines the fraction-field map.  $\square$

So the rigidity of automorphisms does not pass to embeddings:  $(\mathbf{No}, +, \cdot, \mathbf{Oz})$  and  $\mathbf{Oz}$  have proper self-embeddings, and an ordered-field embedding of  $\mathbf{No}$  preserving and reflecting  $\mathbf{Oz}$  need not preserve  $\mathbb{R}$  setwise, even when it is strongly additive and preserves constant terms exactly [12]. 11 proves the theorem for  $\mu_{11}$  through its classification; 12 (for  $\mu_{12}$ ) and 13 (for  $\mu_{13}$ ) give the same block argument directly. The three maps differ only in the order isomorphism: 11's and 13's have the same value image and send  $\omega \mapsto \omega^\omega$ . For  $\mu_{12}$ ,  $\mathbf{G} = C_\omega$ , so even  $\delta = \omega$  would be admissible (lemma 27.2). For example

$$J_{\mu,d}(bt^g) = bt^{\iota_\mu(g)} + t^{\iota_\mu(g)+1} \quad (g < 0), \quad J_{\mu_{11},d}(b\omega) = b\omega^\omega + \omega^{\omega-1} \in \mathbf{Oz},$$

both terms purely infinite although  $J(b)$  is not real; and  $J(b^2) = b^2 + 2bt + t^2$ ,  $J(b^{-1}) = \sum_n (-1)^n b^{-n-1} t^n$ ,  $J(t^g) = t^{\iota_\mu(g)}$  [11, 12]. The ordinary Taylor section alone is not a map on arbitrary normal forms, since supports could meet its integer shifts; the proper exponent embedding separates the scales [12, 13].

**Corollary 27.4** (Image, reals and a missing test [12]; for all three [merge]). *For each  $J = J_{\mu,d}: J(\mathbf{No}) \cap \mathbb{R} = \ker d$ , and  $\text{ct}(tJ(x)) = 0$  for every  $x \in \mathbf{No}$ , although  $t \neq 0$ . So the image does not supply all constant-term tests.*

*Proof.* If  $J(x)$  is constant, every block at  $g \neq 0$  vanishes, so  $x = r$  is real, and  $\rho_{d,1}(r)$  is constant exactly when  $d(r) = 0$  (see also proposition 29.2). An exponent  $\iota_\mu(g) + n = -1$  would make  $\iota_\mu(g)$  a negative integer; but  $\iota_\mu(g)$  is 0 or of absolute value above every integer (lemma 27.2). So no image has a term at  $-1$ .  $\square$

12 proves the corollary for  $\mu_{12}$ ; the merge notes that its proof uses only that nonzero values of  $\iota_\mu$  are infinite, true for all three. The annihilator is the missing detector in the isomorphism proof: tests from the image cannot distinguish  $t$  from 0, whereas the ambient test  $t^{-1}$  does [12].

**Corollary 27.5** (Failure of pair homogeneity [13]). *The ordered embedding  $\rho_{d,1}: \mathbb{R} \rightarrow \mathbf{No}$  is an embedding of the induced pairs  $(\mathbb{R}, \mathbb{Z}) \rightarrow (\mathbf{No}, \mathbf{Oz})$  that extends to the proper pair embedding  $J_{\mu,d}$  but to no automorphism of  $(\mathbf{No}, \mathbf{Oz})$ . So  $(\mathbf{No}, \mathbf{Oz})$  is not strongly homogeneous for embeddings of set-sized induced substructures.*

*Proof.* Restrict the reflection property of theorem 27.3 to  $\mathbb{R}$ ; every  $\mathbf{Oz}$ -preserving automorphism fixes  $b$ , whereas  $\rho_{d,1}(b) = b + t$ .  $\square$

“Strongly homogeneous” means extension of isomorphisms between set-sized induced substructures, not extension of elementary partial maps in a saturated expansion. The pure ordered field  $\mathbf{No}$  has this extension property for set-sized ordered subfields [9, Remark 2.10], so  $\rho_{d,1}$  does extend to an ordered-field automorphism of  $\mathbf{No}$ , and every such extension fails to preserve  $\mathbf{Oz}$ ; the obstruction lies in the integer part [13]. The map also satisfies  $\text{ct} \circ J = \text{ct}$  but not  $J \circ \text{ct} = \text{ct}$  nor  $J \circ \text{ct} = \text{ct} \circ J$ : for  $b$  these compare  $b$  with  $b + t$ . It violates no hypothesis of [theorem 22.1](#), since it is not onto [13]. It does not commute with the omega-map:  $J(\omega^1) = \omega^\omega$  for  $\mu_{11}$  whereas  $\omega^{J(1)} = \omega$ ; no question of [9] on the omega-map is addressed [11].

## 28 Closed images and the two ends of the value group

Use the valuation topology with basic neighbourhoods  $\mathcal{V}_\lambda = \{x : v(x) > \lambda\}$ . On a real Hahn field with nonzero value group it is the order topology; for  $\mathbf{No}$  it is the fine topology. No countable basis is assumed, and for proper classes the topological statements are local conditions on balls; no object containing all open subclasses is formed [11].

**Theorem 28.1** (Closed images [11]). *Every image  $F = J_{\tau,\rho}(K_\Gamma)$  of [theorem 26.3](#) is closed in  $K_\Delta$ ; if  $J$  is proper,  $F$  is nowhere dense.*

*Proof.* Put  $\mathbf{G} = \mathbf{G}_\tau$  and  $D = \tau(\Gamma) + \mathbf{G}$ ; image series are supported in  $D$ . If  $x$  has a nonzero coefficient at  $\lambda \notin D$ , then  $v(x - y) \leq \lambda$  for all  $y \in F$ . If  $\text{supp } x \subseteq D$ , split  $x = \sum_{h \in S} t^h x_h$  with  $x_h \in \mathbb{R}((t^\mathbf{G}))$  by the unique decomposition; the projection  $D \rightarrow \tau(\Gamma)$  is nondecreasing, so  $S$  is well ordered. Put  $r_h = \text{ct}(x_h)$ . If  $x_h = \rho(r_h)$  for all  $h$ , then  $x = J_{\tau,\rho}(\sum_h r_h t^{\tau^{-1}(h)}) \in F$ . Otherwise pick  $h$  and  $c \in \mathbf{G}$  where  $x_h - \rho(r_h)$  has a nonzero coefficient, and put  $L = \max\{h, h + c\}$ . The block of any  $y \in F$  at  $h$  is  $\rho(s)$  for one  $s \in \mathbb{R}$  (possibly 0); if  $s \neq r_h$  then  $x - y$  has a nonzero coefficient at  $h$ , and if  $s = r_h$  at  $h + c$ . So  $x + \mathcal{V}_L$  misses  $F$ , and  $F$  is closed. A subfield containing a ball  $\mathcal{V}_L$  contains every  $z \neq 0$ , as  $z = (zt^\gamma)/t^\gamma$  with both in  $\mathcal{V}_L$  for large  $\gamma$ ; so a proper subfield has empty interior, and a closed one is nowhere dense.  $\square$

**Lemma 28.2** (Induced value embedding [11]). *Every field embedding  $f : \mathbf{No} \hookrightarrow \mathbf{No}$  is ordered and induces an ordered additive embedding  $\tau_f$  of the natural value group with  $v(f(x)) = \tau_f(v(x))$ . No monomial or strongness hypothesis is needed.*

*Proof.* Positivity is being a nonzero square. Archimedean equivalence and strict domination are expressed by inequalities with ordinary integers, which  $f$  preserves and reflects; multiplicativity gives additivity.  $\square$

**Theorem 28.3** (Continuity or external discreteness [11]). *Let  $f : \mathbf{No} \hookrightarrow \mathbf{No}$  be any field embedding. Exactly one of the following holds.*

- (i)  $\tau_f(\mathbf{No})$  is cofinal in  $\mathbf{No}$ . Then  $f$  is continuous, a homeomorphism onto its image, and the image is nondiscrete.
- (ii)  $\tau_f(\mathbf{No})$  is not cofinal. Then there is  $L$  with  $v(y - z) < L$  for all distinct  $y, z \in f(\mathbf{No})$ ; the image is closed and uniformly discrete, and  $f$  is nowhere continuous.

*Proof.* (i) Given  $L$ , choose  $\gamma$  with  $\tau_f(\gamma) > L$ ; then  $f(\mathcal{V}_\gamma) = f(\mathbf{No}) \cap \mathcal{V}_{\tau_f(\gamma)} \subseteq \mathcal{V}_L$ . (ii) Take  $L$  above  $\tau_f(\mathbf{No})$ : nonzero differences in the image have value below  $L$ , so  $f^{-1}(\mathcal{V}_L) = \{0\}$ , not open, and additivity gives discontinuity everywhere. A ball of radius  $L$  contains at most one image point, which gives closedness and uniform discreteness.  $\square$

There is no contradiction in an order-isomorphic copy of the nondiscrete **No** being discrete: the intrinsic order topology of a non-cofinal ordered subfield can be coarser than its subspace topology; the inverse from the image stays continuous [11].

**Theorem 28.4** (The two ends are independent [11]). *For the order embeddings*

$$\mu^{++}(a) = \begin{cases} a, & a \leq 0, \\ a + 1, & a > 0, \end{cases} \quad \mu^{+-} = \mu_{11},$$

$$\mu^{-+}(a) = -\mu_{11}(-a), \quad \mu^{--}(a) = \frac{\mu_{11}(a)}{1 + \mu_{11}(a)},$$

with ranges  $(-\infty, 0] \cup (1, \infty)$ ,  $(0, \infty)$ ,  $(-\infty, 0)$  and  $(0, 1)$ , the value group  $\iota_\mu(\mathbf{No})$  is respectively cofinal and coinital, cofinal and not coinital, coinital and not cofinal, neither; here “coinital” refers to the positive part. In each case there is a proper strong exact-monomial self-embedding preserving and reflecting **Oz**:  $M_{1,\iota_\mu}$  in the coinital cases, where coefficient motion is impossible, and  $\mathcal{T}_{d,1} \circ M_{1,\iota_\mu}$ , moving  $b$  to  $b + \omega^{-1}$ , in the others. The embeddings are continuous with closed nondiscrete image in the cofinal cases and nowhere continuous with discrete image in the others; all four images are nowhere dense.

*Proof.* By lemma 27.1 the value image is the class of normal forms supported in the range. The range of  $\mu^{++}$  has arbitrarily high and low levels and misses  $(0, 1]$ , so the image is proper, cofinal and coinital.  $\mu^{+-}$  is lemma 27.2. For  $\mu^{-+}$  all values are infinitesimal, so the group is bounded by 1, but monomials at arbitrarily negative levels make it coinital. For  $\mu^{--}$  all nonzero values are purely infinite and below  $\omega$  in absolute value, so the group is neither; its gap is  $C_1$ , since a finite value is dominated by every positive image value and an infinite value with leading level  $a > 0$  dominates  $\omega^c$  for some  $0 < c < \min\{a, 1\}$ . Apply corollary 26.7 and theorems 28.1 and 28.3.  $\square$

So cofinality must not replace coinitality in the rigidity criterion:  $J_{\mu_{11},d}$  is continuous with cofinal values and still moves  $\mathbb{R}$  [11].

## 29 Constants, elementarity, conjugacy and parameters

### 29.1 Elementary reduct, nonelementary expansion

$\mathbb{R}$  is parameter-free definable in  $(K_\Gamma, \mathcal{R}_\Gamma)$  by the existential test, the multiplier condition and its units (`odg:def:thm:ideal`, `odg:def:thm:multiplicier`; the internal form is `odg:def:cor:internal`); 11 reproves this with explicit formulas  $D$ ,  $M$ ,  $\text{Const}$ , and derives the rigidity of automorphisms (corollary 5.2).

**Theorem 29.1** (Logical separation [11]; [12] for its example). *Every field embedding  $\mathbf{No} \hookrightarrow \mathbf{No}$  is elementary as an embedding of ordered fields. An embedding of theorem 26.3 (with  $\Gamma = \Delta = \mathbf{No}$ ) that moves some real is not elementary as an embedding of  $(\mathbf{No}, \mathbf{Oz})$ , although it preserves and reflects **Oz**, and its restriction to **Oz** is not elementary as an embedding of rings.*

*Proof.* The first assertion is quantifier elimination for real closed ordered fields [12], read formula by formula for the class structure. If  $\Phi(b) \notin \mathbb{R}$ , the formula  $\text{Const}(x)$  defining  $\mathbb{R}$  holds of  $b$  and fails of  $\Phi(b)$ . For  $\Phi(b) = b + t$  the new target witness is visible:  $\omega \in \Pi$  but  $(b + t)\omega = b\omega + 1 \notin \Pi$ ; the universal multiplier condition quantifies over more elements in the target. For the ring:  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  (`odg:thm:fractions`), the fraction field is uniformly

interpretable in the ring by pairs modulo  $ad = bc$ , with the ring as the class of fractions  $(z, 1)$ , so an elementary ring embedding would induce an elementary embedding of the interpreted pair; represent  $b$  as  $(b\omega)/\omega$ .  $\square$

The first assertion is classical. The new point is that a strong, exact-monomial, predicate-reflecting embedding can lose elementarity in the omnific expansion; coefficient fixing is not proved sufficient for elementarity ([question 32.6](#)) [11]. 12 records the same failure: the multiplier definition of  $\mathcal{A}_k$  quantifies over all of  $\Pi$ , which an embedding need not preserve. This sharpens the Diophantine report's caution after `odg: def: thm: autreal` that its argument does not apply unchanged to endomorphisms: there are proper ring embeddings  $\mathbf{Oz} \hookrightarrow \mathbf{Oz}$  whose fraction-field extensions move a real.

**Proposition 29.2** (Ordinary overlap [11]; [12] for Taylor sections). *For  $F = J_{\tau, \rho}(K_\Gamma)$ :  $F \cap \mathbb{R}((t^{\mathbb{G}_\tau})) = \rho(\mathbb{R})$  and  $F \cap \mathbb{R} = \{r : \rho(r) = r\}$ , a real closed subfield of  $\mathbb{R}$ ; for  $\rho = \rho_{d, \delta}$  it is  $\ker d$ .*

*Proof.* A nonzero block at  $h \neq 0$  lies in  $h + \mathbb{G}_\tau$ , disjoint from  $\mathbb{G}_\tau$ , so an image in  $\mathbb{R}((t^{\mathbb{G}_\tau}))$  is its zero block  $\rho(r)$ ; if that is a real constant, its standard part gives  $r$ . If  $r$  is algebraic over the fixed field  $k$ , then  $\rho(r)$  is a root of the same polynomial, all real roots lie in  $\mathbb{R}$ , and the residue singles out  $r$ ; so  $k$  is relatively algebraically closed in the real closed  $\mathbb{R}$ , hence real closed.  $\rho_{d, \delta}(r) = r$  exactly when  $d(r) = 0$ .  $\square$

## 29.2 Continuum many copies separated by the omnific integers

**Lemma 29.3** (One-sided infinitesimal neighbourhoods [11]). *Let  $F = J_{\tau, \rho}(K_\Gamma)$  and  $\rho(r) \neq r$ . Every  $y \in F$  with  $\text{st}(y) = r$  has  $\text{sgn}(y - r) = \text{sgn}(\rho(r) - r)$ , and  $\rho(r)$  is such a  $y$ . For  $\rho = \rho_{d, \delta}$  with  $d(r) \neq 0$  the sign is that of  $d(r)$ .*

*Proof.*  $y = J(x)$  with  $x = r + \varepsilon$ ,  $v(\varepsilon) > 0$  (by the valuation and constant-term formulas). The nonzero  $\rho(r) - r$  has value in  $\mathbb{G}_\tau^{>0}$ , while  $v(J\varepsilon) \in \tau(\Gamma)^{>0}$  is larger; so  $y - r = (\rho(r) - r) + J(\varepsilon)$  has the leading term of  $\rho(r) - r$ , which is  $d(r)t^\delta$  for a Taylor section.  $\square$

The sign is invariant under every field automorphism of  $\mathbf{No}$  fixing  $\mathbb{R}$  pointwise, in particular under every  $\mathbf{Oz}$ -preserving one ([corollary 5.2](#)), since these preserve order and standard part.

**Theorem 29.4** (One field-conjugacy class, continuum many pair classes [11]). *There are proper strong exact-monomial embeddings  $J_a : \mathbf{No} \hookrightarrow \mathbf{No}$ ,  $a \in \mathbb{R}$ , with images  $F_a$ , such that*

- (i)  $J_a^{-1}(\mathbf{Oz}) = \mathbf{Oz}$ ;
- (ii) all  $F_a$  have the cofinal value image  $\iota_{\mu_{11}}(\mathbf{No})$  and are closed, nondiscrete and nowhere dense;
- (iii) all  $F_a$  are conjugate under strong valuation-fixing field automorphisms of  $\mathbf{No}$ ;
- (iv) for  $a \neq a'$  no field automorphism of  $\mathbf{No}$  fixing  $\mathbb{R}$  pointwise, in particular none preserving  $\mathbf{Oz}$ , maps  $F_a$  onto  $F_{a'}$ .

*So the pairs  $(F_a, F_a \cap \mathbf{Oz})$  are all isomorphic to  $(\mathbf{No}, \mathbf{Oz})$  but lie in pairwise inequivalent positions.*

*Proof.* Choose distinct  $b_1, b_2$  in a transcendence basis of  $\mathbb{R}/\mathbb{Q}$  and let  $d_a$  be the derivation with  $d_a(b_1) = 1$ ,  $d_a(b_2) = a$  and  $d_a = 0$  on the other basis elements; no independence of named constants is assumed. Put  $J_a = \mathcal{T}_{d_a, 1} \circ M_{1, \iota_{\mu_{11}}}$ . (i) and (ii) are [theorems 27.3, 28.1](#)

and 28.3. For (iii),  $\mathcal{T}_{d_{a'},1}\mathcal{T}_{d_a,1}^{-1}$  maps  $F_a$  onto  $F_{a'}$ ; the derivations need not commute, and only the individual inverses are used. For (iv) let  $a < q < a'$  with  $q \in \mathbb{Q}$  and  $r = b_2 - qb_1$ :  $d_a(r) = a - q < 0 < a' - q = d_{a'}(r)$ , so by lemma 29.3 the points of  $F_a$  with standard part  $r$  lie below  $r$  and those of  $F_{a'}$  above, and both sets are nonempty. Finally  $F_a \cap \mathbf{Oz} = J_a(\mathbf{Oz})$  by reflection.  $\square$

This is a lower bound of continuum on the pair-conjugacy types inside one field-conjugacy class, not a classification or an upper bound; the images have identical value data, so no classification can use the value image and its cuts alone [11]. It partly answers 13's question on the conjugacy of its drift images (question 32.8).

### 29.3 Proper copies fixing prescribed parameters

**Lemma 29.5** (Two-level support control [11]). *For a set  $A \subseteq \mathbf{No}$  let  $B$  be the union of the inner supports of the outer exponents of the members of  $A$ , a set. Every order embedding  $\mu$  fixing  $B$  pointwise has  $M_{1,\iota_\mu}(a) = a$  for all  $a \in A$ .*

*Proof.*  $\iota_\mu$  fixes every outer exponent of every member of  $A$ ; this is the inner/outer distinction of lemma 7.1.  $\square$

**Theorem 29.6** (Parameter-fixed discrete and nondiscrete copies [11]). *For every set  $A \subseteq \mathbf{No}$  there are proper strong exact-monomial embeddings  $f_A, g_A : \mathbf{No} \hookrightarrow \mathbf{No}$  fixing  $A \cup \mathbb{R}$  pointwise with  $f_A^{-1}(\mathbf{Oz}) = g_A^{-1}(\mathbf{Oz}) = \mathbf{Oz}$ . The image of  $f_A$  is closed and uniformly discrete and  $f_A$  is nowhere continuous; the image of  $g_A$  is closed, nondiscrete and nowhere dense and  $g_A$  is a homeomorphism onto it; no field automorphism of  $\mathbf{No}$  maps one image onto the other. There is an explicit  $\Omega > 1$  with  $\Omega^{-1} < |y| < \Omega$  for every  $0 \neq y \in f_A(\mathbf{No})$ .*

*Proof.* Take  $B$  as in lemma 29.5 and  $u$  above  $B$ . The map  $\mu^{\text{cmp}}(x) = x$  for  $x \leq u$ ,  $u + \frac{x-u}{1+x-u}$  for  $x > u$ , is an order isomorphism onto  $(-\infty, u+1)$  fixing  $B$ , so every value  $h$  of  $\iota_{\mu^{\text{cmp}}}$  has  $|h| < \beta = \omega^{u+1}$ ; put  $f_A = M_{1,\iota_{\mu^{\text{cmp}}}}$ . The map  $\mu^{\text{imp}}(x) = x$  for  $x \leq u$ ,  $x+1$  for  $x > u$ , fixes  $B$ , omits  $(u, u+1]$  and has cofinal range; put  $g_A = M_{1,\iota_{\mu^{\text{imp}}}}$ . Theorems 26.3, 28.1 and 28.3 and lemma 29.5 give the properties. A field automorphism of  $\mathbf{No}$  is a homeomorphism (its value map is onto; lemma 28.2 and theorem 28.3) and cannot exchange a discrete and a nondiscrete subspace. For  $\Omega = \omega^\beta$ : a nonzero  $y \in f_A(\mathbf{No})$  has natural value  $h$  with  $|h| < \beta$ , so its leading monomial has exponent below  $\beta$  and  $|y| < \Omega$ ; apply this to  $1/y$ .  $\square$

The annulus bounds are external:  $\Omega$  is not in the image, which is internally isomorphic to  $\mathbf{No}$  and not Archimedean [11].

**Theorem 29.7** (Relative coefficient motion [11]). *Let  $A \subseteq \mathbf{No}$  be a set and  $k_A$  the field generated over  $\mathbb{Q}$  by the real coefficients of the members of  $A$ . If  $b \in \mathbb{R}$  is transcendental over  $k_A$ , there is a proper strong exact-monomial embedding fixing  $A$  pointwise, preserving and reflecting  $\mathbf{Oz}$ , with cofinal value image and  $b \mapsto b + t^\delta$  for some  $\delta > 0$ .*

*Proof.* Choose  $d$  zero on  $k_A$  with  $d(b) = 1$ , through a transcendence basis of  $\mathbb{R}/k_A$  containing  $b$ . With  $B$  as in lemma 29.5, choose  $v$  below  $B$  and  $c < v$ , and let  $\mu^{\text{rel}}(x) = v - (v - c)\frac{v-x}{1+v-x}$  for  $x < v$  and  $x$  for  $x \geq v$ , an order isomorphism onto  $(c, \infty)$  fixing  $B$ . Positive values of  $\iota_{\mu^{\text{rel}}}$  have leading exponent above  $c$ , so  $\delta = \omega^c$  lies in the gap. Put  $\mathcal{T}_{d,\delta} \circ M_{1,\iota_{\mu^{\text{rel}}}}$ ; it fixes  $A$  since the exponent lift fixes each member and  $d$  kills its coefficients. Apply theorem 26.6.  $\square$

An ordered embedding fixing a real field fixes the reals algebraic over it, so one cannot fix every real and move another; the transcendence hypothesis is needed [11].

### 30 Surcomplex and Gaussian versions

Identify  $\mathbf{No}(i) = \mathbb{C}((t^{\mathbf{No}}))$  coefficientwise with conjugation  $\overline{x + iy} = x - iy$ ; the union of the real and imaginary supports is well ordered, and a family of surcomplex series is summable if and only if its real and imaginary families are. The general Gaussian problem is kept separate from two restricted ones: with conjugation named, and with the natural valuation assumed preserved. They must not be merged into an unconditional classification [12, 13].

**Theorem 30.1** (Conjugation-compatible automorphisms [12, 13]). *Every field automorphism  $F$  of  $\mathbf{No}(i)$  with  $F(\mathbf{Oz}[i]) = \mathbf{Oz}[i]$  that commutes with conjugation has the form  $F(x + iy) = \sigma(x) + \varepsilon i \sigma(y)$  with  $\sigma$  an  $\mathbf{Oz}$ -preserving automorphism of  $\mathbf{No}$  and  $\varepsilon = \pm 1$ , and  $F$  and  $F^{-1}$  are strongly additive. Conversely every such map has these properties, so this stabilizer is  $\text{Aut}(\mathbf{Oz}) \times C_2$ ; with  $F(i) = i$  imposed, only  $\varepsilon = 1$  occurs and  $F$  is strongly  $\mathbb{C}$ -linear.*

*Proof.* Commuting with conjugation preserves its fixed field  $\mathbf{No}$ , and  $\mathbf{Oz}[i] \cap \mathbf{No} = \mathbf{Oz}$ .  $F(i)$  is a root of  $X^2 + 1$ . [Theorem 23.2](#) makes  $\sigma$  strong, and summability is coordinatewise. Real lifts commute with conjugation, which makes the product direct.  $\square$

**Theorem 30.2** (Valuation-compatible Gaussian automorphisms [12, 13]). *Let  $F$  be a field automorphism of  $\mathbf{No}(i)$  with  $F(\mathbf{Oz}[i]) = \mathbf{Oz}[i]$  that maps the Hahn valuation ring onto itself (equivalently  $v(Fx) = \tau(v(x))$  for an ordered automorphism  $\tau$  of  $\mathbf{No}$ ; values need not be fixed). Then  $F(\mathbb{C}) = \mathbb{C}$ ; with  $\alpha = F|_{\mathbb{C}}$ ,  $F$  is  $\alpha$ -semilinear,  $\text{ct}(Fx) = \alpha(\text{ct} x)$ , and  $F$  and  $F^{-1}$  are strongly additive. The same holds for the fraction-field extension of a ring automorphism of  $\mathbf{Oz}[i]$  whenever that extension preserves the valuation, and for valuation-compatible isomorphisms between full complex Hahn fields with nonzero exponent groups preserving their Gaussian integer parts [13].*

*Proof.*  $\Pi_{\mathbb{C}} = \bigcap_n n\mathbf{Oz}[i]$  ([lemma 2.3](#)); its multiplier ring is  $\mathbb{C} \oplus \Pi_{\mathbb{C}}$  with units  $\mathbb{C}^{\times}$  (`odg:def:thm:multiplier`), so  $F(\mathbb{C}) = \mathbb{C}$ . Valuation compatibility preserves  $\mathfrak{m}_{\mathbb{C}}$ , and  $\mathbf{No}(i) = \Pi_{\mathbb{C}} \oplus \mathbb{C} \oplus \mathfrak{m}_{\mathbb{C}}$  gives  $\text{ct}(Fx) = \alpha(\text{ct} x)$ . Apply [theorem 22.1](#). The fraction-field statement uses  $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$  (`odg:thm:fractions`); the isomorphism statement uses only the ambient fields.  $\square$

$\alpha$  may be a wild automorphism of  $\mathbb{C}$ : coefficientwise application of any automorphism of  $\mathbb{C}$  is strong and preserves  $\mathbf{Oz}[i]$ , since it fixes  $\mathbb{Q}$  and sends  $i$  to  $\pm i$  [13]. This is consistent with the wild lift of [proposition 10.3](#) and with `odg:def:prop:coeffaut`, which are strong. Neither theorem shows that every automorphism of  $\mathbf{Oz}[i]$  has a valuation-compatible extension; conjugation is not assumed intrinsic, and maps moving the real axis are already in the collection ([theorem 10.2](#), `odg:def:lem:twist`). Since  $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ , the open issue is compatibility, not existence of the extension [12, 13] ([question 32.1](#)).

**Theorem 30.3** (Conjugation-compatible embeddings [11]). *The strong field embeddings  $\Psi : K_{\Gamma}^{\mathbb{C}} \hookrightarrow K_{\Delta}^{\mathbb{C}}$  that commute with conjugation, satisfy  $\Psi(t^{\gamma}) = t^{\tau(\gamma)}$  and preserve  $\mathcal{R}_{\Gamma}^{\mathbb{C}}$  are exactly  $\Psi(x + iy) = J_{\tau, \rho}(x) + \varepsilon i J_{\tau, \rho}(y)$ ,  $\varepsilon = \pm 1$ , with  $(\tau, \rho)$  as in [theorem 26.3](#); each satisfies  $\Psi^{-1}(\mathcal{R}_{\Delta}^{\mathbb{C}}) = \mathcal{R}_{\Gamma}^{\mathbb{C}}$ , and  $\Psi(i) = i$  forces  $\varepsilon = 1$ .*

*Proof.* Commuting with conjugation maps  $K_{\Gamma}$  into  $K_{\Delta}$ , and  $\mathcal{R}_{\Gamma}^{\mathbb{C}} \cap K_{\Gamma} = \mathcal{R}_{\Gamma}$ , so the restriction is classified;  $\Psi(i) = \pm i$ . The converse is coordinatewise, and  $\mathcal{R}^{\mathbb{C}}$ -membership is membership of both coordinates.  $\square$

12's and 13's surcomplex embeddings  $x + iy \mapsto J(x) + iJ(y)$  are the case  $\varepsilon = 1$  of [theorem 27.3](#); they are proper and strong, commute with conjugation, reflect  $\mathbf{Oz}[i]$ , move  $b$  to  $b + t$ , and their image meets  $\mathbb{C}$  in  $\ker d + i \ker d$  [12, 13].

**Proposition 30.4** (Complex constants and nonelementarity [11]).  *$\mathbb{C}$  is parameter-free definable in  $(K_\Gamma^\mathbb{C}, \mathcal{R}_\Gamma^\mathbb{C})$ . An embedding of [theorem 30.3](#) with nontrivial  $\rho$  is elementary as an embedding of algebraically closed fields, but not in the Gaussian expansion; for  $\Gamma = \Delta = \mathbf{No}$  its restriction to  $\mathbf{Oz}[i]$  is not elementary as a ring embedding.*

*Proof.* For  $z \in \mathcal{R}_\Gamma^\mathbb{C}$ ,  $z \in \Pi_\Gamma^\mathbb{C}$  if and only if  $z^2 = 2w^2$  for some  $w \in \mathcal{R}_\Gamma^\mathbb{C}$ , since  $\sqrt{2} \notin \mathbb{Q}(i)$ ; the multiplier ring and its units define  $\mathbb{C}$  as in the real case (`odg:def:cor:internal`). A moved real  $b$  has  $\rho(b) \notin \mathbb{C}$ . Pure-field elementarity is quantifier elimination for algebraically closed fields [12]; for the ring use  $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$  and the interpreted fraction field as in [theorem 29.1](#).  $\square$

**Corollary 30.5** (Transferred copies [11]). *The explicit coefficient-moving embedding, the four topological cases and the parameter-fixed copies extend to  $\mathbf{No}(i)$ , preserving and reflecting  $\mathbf{Oz}[i]$ ; parameters in a set  $A \subseteq \mathbf{No}(i)$  are handled through their real and imaginary parts. The family of [theorem 29.4](#) gives field-conjugate surcomplex images that are pairwise nonconjugate under automorphisms preserving both conjugation and  $\mathbf{Oz}[i]$ .*

*Proof.* Use  $\varepsilon = 1$ ; the valuation of  $x + iy$  is the minimum of the coordinate valuations, so the topology is the product topology. An automorphism preserving conjugation and  $\mathbf{Oz}[i]$  restricts to an  $\mathbf{Oz}$ -preserving automorphism of  $\mathbf{No}$ , and each extended image meets  $\mathbf{No}$  in the real image; apply [theorem 29.4](#).  $\square$

No nonconjugacy is claimed for automorphisms preserving  $\mathbf{Oz}[i]$  but not conjugation, which need not fix the real axis pointwise, and non-conjugation-compatible surcomplex embeddings are not classified [11].

## 31 Boundaries of the method

**Proposition 31.1** (Regular cardinal support bounds [12]). *Let  $\kappa$  be a regular uncountable cardinal,  $\Gamma$  a set, and  $K_{<\kappa}$  the field of series in  $k((t^\Gamma))$  with support of size  $< \kappa$ , where summability also requires total support of size  $< \kappa$ . A family in  $K_{<\kappa}$  is summable there if and only if fewer than  $\kappa$  members are nonzero and for every  $y \in K_{<\kappa}$  only finitely many  $\text{ct}(x_j y)$  are nonzero. Consequently a constant-term-covariant field isomorphism between two such fields is strong for this restricted summation.*

*Proof.* A summable family has fewer than  $\kappa$  nonzero members, since each contributes at an exponent of a total support of size  $< \kappa$  and each exponent receives finitely many; pairing finiteness is [theorem 21.4](#). Conversely, regularity bounds the union of fewer than  $\kappa$  supports of size  $< \kappa$ ; monomials test pointwise finiteness, and the countable witness of [lemma 21.2](#) lies in  $K_{<\kappa}$  because  $\kappa$  is uncountable. An isomorphism preserves the number of nonzero members, and the proof of [theorem 22.1](#) carries over.  $\square$

The cardinal condition cannot be dropped: a strictly increasing  $(g_\alpha)_{\alpha < \kappa}$  in  $\Gamma$  gives  $(t^{g_\alpha})$  with finite pairings against every  $y \in K_{<\kappa}$  ([lemma 21.1](#)) but total support of size  $\kappa$  [12]. For singular  $\kappa$  the union argument fails, and no extension is claimed; the proposition identifies no birthday cutoff with a support-bounded Hahn field [12]. 13 records the regular case only inside its singular-cardinal question ([question 32.12](#)).

**Example 31.2** (Small integer parts [13]). The proof of  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  uses an upper bound in  $\mathbf{No}$  for every set of exponents, and does not transfer to set-sized groups. For  $\Gamma = \mathbb{Z}$  and  $u = t^{-1}$ ,  $\mathcal{R}_\mathbb{Z} = \mathbb{Z} \oplus u\mathbb{R}[u]$  and  $\text{Frac } \mathcal{R}_\mathbb{Z} = \mathbb{R}(u) \neq \mathbb{R}((t))$ :  $ru/u = r$  gives  $\mathbb{R}(u)$ , and

$f = \sum_{n \geq 1} t^{n!}$  is not rational, since a rational power series has coefficients obeying an eventual linear recurrence, and arbitrarily long zero blocks followed by a 1 contradict one.

This is the phenomenon of [proposition 18.1](#) and of [11, Proposition 2.4.5]; it is why [corollary 23.4](#) assumes an ambient isomorphism [11, 12, 13].

*Remark 31.3* (Distinctions kept [12, 13]). The detector must lie in the field tested: automatic for full Hahn fields and for the countable witness in  $K_{<\kappa}$ , not for arbitrary subfields. Strongness is not  $\mathbb{R}$ -linearity:  $J_{\mu,d}$  is strong but not  $\mathbb{R}$ -linear, so the adjoint theorem, which is for  $k$ -linear maps, does not apply to it by treating its coefficient lift as scalar multiplication. The retraction  $\text{ret}_\phi$  is linear, never a field retraction for a proper extension. The adjoint theorem needs a set of exponents; automatic strongness of isomorphisms does not. No result of this part concerns the Gonshor exponential, logarithm, the Berarducci–Mantova derivation, the omega-map, birthdays or simplicity; the square roots in (27.2) are ordered-field operations reindexing an additive group. No proper-class Zorn lemma and no basis of the class **No** is used; the only bases chosen are a transcendence basis and a Hamel basis of the set  $\mathbb{R}$  over  $\mathbb{Q}$  [13].

## 32 Status of the questions, and questions of this part

### 32.1 Answered and re-scoped questions

- [Question 11.1](#) is *answered*. Every ring automorphism of **Oz** extends to a strong automorphism of **No** ([theorem 23.2](#)), so preservation of **Oz** forces strongness and there is no nonstrong **Oz**-preserving example; 09’s characterization of the nonstrong pair automorphisms is vacuous, and 10’s question whether every **Oz**-preserving 1-automorphism is strong has the answer yes. The analogous question for **Oz**[ $i$ ] has a positive answer under conjugation or valuation compatibility and is open otherwise ([question 32.1](#)).
- `osq:q:invisible` of `docs/surreal/set-sized-quotients-of-omnific-integers`: its second clause is *answered* by [corollary 24.4](#); the nonstrong case left open in [section 11.1](#) does not exist. Its first clause, on class quotient domains, is untouched. The quotient report is not edited by this merge.
- [Question 11.2](#): unchanged. 12 and 13 ask it again, in the forms “for which support-controlled families of logarithmic characters does  $a_\delta(C_\delta) = 0$  imply two-sided substitution admissibility?” and “give necessary and sufficient support conditions on additive  $\gamma \mapsto L_\gamma \in \mathfrak{m}$  for  $t^\gamma \mapsto t^\gamma \exp L_\gamma$  to extend to a strong automorphism, assuming  $a_\delta(C_\delta) = 0$ ”, suggesting common-shift and finite-rank triangular classes. Automatic strongness settles the recognition of existing automorphisms, not their construction.
- [Question 11.4](#) now concerns every automorphism of **(No, Oz)** with trivial monomial factor, not only the strong ones; it is otherwise unchanged.
- [Question 11.7](#) is related to 12’s question on more intrinsic scalar tests, merged into [question 32.13](#).
- Scope of the answers: **Oz**, the full real Hahn pairs, and **Oz**[ $i$ ] with conjugation or valuation data. For general characteristic-zero  $k$  and unital  $\mathfrak{o}$  the non-claims of 04, 09 and 10 that strongness is not shown ([section B](#)) still apply ([question 32.13](#)).

The novelty statements of 11, 12 and 13 were checked against the tree at the merge. No report proves automatic strongness, the scalar detection of summability, the adjoint theorems, or coefficient-moving embeddings of **(No, Oz)**; their statements about this report and the Diophantine report were accurate at their pins. What they reprove (reconstruction, the

criterion, the strong factorization, the Taylor automorphism) is credited above. Priority remains provisional, as each source says.

## 32.2 Questions

Each question is a proposed research direction, not a published open problem [11, 12, 13]. Duplicates across the three sources are merged, with every source named.

**Question 32.1** (The unrestricted Gaussian stabilizer [12, 13]). Does every ring automorphism of  $\mathbf{Oz}[i]$  have a valuation-compatible extension to  $\mathbf{No}(i)$ , that is, is  $\mathfrak{m}_{\mathbb{C}}$  invariant? Can the positive-support component be recovered from  $\mathbf{Oz}[i]$  and its fraction field without naming conjugation or a valuation?

$\mathbb{C}$  and  $\Pi_{\mathbb{C}}$  are already intrinsic, so the missing object is  $\mathfrak{m}_{\mathbb{C}}$ ; a positive answer with [theorem 30.2](#) would give automatic strongness for all automorphisms of  $\mathbf{Oz}[i]$ . A negative answer should exhibit an automorphism of the ring violating a valuation inequality, not merely moving the real axis or being an unrelated automorphism of the algebraically closed fraction field [12, 13].

**Question 32.2** (Strongness of omnific embeddings [11, 12, 13]). Let  $f : \mathbf{No} \hookrightarrow \mathbf{No}$  be a field embedding with  $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$ . Must  $f$  be strongly additive? What changes if also  $\text{ct} \circ f = \text{ct}$ ? Without strongness, which coefficient and image restrictions survive; can such an  $f$  have a proper dense image, or one that is neither closed nor discrete?

The embeddings of [section 27](#) are strong and do not answer the first clause; a counterexample needs a summable family whose image is not summable or has the wrong sum, and the missing step in the isomorphism proof is access to all target tests ([corollary 27.4](#)) [12, 13]. *Status.* The first two clauses have a negative answer if there is a measurable cardinal, more generally an elementary embedding  $j : V \rightarrow M$  into a transitive class with a critical point, available as a class. The large-cardinal report ([docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms](#)) shows that the map  $\hat{j}$  obtained by applying  $j$  to sign sequences is an ordered field embedding of  $\mathbf{No}$  that fixes  $\mathbb{R}$ , preserves and reflects  $\mathbf{Oz}$  and preserves constant terms, while it sends a summable family to a summable family whose sum is not the image of the sum; so it is not strongly additive. It is not onto, so it does not conflict with [theorem 23.2](#). Consequently a positive answer is not provable in the base theory unless that theory with a measurable cardinal is inconsistent; whether a counterexample exists without large cardinals stays open. The necessary gap restriction of [theorem 26.3](#) and the dichotomy of [theorem 28.3](#) remain valid without strongness, whereas the closed-image proof uses the strong expansion formula. The questions on dense images and on images neither closed nor discrete are not settled here [11].

**Question 32.3** (Coefficient-fixing embeddings and target tests [13]). For  $\mathbb{R}$ -linear embeddings preserving the omnific predicate, can every target test  $\text{ct}(y \cdot)$  be represented, after pullback, by a source multiplier?

Such adjoints suffice for strongness and are necessary at set size ([theorem 25.2](#)); the functional  $\lambda_{\mathbf{On}}$  shows that a proof for  $\mathbf{No}$  must use the multiplicativity of the embedding, not a theorem about all strong linear operators [13].

**Question 32.4** (Coefficient-moving embeddings [12]; re-scoped). Which subfields  $C_0 \subseteq \mathbb{R}$  occur as  $f(\mathbf{No}) \cap \mathbb{R}$  for strongly additive embeddings  $f$  with  $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$ ? Which arise from  $\ker d$ , and are higher-derivation coefficient lifts needed?

*Partly answered* by 11: in the exact-monomial strong class the intersections are the fixed fields of the coefficient sections  $\rho$ , which are real closed, and Taylor sections give  $\ker d$  (proposition 29.2). Which real closed subfields occur, and the classes with characters, unit corrections or no monomial condition, stay open. A classification should separate the coefficient homomorphism, the value-group embedding and the admissibility of their combination; block separation is sufficient, not necessary [12].

**Question 32.5** (Removing exact monomial preservation [11]). Classify the strong embeddings  $f : \mathbf{No} \hookrightarrow \mathbf{No}$  preserving  $\mathbf{Oz}$  with  $f(t^\gamma) = c_\gamma t^{\tau(\gamma)}(1 + u_\gamma)$ ,  $c_\gamma > 0$ ,  $v(u_\gamma) > 0$ . Which compatibility and summability conditions are necessary and sufficient, and can every such embedding be normalized by an  $\mathbf{Oz}$ -preserving automorphism?

The criterion of part I is a starting point, but surjectivity is unavailable: corrections on the image need not extend to all target scales. A first step would keep the corrections on one common positive shift set [11].

**Question 32.6** (Elementarity of coefficient-fixing pair embeddings [11]). For an ordered additive  $\tau : \mathbf{No} \hookrightarrow \mathbf{No}$ , when is  $M_{1,\tau} : (\mathbf{No}, \mathbf{Oz}) \hookrightarrow (\mathbf{No}, \mathbf{Oz})$  elementary? Is fixing  $\mathbb{R}$  sufficient in the exact-monomial strong category?

Theorem 29.1 gives the necessary condition  $\rho = \text{id}$ ; quantifier elimination for real closed fields cannot decide the expanded structure [11].

**Question 32.7** (Coefficient sections [11]). Describe the ordered sections  $\rho : \mathbb{R} \hookrightarrow \mathbb{R}((t^C))$  with residue the identity. Which come from one Taylor derivation, compositions of such, or summable families of higher derivations, and what distinguishes sections with the same fixed subfield?

The classification reduces the coefficient part of the embedding problem exactly to this question; the one-sided neighbourhood invariant is only a first invariant [11].

**Question 32.8** (Conjugacy of proper copies [11, 13]). Classify the images  $J_{\tau,\rho}(\mathbf{No})$  up to  $\mathbf{Oz}$ -preserving automorphisms of  $\mathbf{No}$ . How much do the value image, its cuts, the fixed coefficient field and the one-sided neighbourhood data determine? When are two images  $J_{\mu,d}(\mathbf{No})$  conjugate, and which scale embeddings admit Taylor deformations with several independently prescribed motions?

*Partly answered* by theorem 29.4: the value image alone is insufficient, and continuum many types occur in one field-conjugacy class. 13 notes that since every omnific automorphism fixes  $\mathbb{R}$ , the positions of the deformed coefficient fields may give finer invariants; a restricted target is  $\mathcal{T}_{d,\delta} \circ M_{1,\tau}$  for fixed  $\tau$  and one shift [11, 13].

**Question 32.9** (Homogeneity of the pair [12, 13]). Which isomorphisms between set-sized substructures of  $(\mathbf{No}, \mathbf{Oz})$  extend to automorphisms of the pair?

Fixing the reals of the domain is necessary (corollary 27.5); no sufficiency theorem is proved. The question differs from extension of elementary partial maps in a first-order language. A first target is a back-and-forth theorem for truncation-closed Hahn subfields with prescribed real coefficients and compatible integer parts [12, 13].

**Question 32.10** (Surcomplex embeddings without a real form [11]). Classify the embeddings of  $(\mathbf{No}(i), \mathbf{Oz}[i])$  without assuming commutation with conjugation, and decide which of the nonconjugate copies of corollary 30.5 stay nonconjugate under the larger group.

$\mathbf{Oz}[i]$  reconstructs  $\mathbb{C}$  but gives no pointwise real rigidity for the sign invariant [11].

**Question 32.11** (The proper-class strong dual [12]). Can the strong  $\mathbb{R}$ -linear dual of **No** be described by class coefficient functionals meeting every well-ordered set finitely, and which maps have transposes that stay set-supported on every argument?

Such a description must avoid unlicensed collections of classes; the issue is foundational as well as algebraic [12].

**Question 32.12** (Singular support bounds [12, 13]). For singular uncountable  $\kappa$ , when does a coefficient- and constant-term-compatible automorphism of  $K_{<\kappa}$  preserve summability with total support of size  $< \kappa$ ?

Countable binary witnesses lie in  $K_{<\kappa}$  and detect well-ordering and local finiteness, but not the cardinal bound on unions; a sharp theorem may need uniform bounds on supports of monomial images, and a counterexample must preserve the restricted fields themselves [12, 13].

**Question 32.13** (Other rings and coefficient fields [12]; [merge]). For which integer parts, truncation subrings or distinguished subrings of a real closed Hahn field can one reconstruct a scalar pairing that forces automatic strongness of their stabilizers? In particular, for which characteristic-zero  $k$  and unital  $\mathfrak{o} \subseteq k$  does  $\sigma(\mathcal{R}_{\mathfrak{o}}) = \mathcal{R}_{\mathfrak{o}}$  force a field automorphism of  $k((t^{\Gamma}))$  to be strong?

The omnific proof uses a divisible ideal, its multiplier ring and units to recover coefficients, then the order to recover the infinitesimals; it would be useful to find the weakest hypotheses under which this works and examples of failure [12]. The merge adds the second sentence: for  $\mathfrak{o} = k$  the divisible-ideal step fails, and for non-real  $k$  without valuation data the order step is missing; compare [question 11.7](#).

**Question 32.14** (Isomorphisms of small integer parts [13]). For which ordered groups  $\Gamma, \Delta$  does every ring isomorphism  $\mathcal{R}_{\Gamma} \cong \mathcal{R}_{\Delta}$  extend to an isomorphism of the full real Hahn fields preserving order and integer parts?

[Example 31.2](#) shows the fraction field is not the Hahn field in general. One route is an intrinsic summation completion compatible with the pairing; another is to find two Hahn fields with indistinguishable small integer parts [13].

**Question 32.15** (Exponential, omega-map and differential structure [11, 12, 13]). Which proper embeddings preserving **Oz**, and which gap sections and value embeddings, are compatible with the surreal exponential, the omega-map, a surreal derivation or a simplicity relation? Intersect the full omnific stabilizer with the automorphisms preserving such a structure.

The Taylor block embeddings give flexibility only in the algebraic pair and fail omega-map compatibility at 1; the questions must be compared with those of [9], not inferred from the construction, and the exponential rigidity already in the collection is not to be re-solved [11, 12, 13].

**Question 32.16** (Restricted workspaces [11]). For a support-restricted subfield of a full Hahn field, when does the bottom-gap extension stay inside it, and which restrictions make the canonical integer part have the whole workspace as fraction field?

The block construction preserves well-ordering but not bounds on support cardinality, order type or boundedness; this matters for birthday cutoffs, computable presentations and transseries subfields [11].

**Question 32.17** (Compositions [11]). Describe composition and image inclusion of classified maps in terms of the data  $(\tau, \rho)$ ; the composite of  $(\tau_1, \rho_1)$  and  $(\tau_2, \rho_2)$  has exponent map  $\tau_2\tau_1$  and section  $J_{\tau_2, \rho_2} \circ \rho_1$ . Can image inclusion and ambient extendability be decided from smaller invariants?

**Question 32.18** (Formalization [11, 12, 13]). Formalize the finite-intersection lemma, the triangular or binary detector, the detection theorem and the constant-term isomorphism theorem for arbitrary full Hahn fields; then the multiplier reconstruction and the omnific specialization; and, separately, the bottom-gap classification for set-sized Hahn fields, the closed-image proof and the two-level index lift, isolating the class-size assumptions for **No**.

The sources propose this order; finite checks remain a separate test layer. Nothing in this part is formalized in Lean [11, 12, 13].

## A Provenance ledger

### A.1 Section by section

| Section | Sources and choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Structure from 09; imported results from all three automorphism sources; the Fact 4.2 recheck is the merge's (remark 1.2). 10's provenance and symbols were added with it (batch 28); item (8) of section 1.1 and the rows and paragraph for 11, 12 and 13 in section 1.2 were added in batch 30.                                                                                                                                                                                                                                                                                                                                                             |
| 2       | 04's one-shift lemma and coefficient polynomial; 09's and 04's ring decompositions; 04's and 08's divisible part. Units and constant products are cited from the Diophantine report, not reprinted. 10's one-shift lemma is lemma 2.1.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3       | The criterion is printed once. <i>Choice:</i> 09's generality ( $k$ of characteristic zero, unital $\mathfrak{o}$ ) with 04's import-free proof; the first implication is 09's, which removes 04's use of $\bigcap n\mathfrak{o} = 0$ . 09's derivation form is printed as theorem 3.4; its operator form, which uses cited theorem 1.1, as corollary 3.5. Example 3.2 is from the placement dossier's check. Section 3.3 is 10's: its support-cut form printed as further equivalent conditions, its least-forbidden-shift proof as a marked second route, necessity without strongness, and Archimedean blocks.                                             |
| 4       | Shears from 04 and 09 (after <code>saut:thm:shiftflow</code> ); exact displacement from 04 (08 and 09 have the same computation); 09's uniform shifts; 04's common-shift correspondence as a second, proved route; 08's divisibility-free threshold as a marked case; the translation example from 04, 08, 09 and the surcomplex report; rank dichotomy from 04 and 09 (08's for single flows); bottom subgroup from 04; phase shears from 08. Section 4.7 is 10's: twists with a general profile, the binomial family, their fixed fields, finite orbits and algebraic independence; 10's rank witness and divisibility example are recorded in remark 4.23. |

| Section                 | Sources and choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5                       | Factorization from 04 and 09 (credited to [10] and <code>saut:thm:decomp</code> ); all pair automorphisms from 08 and 09 through <code>odg:def:thm:autreal</code> . 10 re-derives both.                                                                                                                                                                                                                                                                                                                                        |
| 6                       | Bracket law from 04, 08, 09; filtration, rank bound, balanced brackets, mixed-parameter lemma and sharpness from 09; 04's commutator formula and 08's faithful $\mathbb{Z} \wr \mathbb{Z}$ printed as one group in 04's normalization; the all-times remark is the merge's. <a href="#">Example 6.13</a> is 10's.                                                                                                                                                                                                              |
| 7                       | Fixed field $\mathbb{R}$ from 04, 08, 09 (explicit choice of 08, 09); separation, hull, monomial parameters and closure from 04; $\mathbb{R}^\kappa$ and nonsolvability from 09. <a href="#">Section 7.4</a> is 10's.                                                                                                                                                                                                                                                                                                          |
| 8                       | Reconstruction and definitions of $\mathbb{Z}$ and $\Pi$ from 04, 08, 09 are cited to the Diophantine report; printed extras: 08's order formula, the projections (09), floor (08), truncation and the set-sized version (09).                                                                                                                                                                                                                                                                                                 |
| 9                       | Set-parameter nondefinability from 04, 08, 09 printed once with 04's strengthening; representative rules from 09; twists, one-term class and RV from 04; contrasts added by the merge. <a href="#">Section 9.2</a> is 10's; the outer bound in <a href="#">corollary 9.11</a> and the remark on <code>cor:question54</code> are the merge's.                                                                                                                                                                                   |
| 10                      | Complex criterion and conjugation-compatible witnesses from 09 and 04; real axis inside the kernel from 04; wild lift and Gaussian fixed field from 08 and 09 (09's definition of $\mathbb{C}$ is cited to the Diophantine report); <a href="#">proposition 10.5</a> from 10.                                                                                                                                                                                                                                                  |
| 11                      | Status records and questions: 04's four, 08's first and third, 09's nonstrong question (merged with 04's first), and the merge's non-divisibility question. 10's four questions: nonstrong maps and generation are merged into <a href="#">questions 11.1</a> and <a href="#">11.4</a> , definable closure is recorded under <a href="#">question 11.3</a> , and other integer parts is <a href="#">question 11.7</a> ; the item on <code>dsn:q:languages</code> and the correction of 10's novelty statement are the merge's. |
| <a href="#">part II</a> | Manuscript 02 throughout, with credits to <code>bst:thm:fieldcriterion</code> , <code>bst:prop:localdensity</code> , <code>osq:prop:classder</code> and <code>odg:thm:fractions</code> ; <a href="#">section 19</a> is the merge's.                                                                                                                                                                                                                                                                                            |

| Section  | Sources and choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part III | <p>Manuscripts 11, 12 and 13 (batch 30). 20: the merge's. 21–24: 12's spine with 13's binary witness (second route), set-pair isomorphisms, coefficient matrix and <a href="#">example 23.6</a>; the reconstruction is cited, not reprinted; the convex criterion reproved by 12 and 13 is <a href="#">theorems 3.3</a> and <a href="#">3.8</a>; <a href="#">corollary 24.4</a> is the merge's. 25: 12's and 13's adjoint theory, 12's general retraction and tower law, 13's kernel statement. 26–29: 11's, with the embeddings of 12 and 13 as the instances of <a href="#">theorem 27.3</a>, 12's image corollary (extended to all three by the merge), 13's non-homogeneity; the Taylor automorphism is <code>saut:thm:coeffflow</code> (credit added for 12 and 13); the gap <math>C_\omega</math> in <a href="#">lemma 27.2</a> is the merge's. 30: 12 and 13 for automorphisms, 11 for embeddings. 31: 12's regular cardinals, 13's small integer parts, both sources' distinctions. 32: the thirty questions of the three sources (ten each) merged into eighteen, with 12's and 13's admissibility question recorded under <a href="#">question 11.2</a>; the status records and the note on the large-cardinal report are the merge's.</p> |

## A.2 Printed once, and where the merge chose

- *The classification* (04 thm. criterion, 08 thm. threshold, 09 thms. derivation and stabilizer): [theorem 3.3](#) with 04's proof, [theorem 3.4](#), [corollary 3.5](#), and 08's [theorem 4.10](#) as the divisibility-free single-flow case. 04's  $a_{\sigma,\delta}$  and 09's  $c_\delta$  are kept as different functionals with equivalent conditions ([example 3.2](#)).
- *Rank triviality* (04, 09; 08 for single flows): [theorem 4.13](#). *Common fixed field*  $\mathbb{R}$  (04, 08, 09): [theorem 7.2](#). *Set-parameter nondefinability* (04, 08, 09): [theorem 9.1](#).
- *The metabelian non-nilpotent group*: 04's  $\langle A_s, B_u \rangle$  and 08's  $\langle A_1, B_1 \rangle$  are the same group at  $s = u = -1$ ; printed once as [theorems 6.8](#) and [6.10](#).
- *Strong factorization* (04, 09): [theorem 5.1](#).
- *Not reprinted* (cited to the Diophantine report): units and constant products, Pell rigidity,  $\text{Frac } \mathbf{Oz} = \mathbf{No}$ , the multiplier ring, the reconstruction of  $\mathbb{R}$  and of the order, automorphisms fix  $\mathbb{R}$ , the split restriction to  $\text{Aut } \mathbb{C}$ , the first-order and existential definitions of  $\mathbb{Z}$ ,  $\Pi$  and  $\mathbb{C}$ .
- *Kept as second routes*: 04's common-shift correspondence ([theorem 4.8](#)) beside the imported one; 04's separation lemma beside the explicit choice in [theorem 7.2](#); 09's Pell-divisibility definition of  $\Pi_{\mathbb{C}}$  is mentioned beside the Diophantine report's.
- *Added by the merge* (tagged [merge], complete proofs): the combined statement of [theorem 3.3](#); [example 3.2](#); [remark 6.11](#); [question 11.6](#); with source 10, the converse in [theorem 4.18](#) and the outer bound in [corollary 9.11](#).
- *Source 10, printed once* (batch 28): its criterion (10 thm. criterion, cor. logcut) as [theorem 3.3](#) with the further equivalent conditions of [theorem 3.8](#); its rank theorem as [theorem 4.13](#); its stabilizer splitting as [theorem 5.1](#); its fixed field for  $F = \exp \xi$  as [theorem 4.2](#); its parameter theorem as [theorem 9.1](#) (with the two-term witness added as

theorem 9.8); monomials and the omega-map in its nondefinability theorem as theorem 9.1 and corollary 9.2; its inner-support bound as a consequence of theorem 7.4; its Gaussian theorem as theorem 10.1 with the extras of proposition 10.5; its nonabelian example beside theorem 6.8; its one-shift lemma as lemma 2.1; its floor, fraction-field and coefficient-rigidity statements as corollary 5.2, `odg:thm:fractions`, `odg:def:thm:multiplier` and `odg:def:thm:autreal`, not reprinted.

- *Source 10, kept as a second route*: its least-forbidden-shift proof of necessity (theorem 3.8), beside 04's polynomial argument; its direct proof of the inner-support bound (corollary 9.11).
- *Sources 12 and 13, printed once* (batch 30): scalar detection (12 thm. 3.3, 13 thm. 3.3) as theorem 21.4; the constant-term isomorphism theorem (12 thm. 4.1, 13 thm. 4.1) as theorem 22.1, with 12's weaker hypothesis; automatic strongness of  $\text{Aut}(\mathbf{Oz})$  (12 thm. 5.5, 13 thm. 6.1) as theorem 23.2; the set pairs (12 cor. 5.6, 13 prop. 5.5 and cor. 6.2) as proposition 23.1 and corollary 23.4; monomial determination (12 cor. 5.7, 13 cor. 6.3); the factorization (12 thms. 6.1–6.2, 13 thms. 7.1–7.2) as theorem 24.1; scalar representability and adjoints (12 prop. 7.2, thm. 7.3; 13 thms. 8.1, 8.3); the retraction (12 thm. 7.4, 13 prop. 8.4) as theorem 25.3; monomial adjoints (12 ex. 7.5, 13 ex. 8.5); the class functional (12 thm. 8.1, 13 ex. 11.2); the surcomplex automorphism theorems (12 thms. 10.1–10.2, 13 thms. 10.1–10.2). Their reprinted reconstruction (12 lems. 5.1–5.2 and lem. 5.4, prop. 5.3; 13 lems. 5.1–5.2 and prop. 5.3) is cited from the Diophantine report and corollary 5.2, and their reproofs of the criterion are the two routes of theorems 3.3 and 3.8.
- *Sources 11, 12 and 13, printed once*: the embedding moving a real (11 thm. 6.3, 12 thm. 9.5, 13 thm. 9.5) as theorem 27.3, an instance of 11's classification;  $J(\mathbf{No}) \cap \mathbb{R} = \ker d$  (12, 11 prop. 8.5) as corollary 27.4 and proposition 29.2; nonelementarity (12 rem. 9.8, 11 thm. 8.3) as theorem 29.1; the surcomplex embeddings (12 prop. 10.3, 13 cor. 10.4) as the case  $\varepsilon = 1$  of theorem 30.3. 11's reproof of the Taylor automorphism is `saut:thm:coeffflow`; its constants and automorphism rigidity (11 prop. 8.1, cor. 8.2) are `odg:def:thm:multiplier`, `odg:def:cor:internal` and corollary 5.2.
- *Kept as a second route* (batch 30): 13's binary finite-row proof of detection beside 12's triangular detector (theorem 21.4).
- *Added by the merge* (batch 30): corollary 24.4, the gap  $C_\omega$  for 12's exponent embedding (lemma 27.2), the extension of corollary 27.4 to all three maps, and the second sentence of question 32.13.

### A.3 Corrections and stale statements

1. *Divisibility* (09): its remark that divisibility is essential is correct for  $\mathcal{A}_k$  and not for  $\mathcal{R}_{\mathbb{Z}}$ : the translation does not preserve  $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$  (remark 4.11).
2. *Fact 4.2 of [9]* (09): the placement dossier doubted that it states the bijection; the merge re-read the PDF and found that it does, with only an assertion for proper classes (remark 1.2). 09's claims  $\mathcal{U}_{\mathbf{Oz}} = \text{Exp}(\mathcal{L}_A)$  and the upper half of “exactly  $r - 1$ ” in its nonsolvability theorem rest on that assertion and are stated so (corollary 3.5 and theorem 7.7).
3. *Stale novelty* (04, 08, 09): their reconstruction, first-order recovery of  $\mathbb{R}$  and definitions of  $\Pi$  were written at pins `fb5c4b53` and `befe739b`, before the Diophantine report's reconstruction section was placed; they are cited, not claimed (section 8). 04's remark

that its reconstruction is not first order is superseded by `odg:def:cor:internal`. 08's definition of  $\Pi$  has the structure of `odg:cor:definablect`, which was in the tree at 08's pin.

4. *08's Question 2* is answered ([section 8](#)) and dropped.
5. *Credits added*: `saut:thm:shiftflow` (08 and 09 did not credit it), `saut:thm:decomp` (04 and 09 credited only [\[10\]](#)), the rigidity report's `thm:no` and `cor:question54` (09 cited only [\[9, Proposition 5.2\]](#)), `bst:thm:fieldcriterion`, `bst:prop:localdensity`, `osq:prop:classder` and `odg:thm:fractions` (for 02); and the surcomplex report's  $T_1(t) = t/(1-t)$  is noted not to preserve  $\mathbf{Oz}$  ([example 4.12](#)).
6. *Repository statements checked true*: at both pins the catalogue listed 51 reports (04, 08); the fixed-shift flow is `saut:thm:shiftflow` (04); phase twists preserve  $\mathbf{Oz}[i]$  and move the real axis (04); the exponential-rigidity question is answered in its report (04); Pell rigidity and a definition of  $\mathbb{Z}$  were in the Diophantine report (08);  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  was there (09, 02); no report contained an  $\mathbf{Oz}$ -automorphism classification (04, 08, 09). 04 reports two date fields for [\[9\]](#) v3 (23 April and 24 August 2026); it is cited as version 3 of 23 April 2026, as elsewhere in the collection.
7. *Stale novelty* (10): at its pin `58cd8e1`, which does not contain this report, 10 stated that no inspected report guide contains the support-cut criterion and proposed the criterion, the rank boundary, the specialized fixed fields and the parameter witnesses as new. At the merge these are [theorems 3.3, 4.2, 4.13 and 9.1](#); 10 is credited as an independent fifth source, and only the material listed in [section 1.2](#) is printed as its contribution ([section 11.1](#)).
8. *Checked true* (10):  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  is in the Diophantine report (`odg:thm:fractions`); the multiplier identity, real-coefficient rigidity and the Gaussian phase twists recorded in that report's reconstruction audit are now `odg:def:thm:multiplier`, `odg:def:thm:autreal` and `odg:def:lem:twist`; 10 credits them and its own proofs of them are not claimed as new. No mathematical error was found in 10.
9. *Status notes* (batch 30): [question 11.1](#) is answered by [theorem 23.2](#). The statements that recorded it as open or partly answered now carry status notes: [remark 3.6](#), [remark 3.9](#), the paragraph after [theorem 5.1](#), [corollary 5.2](#), the item on `osq:q:invisible` in [section 11.1](#), and the paragraph after [question 11.1](#). No label, statement number or earlier text was removed; the non-claims of 04, 09 and 10 are kept as source records with a status note ([section B](#)).
10. *A false reason, corrected* (09; merge text of batch 26): after [corollary 10.4](#) the report said that a real automorphism “preserves order and cannot move an ordinary real”. Read literally this is false: the Taylor automorphism `saut:thm:coeffflow` is strong and order preserving and moves a real  $b$  to  $b + t^\delta$ . The correct reason is that an automorphism preserving  $\mathbf{Oz}$  fixes  $\mathbb{R}$ , which is definable in the pair. The sentence is corrected in place; no result depended on it, so this is a correction, not a retraction.
11. *Credits added* (batch 30): `saut:thm:coeffflow` for the Taylor motion used by 12 and 13, which do not credit it (13's audit calls it an established mechanism); 11 credits it.
12. *Repository statements checked* (batch 30): 12's statement that this report's Theorem 5.1 is the strong factorization and that its Corollary 5.2 fixes  $\mathbb{R}$  and commutes with the projections and floor without strongness, and that 10's audit disclaims a classification of nonstrong automorphisms, is accurate at its pin `a6c68ac`. 13's statements on [theorems 3.3 and 5.1](#) and on the Diophantine reconstruction, and 11's on the surcomplex Taylor motions and exponent lifts, the Diophantine ideal test, multiplier and fraction field and

the automorphism rigidity of this report, are accurate at their pins. Their statements that no report proves automatic strongness, scalar detection or coefficient-moving embeddings are still true at the merge.

13. *Wrong statements in delivered files* (batch 30), disclosed and not edited, since delivered files are kept byte-identical: 11's audit (11-coefficient-gaps-SOURCE\_AUDIT.md, Section 2) calls a6c68ac3826ac337762b71a2ef901997d019260c "a tree SHA, not a commit SHA"; it is a commit (it is 12's pin), whose tree is 12bfef3. 11's pin cf56b89, its tree 21d04b0 and its four blob hashes are correct. 13's audit (13-omnific-isomorphisms-SOURCE\_AUDIT.md) gives 71c1710 as the blob of the top-level README.md at its pin 0865f04; that blob is the file at the earlier commit b895e86, and at 0865f04 it is 8c636c7. Neither affects the mathematics.

#### A.4 Symbols renamed from the sources

$\mathcal{J}$  (08, 09) and  $\mathcal{I}$  (04) are  $\Pi$ ;  $B$  (09),  $\mathcal{A}$  (04),  $\mathcal{B}_k$  (08) are  $\mathcal{A}_k$ ;  $A_D$  (09),  $O_D$  (04),  $A_k$  (08),  $R_\Gamma(D, k)$  (02) are  $\mathcal{R}_o$  or  $\mathcal{R}_o(k, \Gamma)$ , with the coefficient ring  $D$  renamed  $o$  because  $D$  names derivations;  $\mathcal{P}_\Gamma(k)$  (02) is  $\Pi_k(\Gamma)$ ;  $H_\delta$  (08) and  $H(a)$  (02) are  $C_\delta$  and, in [part II](#),  $C(a)$ ;  $\mathcal{U}_{A_D}$  (09) and  $\mathcal{U}_D(G)$  (04) are  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ ;  $D_{\varphi, \delta}$  (08) is  $D_{\delta, -\varphi}$ ;  $q_\eta$  (09),  $[\omega^\beta] \cdot$  (04),  $c_a$  (08) and  $\chi_b$  (02) are  $q_\beta$ ; 09's  $H = \omega^\eta$  and  $H_j$  are  $\eta = \omega^\beta$  and  $\eta_j$ ; 08's  $A_1, B_1, u, E$  are  $A_{-1}, B_{-1}, u_0, \exp$ ; 02's covering element  $c$  is  $z$ , its  $K_b$  and  $H_j$  are  $K_{bd}$  and  $\Gamma_{\leq j}$ . 08's  $\omega$ -convention statements are translated to the  $t$ -convention ([section 1.5](#)); [part II](#) keeps 02's large-monomial convention ([section 12.1](#)). Source 10 (large-monomial convention,  $X^g = t^{-g}$ , translated):  $H_h$  is  $C_\delta$  and  $C_g$  is  $H_g$  (letters swapped);  $\text{pr}_+, \text{ct}, \text{pr}_-$  are  $P_-, P_0, P_+$  (subscripts reversed);  $\mathcal{I}$  is  $\Pi$ ;  $R_D$  is  $\mathcal{R}_o$ ;  $B$  is  $\mathcal{A}_k$ ;  $\mathcal{U}_D$  is  $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ ; the shift  $h$  is  $\delta$ ;  $u_g$  is  $1 + \varepsilon_\sigma(-g)$ ;  $\lambda_b$  is  $q_b$ ;  $P$  and  $T$  are  $\Theta$  and  $\xi$ ;  $\sigma_{\lambda, F, h}$  is  $\sigma_{h, -\lambda}^F$ ;  $\mathcal{N}^\dagger$  is  $\mathfrak{S}^\dagger$  ([section 1.5](#)). Sources 11, 12 and 13 (all in the  $t$ -convention; the table of [section 20.3](#)):  $I_\Gamma, A_G, \mathcal{J}_\Gamma$  are  $\mathcal{R}_\Gamma$ ;  $A_\Gamma, B_\mathbb{R}, \mathcal{A}$  are  $\mathcal{A}_\Gamma$ ; 12's  $\ell_g$  and 13's  $L_\gamma$  are  $h_\sigma(g)$ ; 11's  $C_\tau$  is  $\mathcal{G}_\tau$  and its  $H$  is  $\tau(\Gamma)$ , its  $\mathcal{F}$  is  $C_1$ , its  $\Lambda$  is  $\Delta$ , its  $\Phi_{\tau, \rho}$  is  $J_{\tau, \rho}$ , its  $E_{d, \delta}$  is  $\mathcal{T}_{d, \delta}$ , its  $S_\tau$  is  $M_{1, \tau}$ , its  $T_h$  is  $\iota_\mu$ , its  $p$  is  $\mu_{11}$ , its  $h_{jk}$  are  $\mu^{\pm\pm}$ , its  $h_c, h_j, h$  in the parameter theorems are  $\mu^{\text{cmp}}, \mu^{\text{jmp}}, \mu^{\text{rel}}$ , its  $\mathcal{B}_\lambda$  is  $\mathcal{V}_\lambda$ , its  $J_\Gamma, L_\Gamma$  are  $\mathcal{R}_\Gamma^\mathbb{C}, K_\Gamma^\mathbb{C}$ ; 12's  $\theta$  is  $\iota_{\mu_{12}}$ , its  $\psi$  is  $\mu_{12}$ , its  $D, \rho_D$  and  $J_D$  are  $d, \rho_{d,1}$  and  $J_{\mu_{12}, d}$ , its  $j^\dagger(1)$  and  $E_j$  are  $e_\phi$  and  $\text{ret}_\phi$ , its  $\Lambda$  is  $\lambda_{\text{On}}$ , its **Sc** is **No**( $i$ ); 13's second  $\tau$  is  $\iota_{\mu_{13}}$ , its  $p$  is  $\mu_{13}$ , its  $\rho = \sigma|_k$  is  $\alpha$ , its  $D, E_D$  and  $F_D$  are  $d, \rho_{d,1}$  and  $J_{\mu_{13}, d}$ , its  $T^*$  is  $T^\dagger$ , its  $\lambda$  is  $\lambda_{\text{On}}$ , its  $\mathcal{P}$  is  $\Pi$ , and its factorization  $U \circ M$  is  $M \circ u$  with  $U = MuM^{-1}$ . The large-cardinal report's  $J$  is  $\hat{j}$ . No normalization changes.

## B Non-claims, by source

No source claims to solve a named conjecture, and none claims Lean verification, peer review or historical priority; the finite suites check identities only. The report keeps every limitation stated by each source.

#### Source 09.

1. The classification is only within strong, coefficient-fixing, leading-term-preserving automorphisms; it does not assert that an abstract automorphism preserving  $\mathcal{R}_o$  is strong, and nonstrong pair automorphisms are not classified (the definability recovery applies to them; the logarithmic classification does not).
2. Coefficient functionals come from an existing strong derivation; arbitrary functions satisfying the vanishing condition are not asserted to assemble into an operator.

3. The shears are plain-field automorphisms and are not claimed to preserve the Gonshor exponential.
4. No set or NGB class of class-function graphs is formed; actions are encoded by single class relations; supports are sets.
5. The exponential–logarithm correspondence and the factorization pattern are imported, not reproved or claimed; the explicit many-shift constructions avoid the class version of the import.
6. No new factorization theorem for omnific integers is claimed; [11] addresses a different problem.
7. **Oz** is not identified with a model of full ordinary arithmetic.
8. Truncation at a named monomial is not a uniform definable choice of boundary.
9. Nondefinability is relative to the field-and-integer-part language; Conway normal forms remain set-theoretically defined.
10. Hahn-sum formulas are not claims of convergence in the fine topology.
11. Modulus equivariance is not a claim that moduli are fixed.
12. The repository comparison was bounded and targeted; search coverage is not a proof of novelty.
13. The exact-arithmetic program checks finite identities only; it does not decide well-ordering, prove the correspondence, verify class-theoretic statements or certify novelty; the formalization order is a proposal, not implemented modules.

#### Source 04.

1. It does not classify abstract (nonstrong) automorphisms of **Oz**; the fraction-field extension of an abstract automorphism is not known to be strong.
2. It imposes no exponential, omega-map, birthday or simplicity structure, and does not revisit the collection’s exponential-rigidity question as if unanswered.
3. The criterion classifies existing maps; it is not a construction for arbitrary logarithmic characters, and the convex condition alone is not asserted to solve global admissibility.
4. The relative fixed field is computed for the elementary subgroup only; the full stabilizer only for monomial parameters; the Hahn hull is not asserted to be definable closure.
5. Class maps are treated individually; no class of all automorphisms is formed.
6. Nothing is claimed after naming a proper class of parameters.
7. Bare nondefinability of coefficient-one monomials already follows from character twists; the nondefinability statement alone is not advertised as a new mechanism.
8.  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  is a full-class statement; a set-sized Hahn field need not be the fraction field of its integer-part ring.
9. Its intrinsic coefficient invariance was not claimed to be first order (now superseded by the Diophantine report).
10. Real and imaginary shear times preserve different structures; its Gaussian result is not the first surcomplex automorphism moving the real axis.
11. No generation or density theorem; fine-topology convergence is not strong summation.

12. The fixed-shift exponential, the general monomial/kernel decomposition and the exponential–derivation correspondence are prior work.
13. Its finite checks are not proofs and do not implement arbitrary normal forms; no Lean file, no repository checkout or build; historical priority unverified.

#### Source 08.

1. The threshold covers the elementary family only; rank-one rescalings and characters preserve the integer part; not every leading-term-preserving automorphism is claimed to be the exponential of one fixed-shift derivation.
2. Finite-dimensional phase images are a sufficient condition only.
3. No first-order Gaussian reconstruction was claimed (one now exists, [section 8](#)).
4. The automorphisms are not asserted to commute with the surreal exponential, the Berarducci–Mantova derivation or the omega-map; the wreath product does not contradict rigidity for those structures.
5. Parameter avoidance fails for set-sized fields with the whole field named; a universe-relative collection is not identified with the proper class.
6. No classification of the full stabilizer, and no solution of additional-structure problems.
7. The reconstruction is not a homomorphism  $\mathbf{Oz} \rightarrow \mathbb{R}$ .
8. The nondefinability is not a statement about adding an arbitrary class predicate, nor about automorphisms required to commute with the omega-map.
9. Strong summation is not topological convergence.
10. Pell rigidity, the existence of nontrivial Hahn automorphisms, the derivation–exponential construction and the monomial-character viewpoint are not claimed new; some elementary lemmas may be known; priority is provisional.
11. Its questions are not named historical problems.
12. No Lean; not refereed; the checks are regression tests, not proofs of well-ordering, parameter avoidance or definability.

#### Source 02.

1. Root-covered is sufficient, not claimed necessary; failure of the monomial criterion is not used to claim failure of cancellation or rigidity.
2. The finite-type theorems are theorems about sets; no  $\text{Spec}(\mathbf{Oz})$  and no class compactness; “ $\text{Aut}(\mathbf{Oz})$ ” and ML of class rings are shorthand; the finite-type theorem is not a statement that a class cannot inject into a set.
3. It does not classify  $\text{Aut}(\mathbf{Oz})$ , all derivations, or multivariable polynomial automorphisms, and makes no Jacobian-conjecture claim.
4. Not every homomorphism into every target is constant: injectivity and domain targets are required; no noninjective analogue is given.
5. It does not claim that  $R[\mathbf{T}]$  has no locally nilpotent derivation; a Hahn expansion is not a grading.
6. The orbit-field formula concerns monomials; only a lower bound holds for the full formal image.

7. Formal flows involve no analytic convergence and no fine-topology limits,  $s$  is not a surreal, and they are not exponentiation inside a Hahn field.
8. The cancellation theorem does not solve the unrestricted cancellation problem;  $\text{ML}(R[\mathbf{T}]) = R$  is not a solution of a general stable-rigidity conjecture.
9. Not new: character twists [9], strong-invariance terminology [8], the Makar–Limanov framework [4], Euler derivations, Vandermonde arguments, and the example of [11].
10. No named factorization conjecture is claimed solved; the literature search is not a priority certificate.
11. The questions are not presented as published open problems; no Lean, no peer review; the checks are finite identities.

#### Source 10.

1. The classification is within strong, coefficient-fixing, leading-term-fixing automorphisms; nonstrong and coefficient-moving automorphisms are not classified, and strongness is not a harmless omitted hypothesis.
2. The criterion classifies existing strong maps; an abstract character obeying the cut condition is not asserted to give an automorphism, since summability and an inverse must be proved.
3. The rank theorem concerns the leading-term-fixing factor only; it does not say that all rank-one automorphisms are trivial, and monomial maps such as  $t^\gamma \mapsto t^{2\gamma}$  and character rescalings are not counterexamples. Archimedean rank is not the order-unit condition of the holonomic report.
4. The strong valued decomposition, principal-unit characters and their crossed law [10], simplicity rigidity [9], the existence of nontrivial strongly linear 1-automorphisms, Hahn normal forms and the support description of **Oz** [11] are prior work; the splitting is not a new decomposition theorem for  $\text{Aut}(\mathbf{No})$ .
5. Not every truncation is preserved, only signed Archimedean blocks.
6. The fixed field is relatively, not absolutely, algebraically closed; no exact transcendence degree of a class extension is claimed.
7. The nonabelian example gives no presentation or exhaustive description of the stabilizer; no transfinite composition or closure theorem is claimed.
8. No proper-class-indexed sum is formed and no class of automorphisms is formed; class maps are individual, in NBG with global choice.
9. The bounded-birthday automorphism does not preserve birthdays or the simplicity hierarchy globally.
10. The definable-closure statements are upper bounds, not descriptions of definable closure and not quantifier elimination; they do not recover every coefficient from the projections.
11. The formal exponential is not the Gonshor exponential.
12.  $\text{Frac } \mathbf{Oz} = \mathbf{No}$ , the multiplier identity and coefficient rigidity are prior repository results, credited and not claimed; automatic fixing of coefficients is asserted only for the full surreal field, not for arbitrary  $\mathcal{R}_o \subseteq K$ .
13.  $\mathcal{R}_o$  is called integer-part-shaped, not an integer part in every generality; no ordering of the complex field is asserted and no global surcomplex exponential is introduced.

14. The divisibility example is for  $\mathfrak{o} = k$  and does not assert that translation preserves  $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ ; the criterion is not transferred to other integer parts.
15. Initial subdomains, truncation-closed subrings and subrings stabilized by an automorphism are different notions, and none is substituted for another.
16. It does not answer the omega-map question of [9], factorization problems for omnific integers, or the holonomic order-unit question; no named published conjecture is solved, and its questions are research questions.
17. Its finite checks do not prove summability, well-ordering of arbitrary supports, the existence of separating functionals, class-size statements or nondefinability.
18. No Lean, no repository clone or build, no peer review; novelty is provisional, and the repository and literature comparison was targeted, not exhaustive.

#### Source 11.

1. It is an unrefereed research draft with no Lean verification; priority is provisional and the novelty assessment limited; the existence of proper Hahn-field embeddings is not claimed as new.
2. The classification carries the exact-monomial hypothesis: it is not a classification of all surreal embeddings, does not assert that omnific-preserving embeddings are strong, and does not cover characters or unit corrections.
3. Taylor automorphisms, exponent lifts, the reconstruction of constants and the classical inputs (Hahn fields, normal forms, real and algebraic closedness, quantifier elimination, extension of derivations) are prior work.
4. The derivations of  $\mathbb{R}$  are choice-dependent and noncomputable.
5. “Monomial-preserving” does not mean commuting with the omega-map, and “strong” does not include  $\mathbb{R}$ -linearity; no omega-map or exponential question of [9] is claimed, and the explicit map fails omega-map compatibility at 1.
6. Elementarity of pure-field embeddings is classical; fixing the coefficients is not proved sufficient for elementarity of the pair.
7. The continuum of conjugacy types is a lower bound, not a classification.
8. No nonconjugacy is claimed for automorphisms preserving  $\mathbf{Oz}[i]$  but not conjugation, and non-conjugation-compatible surcomplex embeddings are not classified.
9.  $\text{Frac } \mathcal{R}_\Gamma = K_\Gamma$  is not assumed for set-sized  $\Gamma$ .
10. No collection of class automorphisms or of open subclasses is formed; the topological claims are local.
11. The annulus bounds are external.
12. The finite tests prove no infinite, class, conjugacy or model-theoretic statement.
13. No repository was modified or built.

#### Source 12.

1. It is unrefereed, has no Lean verification, solves no named published conjecture and certifies no priority; the repository comparison was limited to four files and five public sources, the full articles and the ledger were not read, and a local clone failed.

2. Support duality and finiteness spaces are not new;  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  is classical; the reconstruction and the convex criterion are prior repository results.
3. Admissibility is retained: arbitrary coefficient arrays are not claimed to define automorphisms.
4. The proper embeddings are not a counterexample to strongness of all embeddings, and are not claimed initial, exponential or omega-compatible; the derivation is not computable.
5. The adjoint theorem is for set-sized groups only, fails for  $\mathbf{No}$ , and does not apply to non- $\mathbb{R}$ -linear maps.
6. No unrestricted Gaussian valuation-preservation theorem is proved, and preservation of the real axis is not assumed.
7. The support-bound result is for regular uncountable  $\kappa$  only; no singular case, and no identification with a birthday cutoff.
8.  $\text{ret}_\phi$  is not a field retraction or a positive conditional expectation.
9. No preservation of the exponential, logarithm, omega-map, birthdays or simplicity is claimed.
10. Invariance under automorphisms is not first-order definability.
11. The set-pair corollary does not assert that automorphisms of  $\mathcal{R}_\Gamma$  extend when  $\text{Frac } \mathcal{R}_\Gamma$  is smaller.
12. Omnific-preserving automorphisms are not claimed to be trivial.
13. The distinction from the three-duals report is not an exhaustive non-overlap claim.
14. Finite checks are not proofs.

### Source 13.

1. It is AI-assisted and unrefereed, not proof-assistant checked, not a resolution of a named conjecture, and makes no priority claim; a failed search is not novelty, and it does not claim to settle every question posed.
2. Earlier repository results are credited: fractions, reconstruction, the strong stabilizer and the shears.
3. The scalar strong-dual representation is set-size only; the automorphism theorem is not.
4. The residue criterion is not valuation-topological convergence and not first-order definability of summation.
5. Not all automorphisms of  $\mathbf{No}$  are strong (the examples of [9] stand), and omnific automorphisms are not trivial.
6. Local convex vanishing is not proved sufficient, and no class-wide operator exponential-logarithm correspondence is used.
7. It does not prove that every automorphism of  $\mathbf{Oz}[i]$  is valuation-compatible, and conjugation is not intrinsic; its two Gaussian theorems must not be merged into an unconditional classification.
8. The full-pair theorem assumes an ambient isomorphism; isomorphisms of the small integer parts are not claimed to extend.
9. No Gonshor exponential, Berarducci–Mantova derivation or omega-map is used; its exponent map is not a field map; no proper-class Zorn lemma or basis is used.

10. The drift embedding comes with no uniqueness claim; the failure of homogeneity concerns set-sized induced substructures, not elementary maps.
11. Finite checks do not verify infinite summability, transcendence bases or surjectivity.
12. No repository build was run, and the formalization ledger was not read.

*Status note* (batch 30). Non-claim 1 of 09, non-claim 1 of 04 and non-claim 1 of 10 say that nonstrong automorphisms are not classified and that strongness is not shown. They are kept as the sources’ records. For **(No, Oz)** and the full real Hahn pairs with divisible exponent group, [theorem 23.2](#) and [corollary 23.4](#) now show that every automorphism is strong, so [theorem 24.1](#) classifies them all; for **Oz**[*i*] without conjugation or valuation data, and for a general  $(k, \mathfrak{o})$ , those non-claims still describe the state of knowledge ([questions 32.1](#) and [32.13](#)).

Counts: 13 non-claims from 09, 13 from 04, 12 from 08, 11 from 02, 18 from 10, 13 from 11, 14 from 12, 12 from 13.

## C Finite checks

Eight suites ship with the report, byte-identical to their delivery, under the prefixes of their manuscripts. They check finite identities in exact arithmetic; they do not prove Hahn summability, class-sized statements, definability or novelty. They were rerun on copies for the merges (10’s for the batch-28 addition, 11’s, 12’s and 13’s for the batch-30 addition) with Python 3.14.4 and SymPy 1.14.0, with the recorded results reproduced.

| Script and record                                                                                                          | Content and behaviour                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code/09-omnific-preservin<br>g-verify_finite_identities.py;<br>data/09-omnific-preservin<br>g-verification_output.txt      | Standard library. Product rule and bracket on finite Laurent polynomials, balanced brackets for ranks 2 to 8, mixed group-commutator coefficients with square-zero parameters, shear iterates, exponential laws, the rank-one escape, Pell growth. Twelve PASS lines and “ALL CHECKS PASSED”. Prints only.                                                                   |
| code/04-preserving-autom<br>orphisms-verification.py;<br>data/04-preserving-autom<br>orphisms-verification_re<br>port.json | SymPy 1.14.0 (data/04-preserving-automorphisms-requirements.txt). 7,062 assertions in 17 groups: exponential group law, bracket formula, coefficient polynomial, first corrections, commutator coefficients in lexicographic $\mathbb{Q}^3$ . <i>Writes</i> its report to <code>--output</code> , by default <code>verification_report.json</code> in the working directory. |
| code/08-integer-part-sym<br>metries-verify.py;<br>data/08-integer-part-sym<br>metries-verification.json                    | Standard library. 9,469 checks in 15 categories: derivation rule, falling-factorial iterates, flow composition, bracket, Vandermonde determinants, the semidirect-product law, commutator phases. <i>Always writes</i> <code>verification.json</code> beside itself.                                                                                                         |
| code/02-parameter-rigidit<br>y-verify.py;<br>data/02-parameter-rigidit<br>y-verification.json                              | Standard library. 7,041 assertions in 13 families: Euler product rule, factorial Leibniz, truncated flow composition, commutation, geometric remainders, Vandermonde determinants, binomial identity. Writes <code>verification.json</code> beside itself unless <code>--output</code> is given.                                                                             |

| Script and record                                                                                                       | Content and behaviour                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>code/10-support-cut-verif<br/>y.py; data/10-support-cut<br/>-verification.json</code>                             | Standard library, exact rational arithmetic. 28,668 assertions in 19 categories: binomial coefficients, formal additive composition and inverses by two independent power algorithms, rank-two composition, inverse and multiplicativity, signed support projections, exact first displacement, the least-bad-shift coefficient and the rational root crossing zero, the three-scale commutator. Writes <code>verification.json</code> beside itself unless <code>--output</code> is given. |
| <code>code/11-coefficient-gap<br/>s-verify_finite_models.py;<br/>data/11-coefficient-gap<br/>s-verification.json</code> | Standard library, exact fractions, seed 20260923. 3,880 assertions in 15 categories: truncated Taylor identities over $\mathbb{Q}[b, b^{-1}]$ , multiplication of finite outer Hahn polynomials with lexicographic shifts, Taylor inversion, integer-part preservation and reflection, signs of negative blocks, rational samples of the index maps, parameter fixing. Prints to standard output; writes a file only with <code>--output</code> .                                           |
| <code>code/12-automatic-strongn<br/>ess-verify.py;<br/>data/12-automatic-strongn<br/>ess-verification.json</code>       | Standard library, exact fractions, seed 20260923. 17,774 assertions: 10,800 triangular-detector checks, 4,502 monomial adjoint and transfer checks, 2,472 rank-two Taylor-block and binomial checks. <i>Writes</i> <code>verification.json</code> beside itself unless <code>--output</code> is given.                                                                                                                                                                                      |
| <code>code/13-omnific-isomorphi<br/>sms-verify.py;<br/>data/13-omnific-isomorphi<br/>sms-verification.json</code>       | Standard library, exact fractions, seed 20260923. 13,885 assertions in 16 categories: binary finite-row choices and their persistence, descending-support detection, coefficient reconstruction, monomial adjoints and retractions, scale-separated Taylor models. <i>Writes</i> <code>../data/verification.json</code> relative to its own directory unless <code>--output</code> is given.                                                                                                |

The placement dossiers also recorded independent checks (281 for the automorphism cluster, including [example 3.2](#), the commutator formula for symbolic  $s, u$ , the leading terms of [theorem 6.8](#) for  $n \leq 4$ , the conjugation lemma and the failure example of [remark 4.11](#); for 02 the quotient-rule identity of [example 17.8](#) to 29 terms; and 8,724 assertions for sources 11–13, including the Taylor-block embedding in a lexicographic  $\mathbb{Z}^2$  model, the failure of the ambient Taylor automorphism to preserve **Oz** ([section 26.1](#)), the binary finite-row algorithm over  $\mathbb{F}_2$  and  $\mathbb{Q}$ , the triangular detector, the least-forbidden-shift coefficient and [example 23.6](#)); these scripts are not shipped.

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